

General Information

Project Name	YS-20240829-053522
Author	YellowScan CloudStation
Mission Start (UTC)	Thu Aug 29 2024 05:35:24
Mission End (UTC)	Thu Aug 29 2024 05:59:46
Mission Duration	00:24:22
Location	Tjodalyng, Vestfold, Norway



System Information

HARDWARE DETAILS

CPU	Intel(R) Core(TM) i9-10885H CPU @ 2.40GHz
RAM	32.00 Go

SOFTWARE DETAILS

Operating System	Windows 10 Version 2009
Computer Name	LT08
Qinertia Version	Qinertia UAV OEM 4.0.25506-stable
License	

Imported Data

Get details about the import process for each data source. Most importantly, you can check for data gaps and correct coverage.

Summary

INERTIAL MEASUREMENT UNIT (IMU)

Begin (UTC)	Thu Aug 29 2024 05:35:24
End (UTC)	Thu Aug 29 2024 05:59:45
Number Epochs	292403
Data Interval (Hz)	200.0
Coverage Ratio	100%
Status	Valid

REAL TIME GNSS POSITION (PVT)

Begin (UTC)	Thu Aug 29 2024 05:35:24
End (UTC)	Thu Aug 29 2024 05:59:46
Number Epochs	7311
Data Interval (Hz)	5.0
Coverage Ratio	100%
Status	Valid

RAW GNSS OBSERVABLES (RAW)

Begin (UTC)	Thu Aug 29 2024 05:35:24
End (UTC)	Thu Aug 29 2024 05:59:45
Number Epochs	1462
Data Interval (Hz)	1.0
Coverage Ratio	99%
Status	Valid

Input Files

EVENTS

Path
E:/20240829_Olberg/LIDAR/YS-20240829-053522/YS-20240829-053522.sbg

IMU

Path
E:/20240829_Olberg/LIDAR/YS-20240829-053522/YS-20240829-053522.sbg

PVT

Path
E:/20240829_Olberg/LIDAR/YS-20240829-053522/YS-20240829-053522.sbg

RAWGNSS

Path
E:/20240829_Olberg/LIDAR/YS-20240829-053522/YS-20240829-053522.sbg

Processing Settings

Summary

This section details the settings to use for the processing such as the GNSS antenna lever arm or how the IMU is oriented within the vehicle. If the report is generated from a real-time solution, and you have changed the settings read from the INS, the following settings could be inconsistent.

IMU Model Identifier	quantaMicroStdV1
IMU Name	Quanta UAV/Micro v1
Processing Mode	Tightly Coupled PPK (Single Base)
Motion Profile	UAV



Description

Quanta UAV/Micro is a very high performance and yet miniature MEMS based Inertial Navigation System (INS). It is ideally suited for high precision survey jobs with a post processed roll/pitch accuracy better then 0.015° and a single antenna true heading accuracy of 0.04°. The small and lightweight OEM package is ideal for UAV Lidar payloads or for tight mobile mapping and backpack OEM integrations.

Details

RAW GNSS OBSERVABLES (RAW)

Receiver Model	Ublox F9
Antenna Type	generic
Antenna Reference	ARP (Antenna Reference Point)
Antenna Height (m)	0.0
Primary Antenna Lever Arm (m)	[0.032, -0.095, -0.464]

INERTIAL MEASUREMENT UNIT (IMU)

Rough Alignment	X: Backward Y: Right Z: Up
Fine Alignment (°)	Roll: 0.0 Pitch: 0.0 Yaw: 0.0
Reference Point	Custom
Reference Point Offset	[-0.179, -0.005, -0.027]
Primary Lever Arm (m)	[0.0, 0.0, 0.0]

Summary

DEFINITIONS

Project CRS	USER:UB4BKBZXT3	Norway - ETRS89 (geographic 2D) + NN2000
Project Datum	EPSG:6258	European Terrestrial Reference System 1989 ensemble (ETRS89)
Project Effective Datum	EPSG:6258	European Terrestrial Reference System 1989 ensemble (ETRS89)
Origin Datum	EPSG:6258	European Terrestrial Reference System 1989 ensemble (ETRS89)
GNSS EphemerisDatum Datum (PPP)	EPSG:1322	International Terrestrial Reference Frame 2020 (ITRF2020)
GNSS Base Station Datum (Virtual RINEX)	EPSG:6258	European Terrestrial Reference System 1989 ensemble (ETRS89)

TRANSFORMATIONS

Source	Target	Code	Method	Accuracy (m)
[EPSG:1165] International Terrestrial Reference Frame 2014	[EPSG:1322] International Terrestrial Reference Frame 2020	EPSG:9991	Time-dependent Position Vector (Helmert 15-params (Inv. sign))	0.001
[EPSG:1165] International Terrestrial Reference Frame 2014	[EPSG:1186] European Terrestrial Reference Frame 2000	EPSG:8405	Time-dependent Position Vector (Helmert 15-params (Inv. sign))	N/D

Details

PROJECT COORDINATE REFERENCE SYSTEM

Datum	[EPSG:6258] European Terrestrial Reference System 1989 ensemble (ETRS89)
Ellipsoid	GRS 1980
Geoid	HREF2018B_NN2000_EUREF89
Projection	Popular Visualisation Pseudo-Mercator

POPULAR VISUALISATION PSEUDO-MERCATOR

False easting	False northing	Latitude of natural origin	Longitude of natural origin
0	0	0	0

TRANSFORMATION FROM ITRF2014 TO ITRF2020 (1) *EPSG:9991*

Translation (m) (m/year)			Scale (ppm) (ppm/year)	Rotation (arcsec) (arcsec/year)			Epoch
0.001400	0.000900	-0.001400	0.000420	0.000000	0.000000	0.000000	2015.000
0.000000	0.000100	-0.000200	0.000000	0.000000	0.000000	0.000000	

TRANSFORMATION FROM ITRF2014 TO ETRF2000 (1) *EPSG:8405*

Translation (m) (m/year)			Scale (ppm) (ppm/year)	Rotation (arcsec) (arcsec/year)			Epoch
0.054700	0.052200	-0.074100	0.002120	0.001701	0.010290	-0.016632	2010.000
0.000100	0.000100	-0.001900	0.000110	0.000081	0.000490	-0.000792	

Base Stations & GNSS Data

Find below the list of all base stations used in the processing.

Base Station VIRTUAL RINEX

INFO

Receiver	Trimble
Providers	
Constellations	GPS GLONASS GALILEO
Data rate (s)	1.0
Observation length (h)	4
Mean Baseline (km)	0.148

USED POSITION

Position Type	Manual
Latitude (dms)	59°0'28.483561"
Longitude (dms)	10°7'53.892839"
Ellipsoid Height (m)	109.502
Datum	ETRS89

ANTENNA

Model	TRM55971.00 NONE
Reference	ARP (Antenna Reference Point)
Height (m)	0.0

IMPORTED FILES

Path	Source
E:/20240829_Olberg/LIDAR/Rinex/virt242e00.rnx/virt242e00.24o	User Import

GNSS Resources

Reports global GNSS resources files used in the processing such as space vehicle ephemeris and clock.

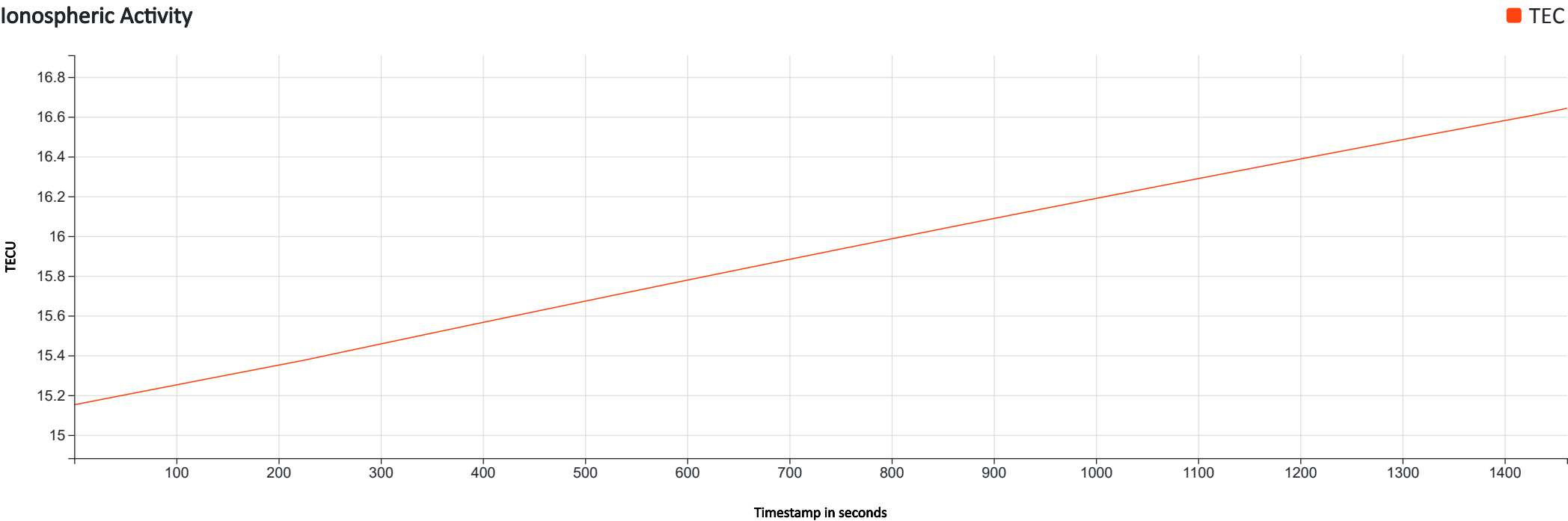
Type	Source	File
Broadcasted Ephemeris	online	ftp://igs-ftp.bkg.bund.de/IGS/BRDC/2024/242/BRDC00WRD_R_20242420000_01D_MN.rnx.gz
Broadcasted Ephemeris	User Import	E:/20240829_Olberg/LIDAR/Rinex/virt242e00.rnx/virt242e00.24g
Broadcasted Ephemeris	User Import	E:/20240829_Olberg/LIDAR/Rinex/virt242e00.rnx/virt242e00.24l
Broadcasted Ephemeris	User Import	E:/20240829_Olberg/LIDAR/Rinex/virt242e00.rnx/virt242e00.24n
Ionosphere Maps	online	https://gnss-api.sbg-systems.com/api/v1/resources/nasa/gnss/products/ionex/2024/242/c1pg2420.24i.Z

Ionospheric Activity

This section provides some info about ionospheric activity in the project area.

Activity	Medium
TEC (TECU)	16.4
IonoShield Activation Threshold (km)	15.0

Ionospheric Activity



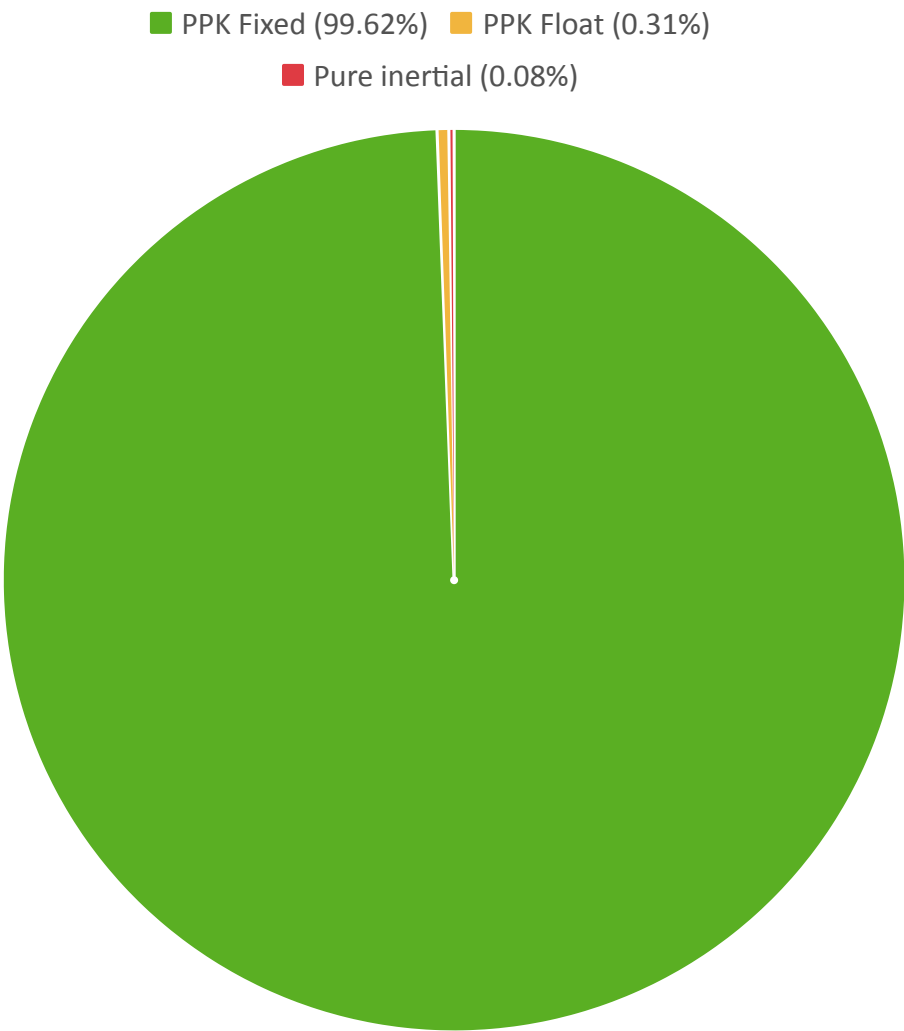
Processed Solution

Summary

This section presents the processed GNSS/INS solution results and quality.
You can review the data (attitude, trajectory, ...) as well as check overall QC indicators and estimated accuracies.

Selected Dataset	Auto (Merged)	Processing Started (Local)	Fri Aug 30 2024 03:35:16
IMU Model	quantaMicroStdV1	Pre-processing Duration (s)	5.4
Motion profile	UAV	Forward Duration (s)	25.3
Processing Mode	Tightly Coupled PPK (Single Base)	Backward Duration (s)	25.8
Passes Count	3	Merge Duration (s)	2.0
First Epoch (UTC)	Thu Aug 29 2024 05:35:23	Total Duration (s)	33.3
Last Epoch (UTC)	Thu Aug 29 2024 05:59:45		

Solution Types



Accuracies

ESTIMATED POSITION ACCURACY

	Average	Std.	RMS	Min	Max
Horizontal (m)	0.01	0.004	0.011	0.009	0.124
Vertical (m)	0.01	0.001	0.011	0.009	0.04

ESTIMATED VELOCITY ACCURACY

	Average	Std.	RMS	Min	Max
Horizontal (m.s ⁻¹)	0.0068	0.0014	0.0069	0.0055	0.0411
Vertical (m.s ⁻¹)	0.0037	0.0004	0.0037	0.003	0.0071

ESTIMATED ATTITUDE ACCURACY

	Average	Std.	RMS	Min	Max
Roll/Pitch (deg)	0.0182	0.0021	0.0183	0.0168	0.0463
Heading (deg)	0.1471	0.0309	0.1503	0.048	0.2224

Separations

POSITION SEPARATION

	Average	Std.	RMS	Min	Max
Horizontal (m)	0.0078	0.0069	0.0104	0.0	0.1104
Vertical (m)	0.0006	0.0073	0.0073	-0.0479	0.0611

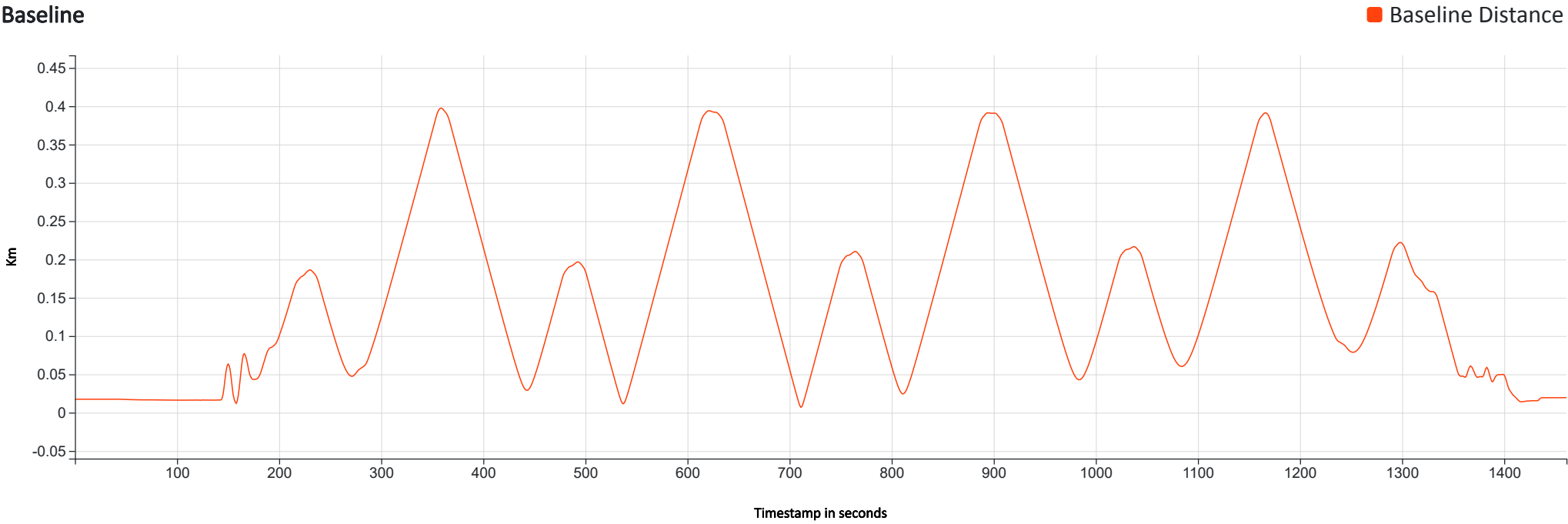
VELOCITY SEPARATION

	Average	Std.	RMS	Min	Max
Horizontal (m.s ⁻¹)	0.0052	0.0049	0.0071	0.0	0.0562
Vertical (m.s ⁻¹)	0.0002	0.0044	0.0044	-0.014	0.0127

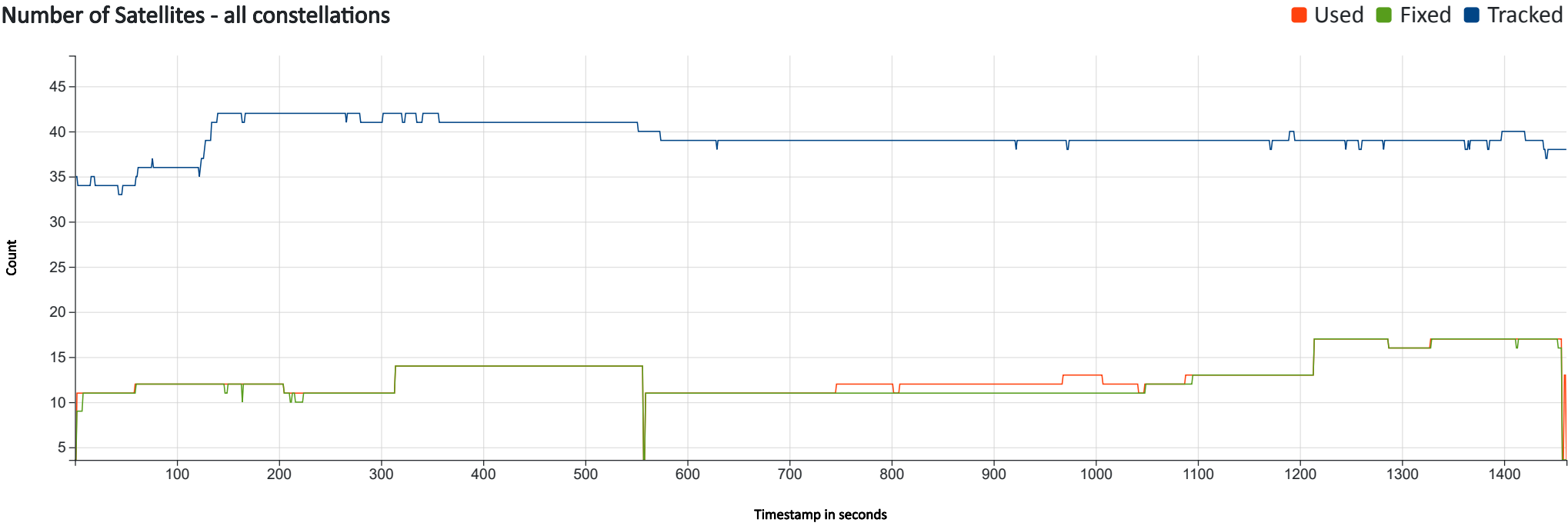
ATTITUDE SEPARATION

	Average	Std.	RMS	Min	Max
Roll/Pitch (deg)	-0.0068	0.0221	0.0231	-0.1175	0.1561
Heading (deg)	0.4681	0.7102	0.8506	-0.4934	1.881

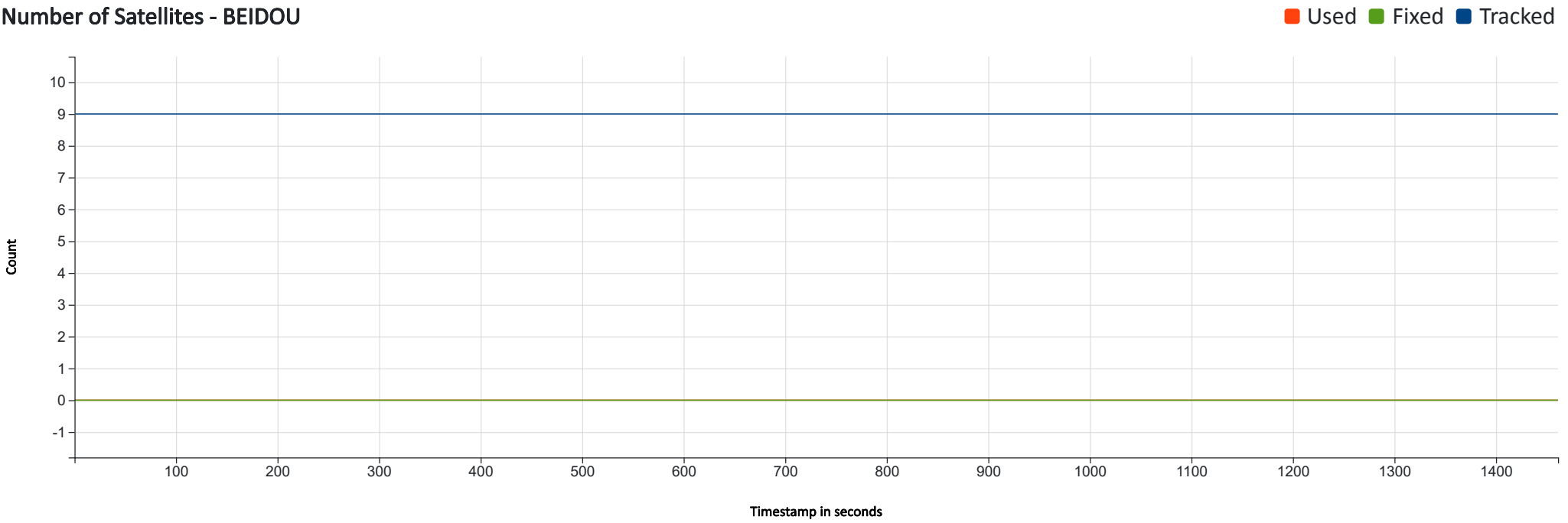
Baseline



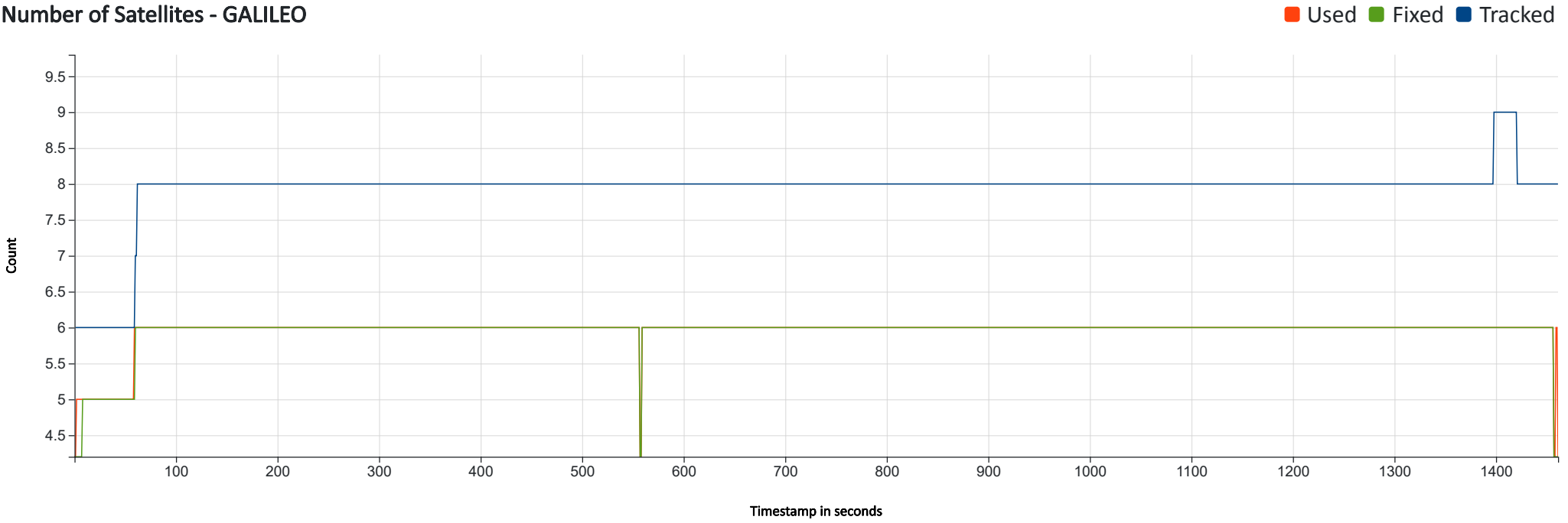
Number of Satellites - all constellations



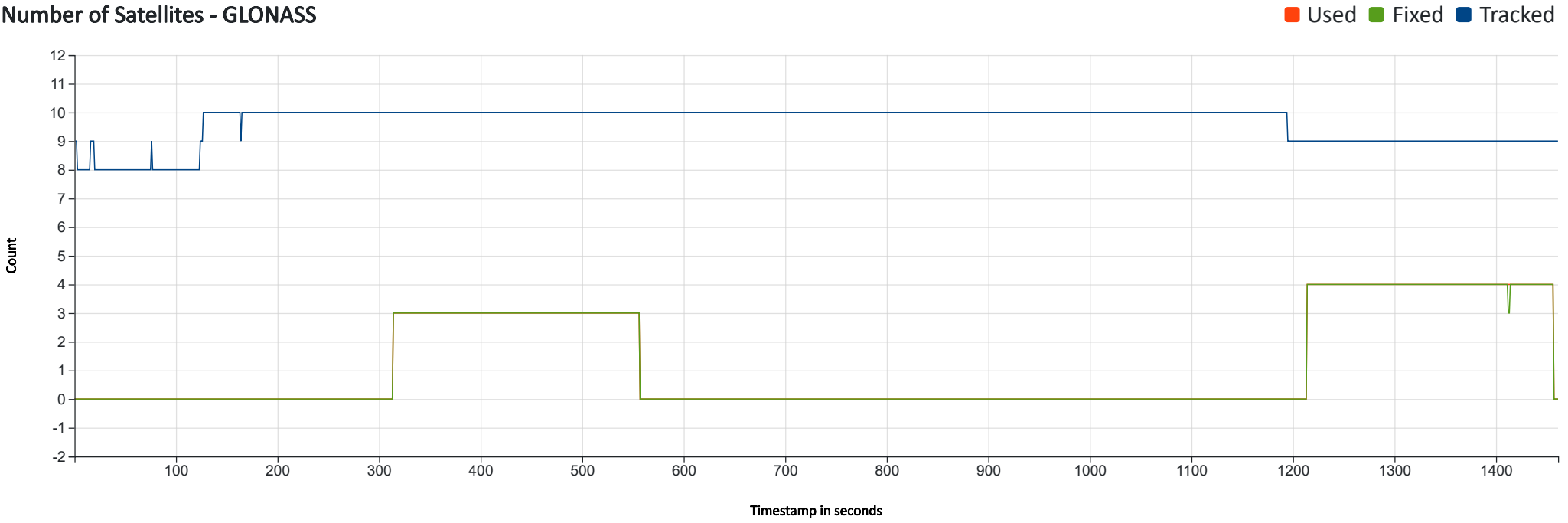
Number of Satellites - BEIDOU



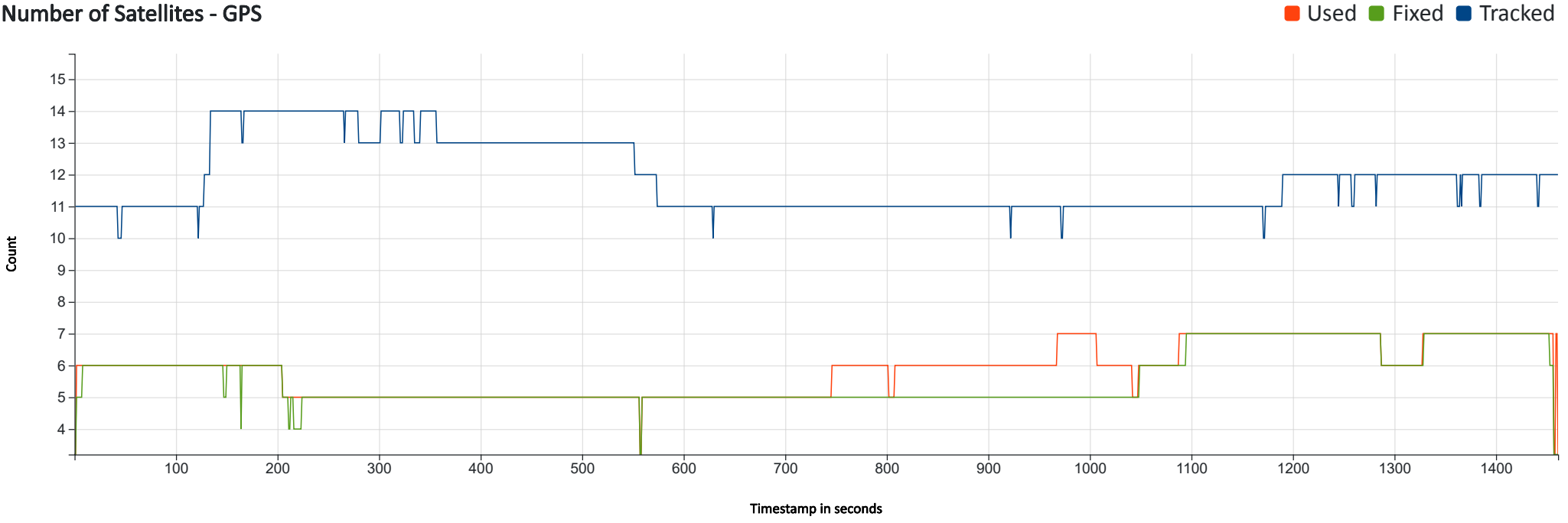
Number of Satellites - GALILEO



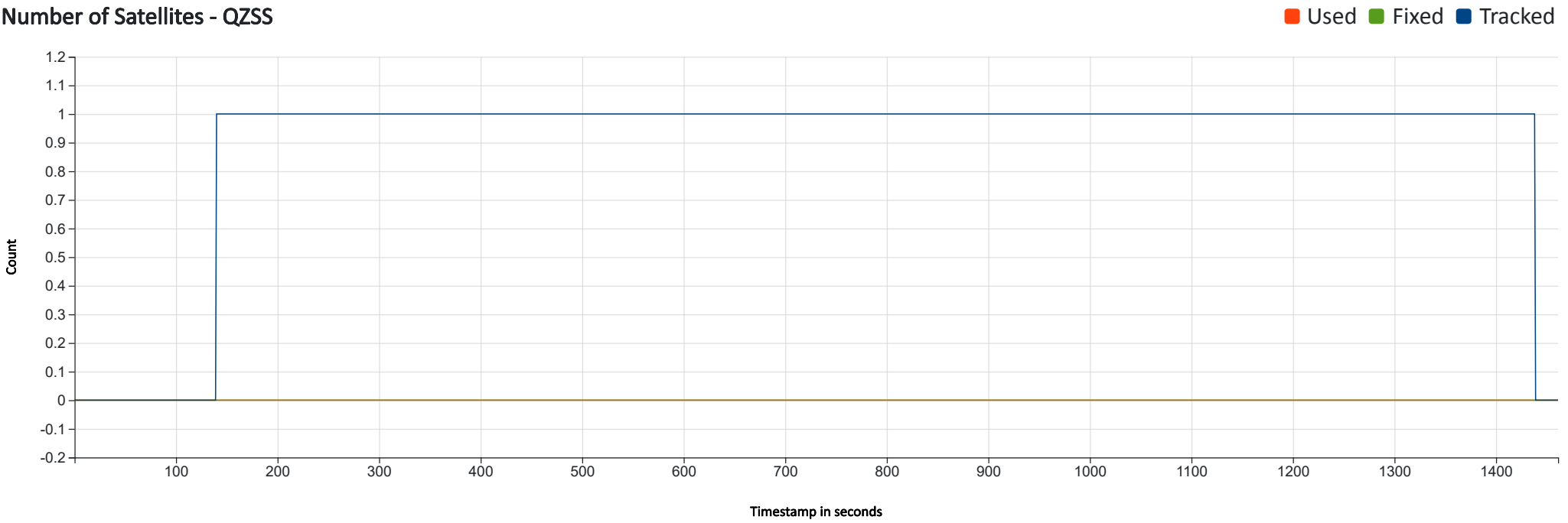
Number of Satellites - GLONASS



Number of Satellites - GPS

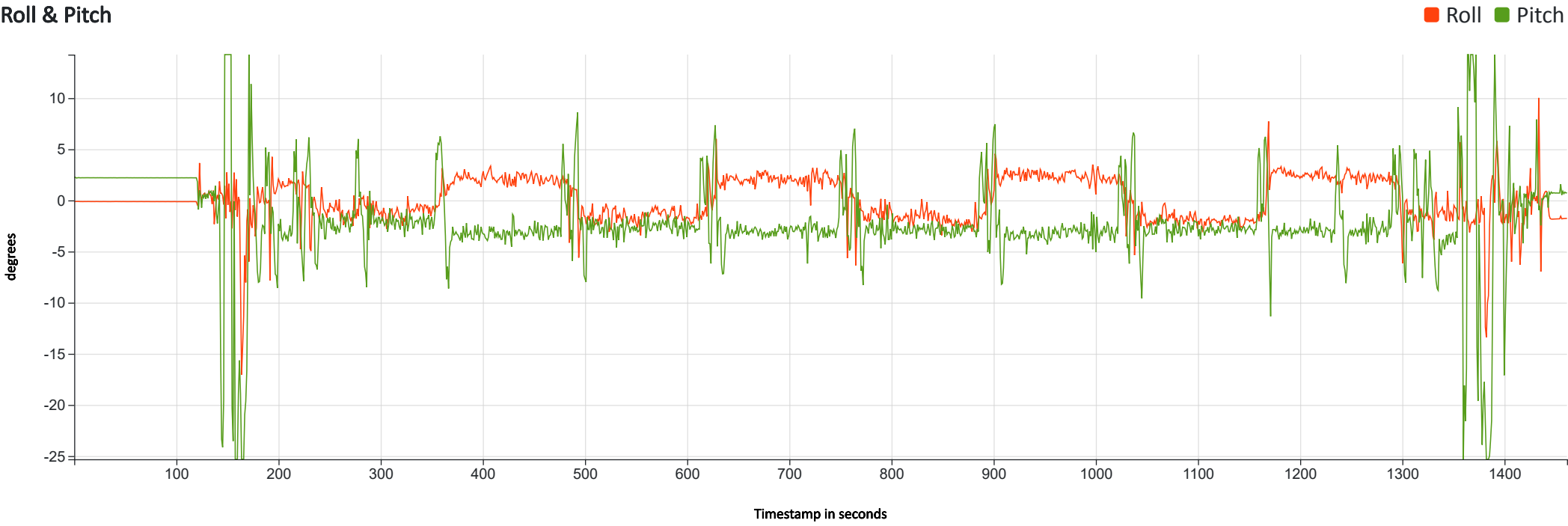


Number of Satellites - QZSS

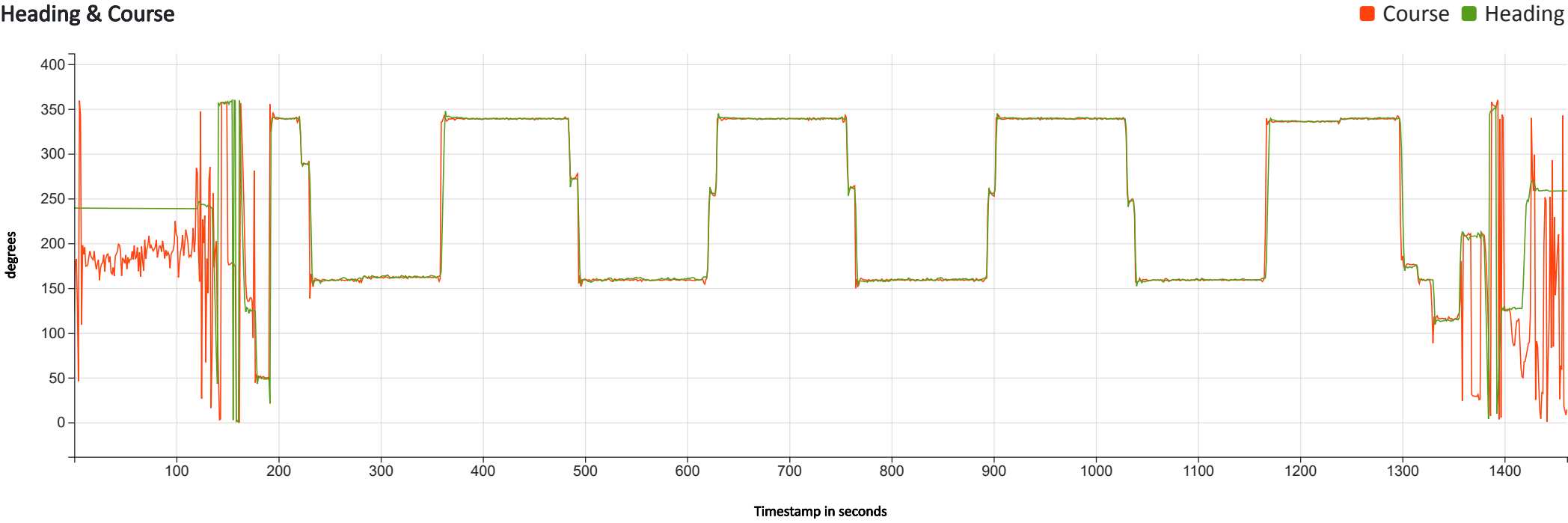


Attitude

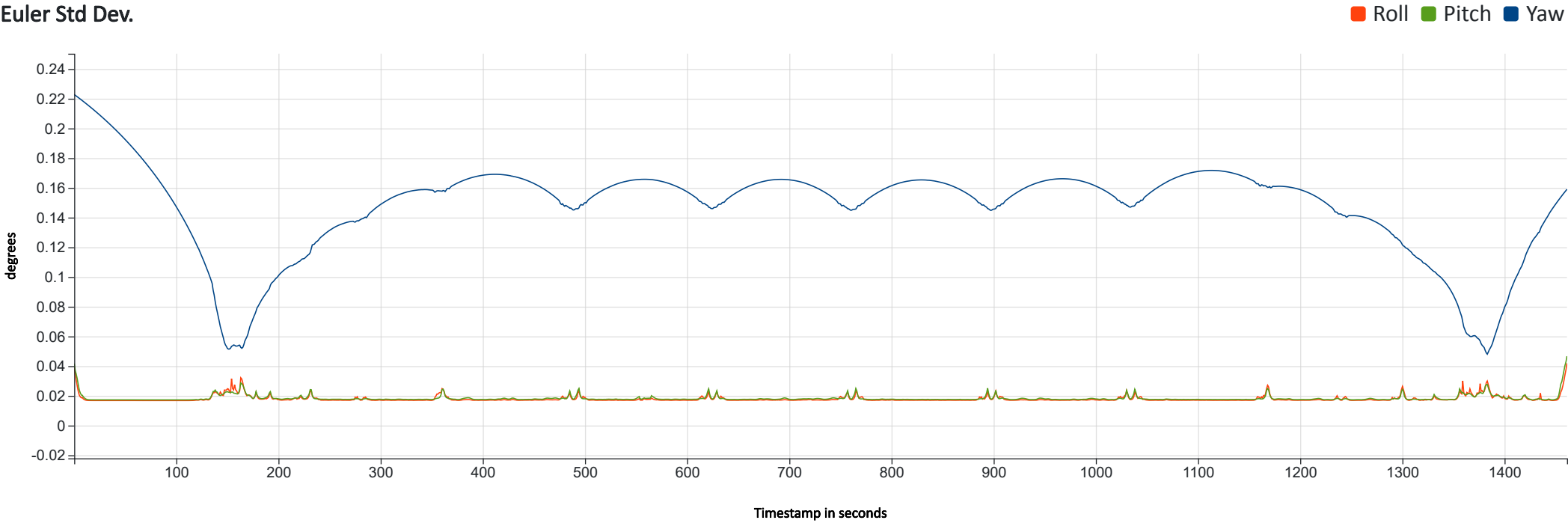
Roll & Pitch



Heading & Course

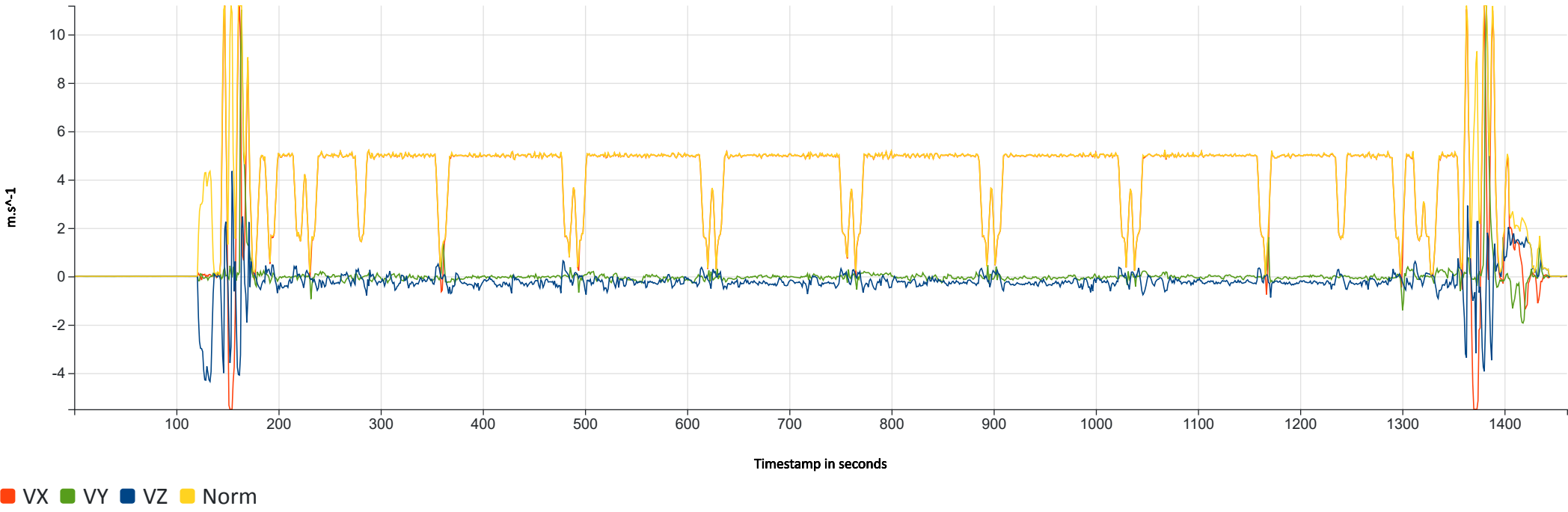


Euler Std Dev.

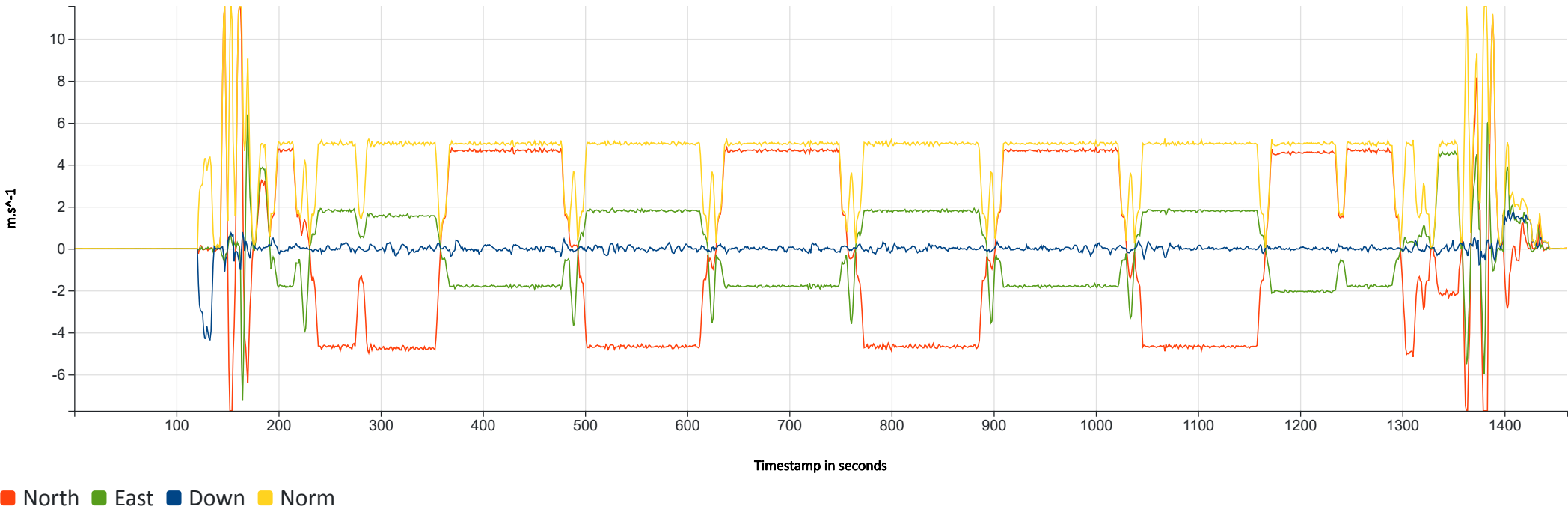


Velocity

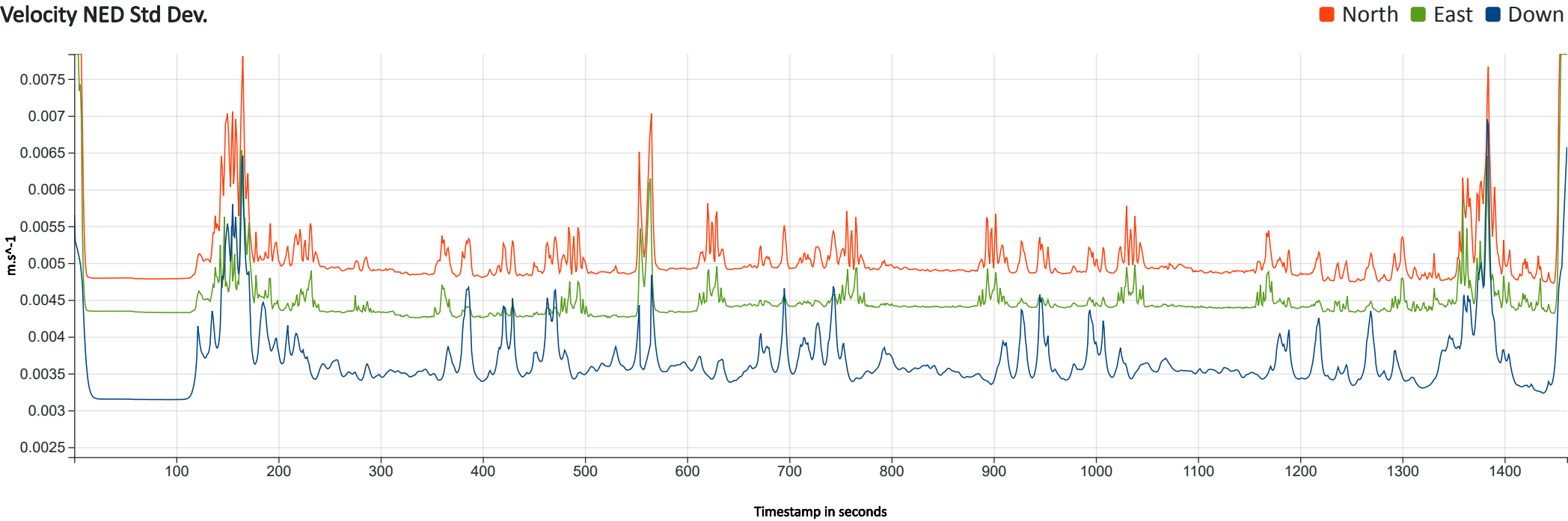
Velocity Body



Velocity NED

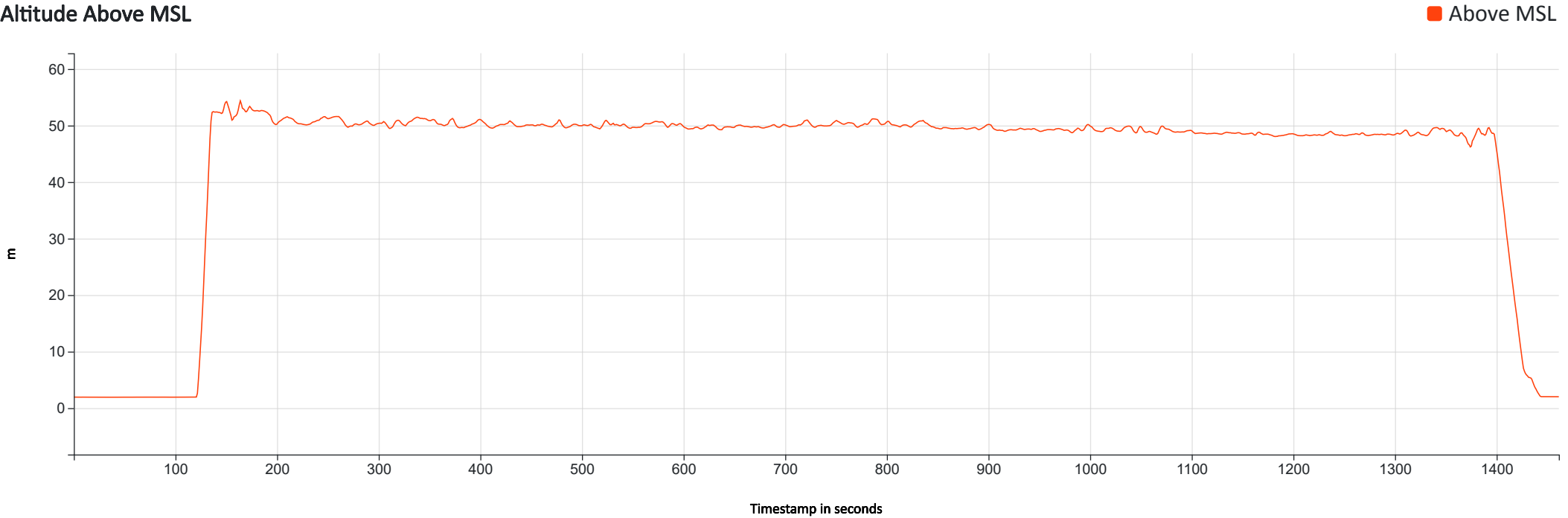


Velocity NED Std Dev.

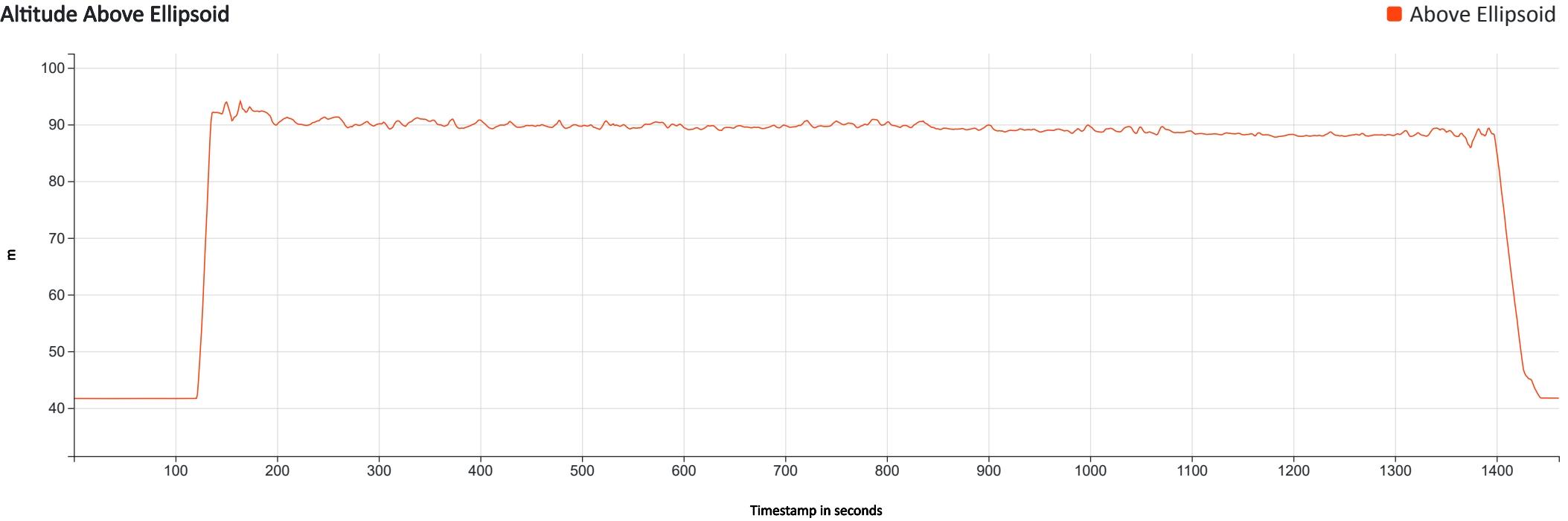


Position

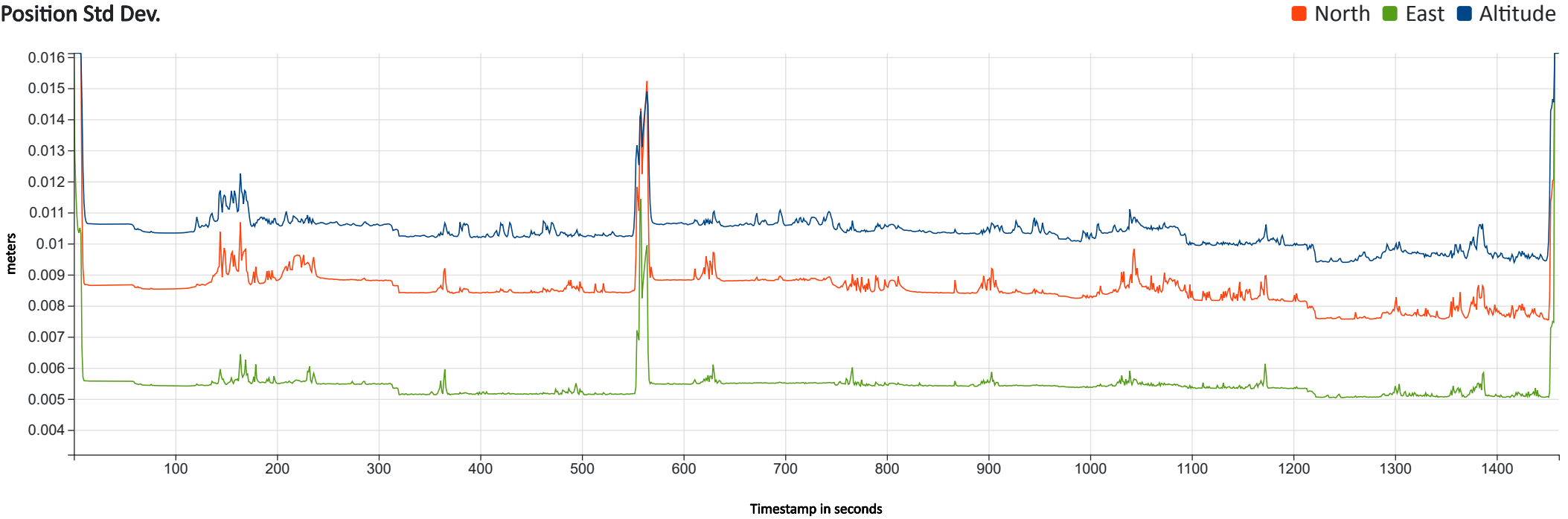
Altitude Above MSL



Altitude Above Ellipsoid



Position Std Dev.



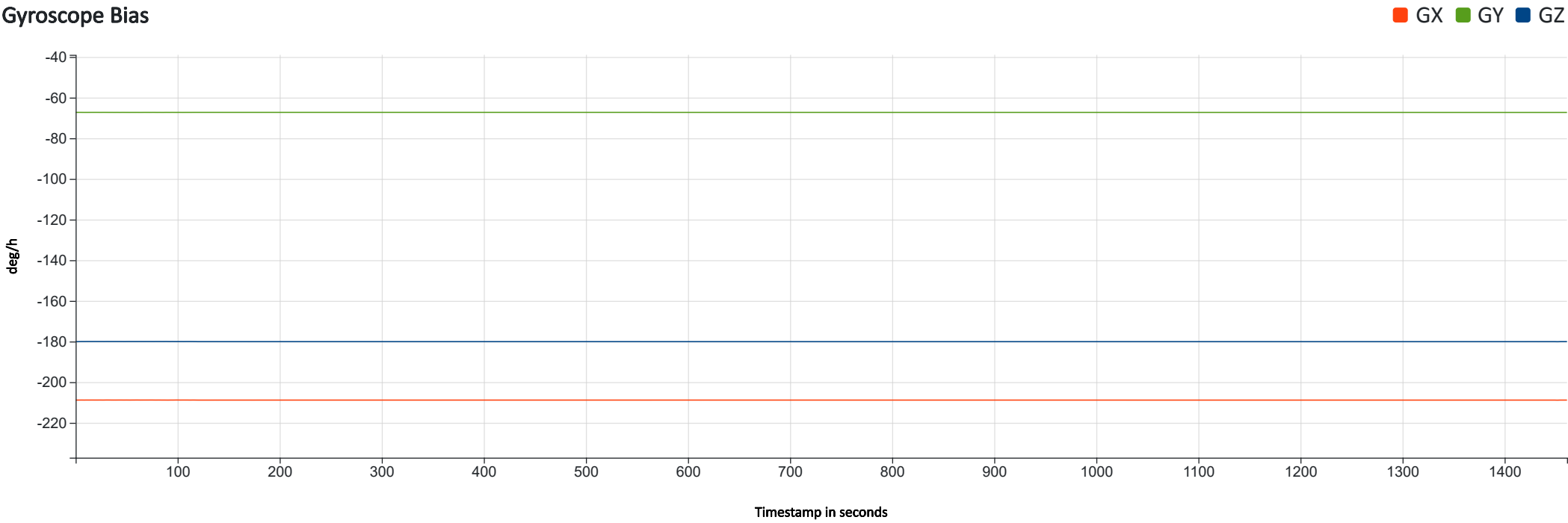
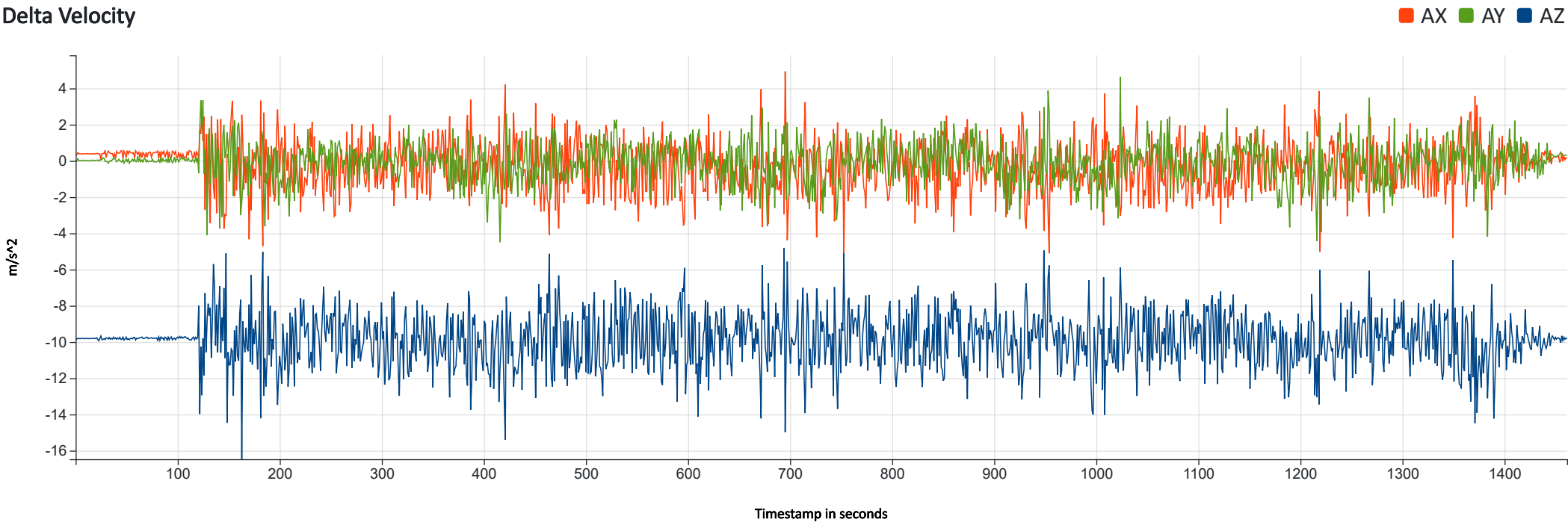
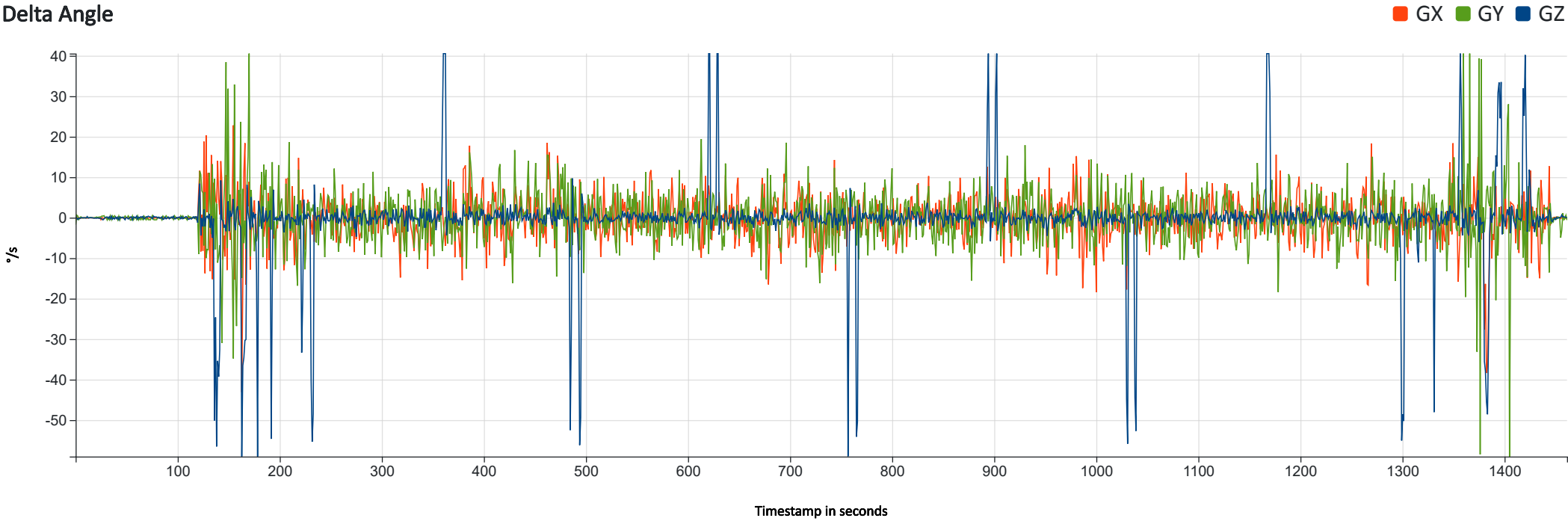
Trajectory (Quality)



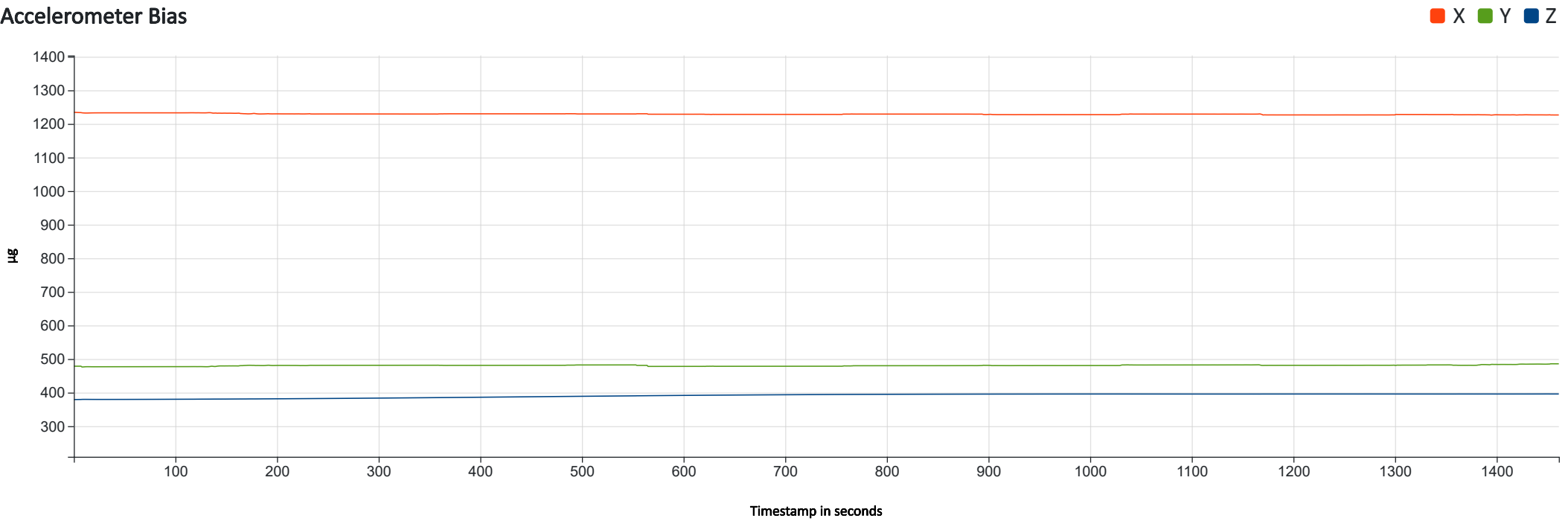
Trajectory (Solution type)



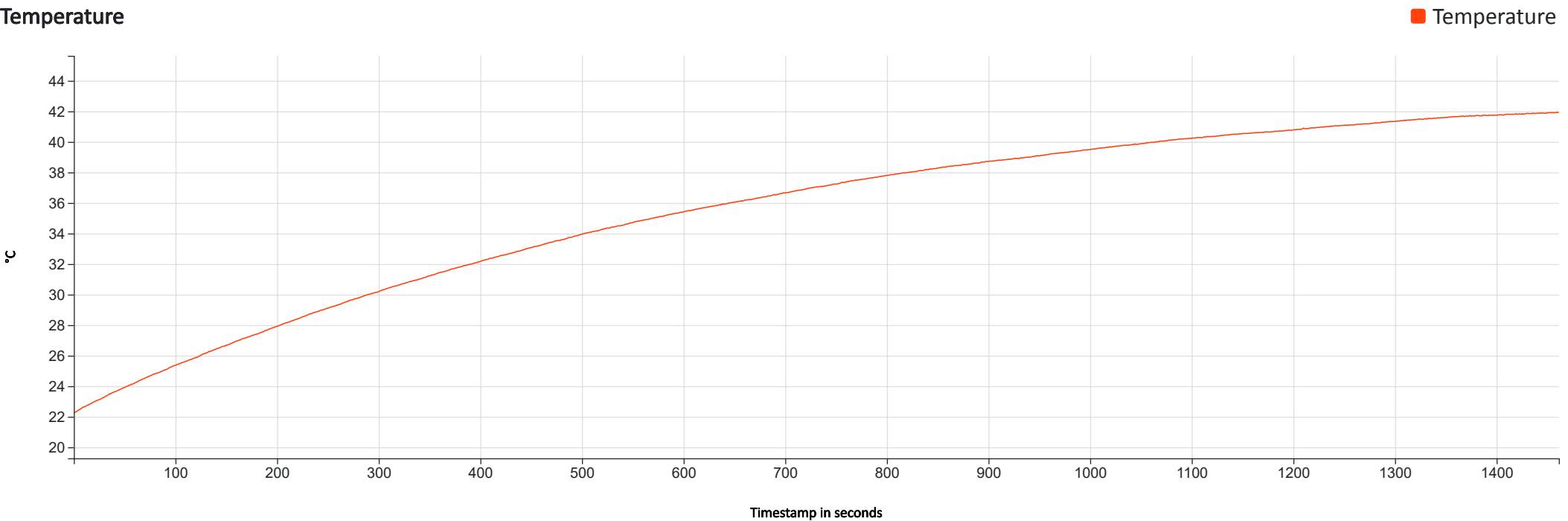
IMU



Accelerometer Bias

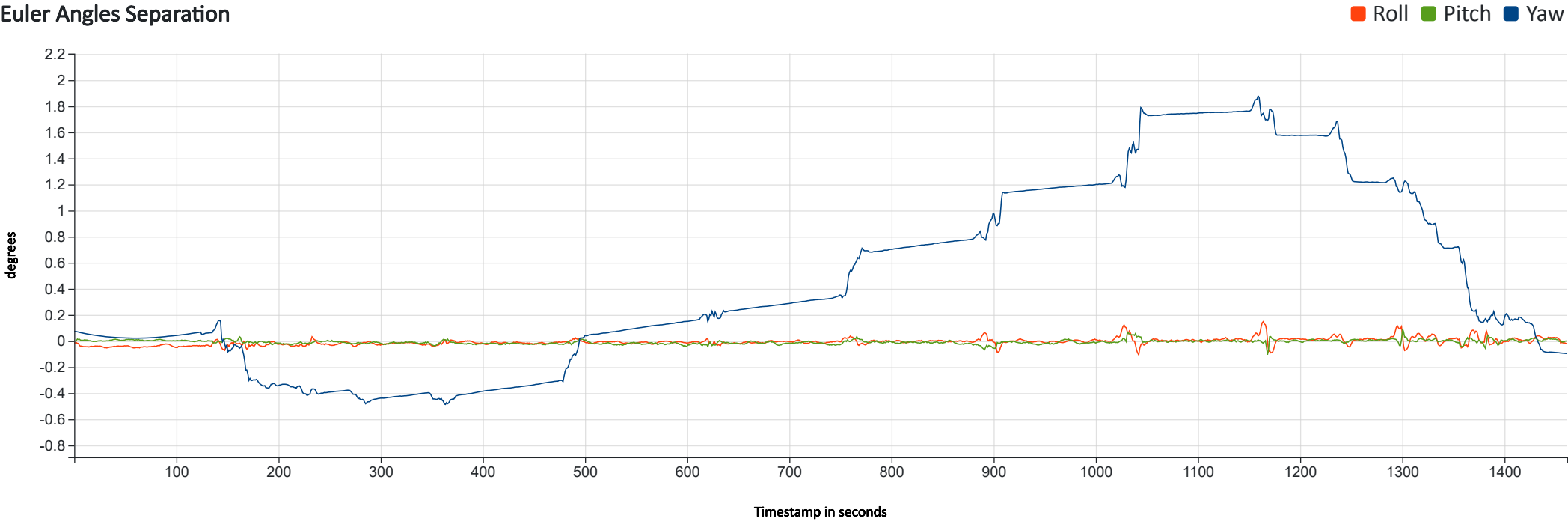


Temperature



Separations

Euler Angles Separation

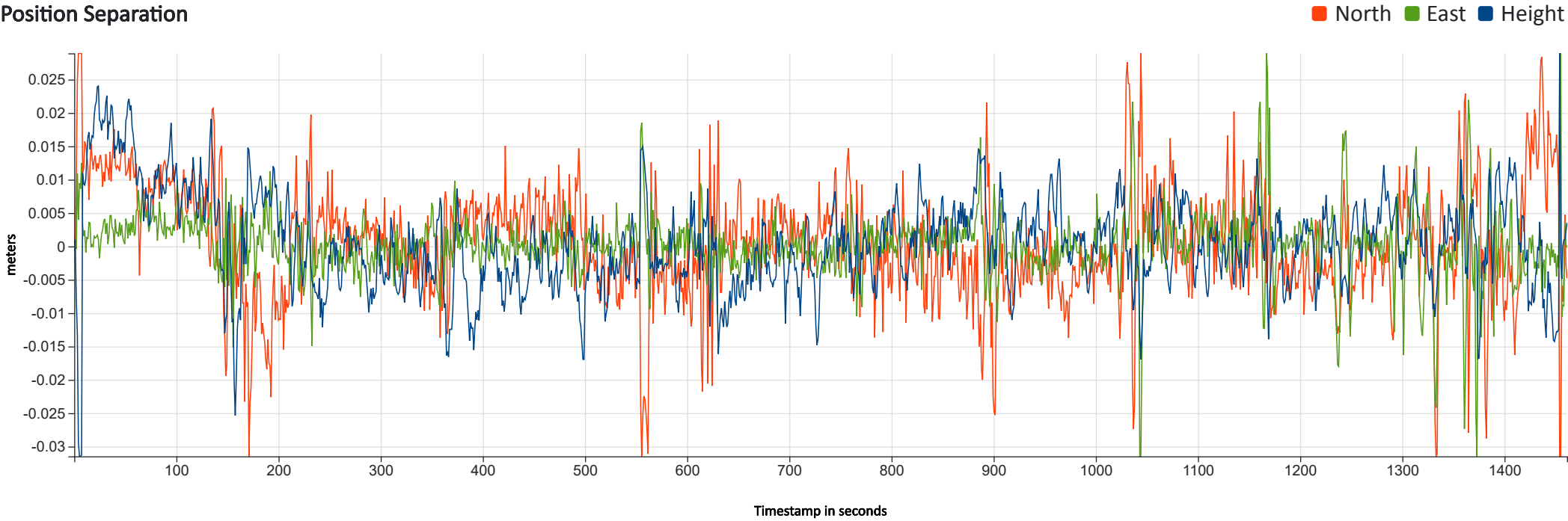


Velocity Separation



North East Down Norm

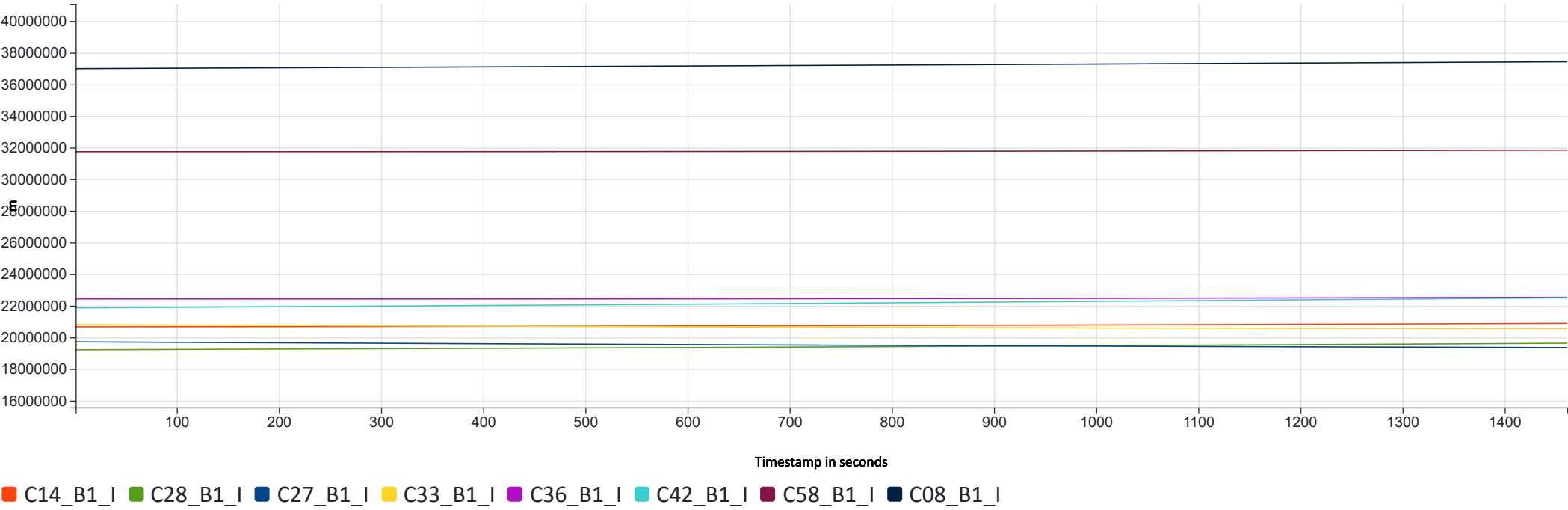
Position Separation



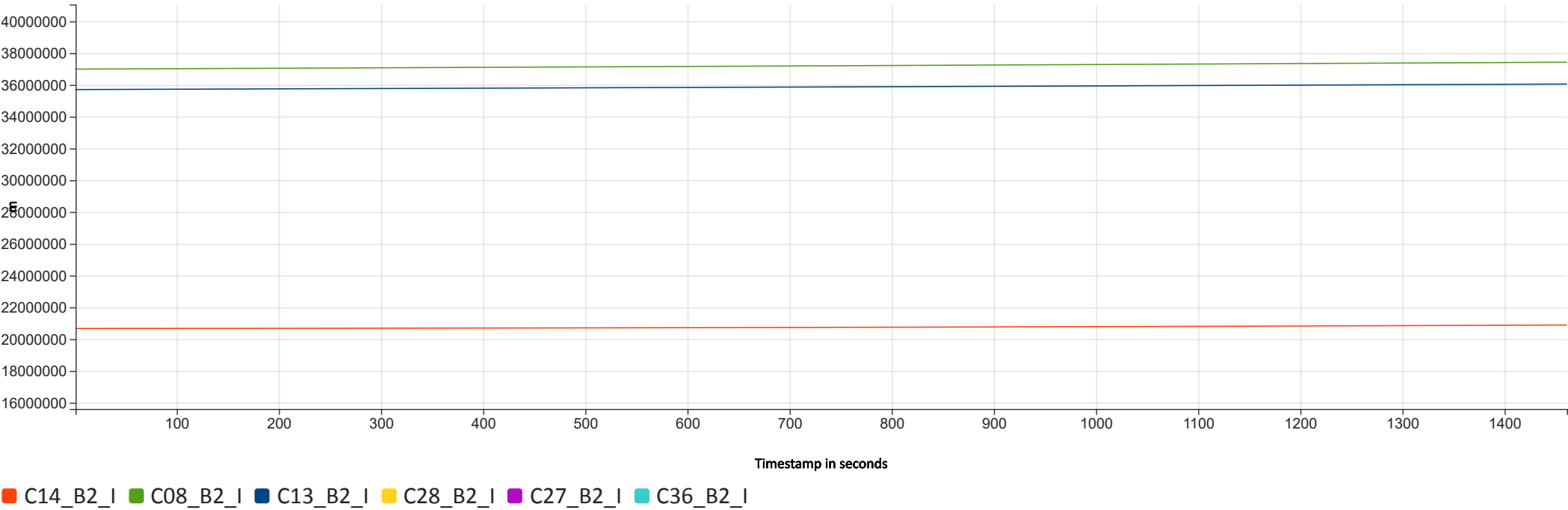
Advanced Rover GNSS QC

Pseudo range

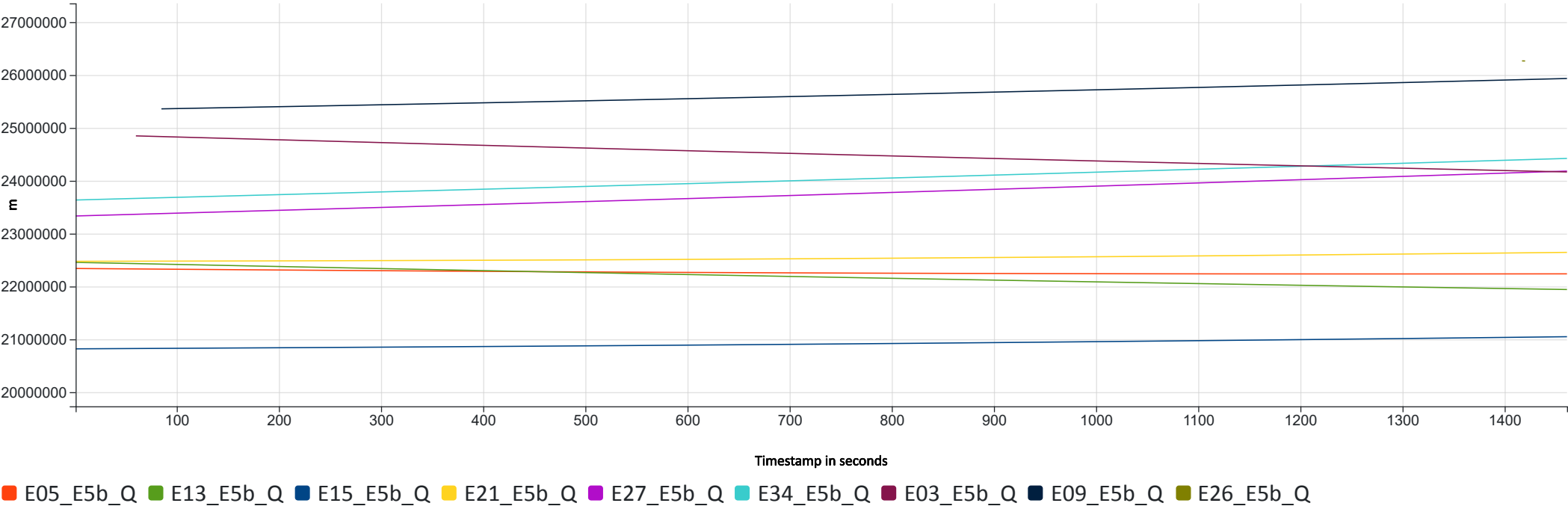
Pseudo range (m) - Beidou B1



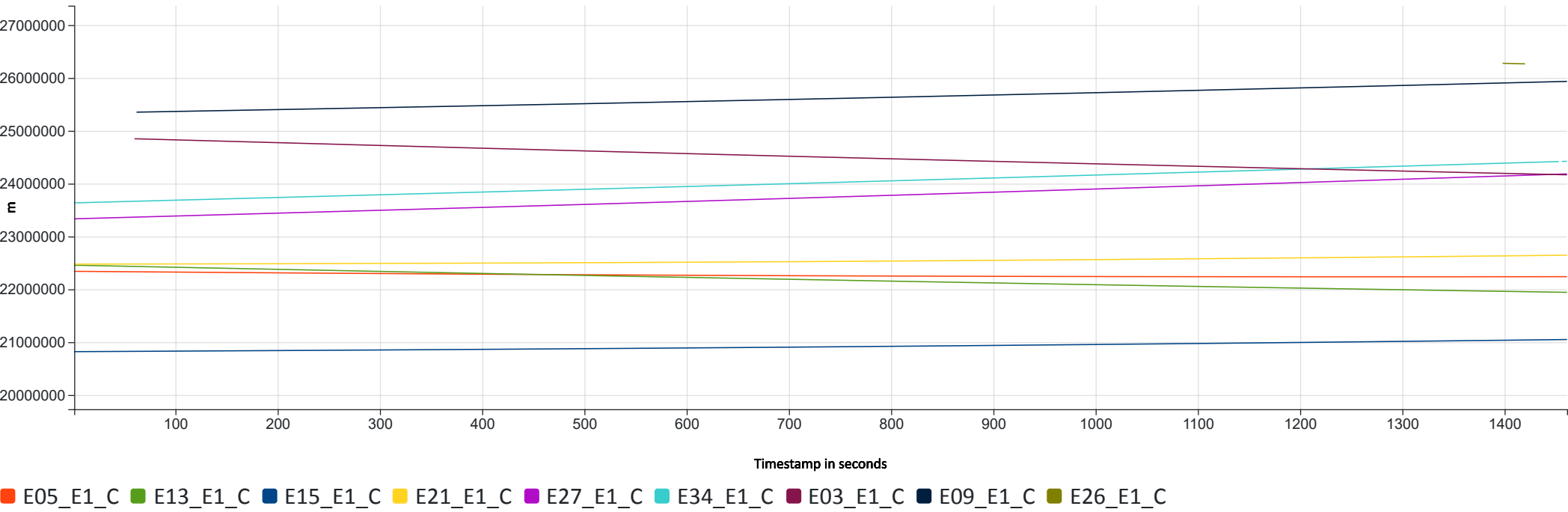
Pseudo range (m) - Beidou B2



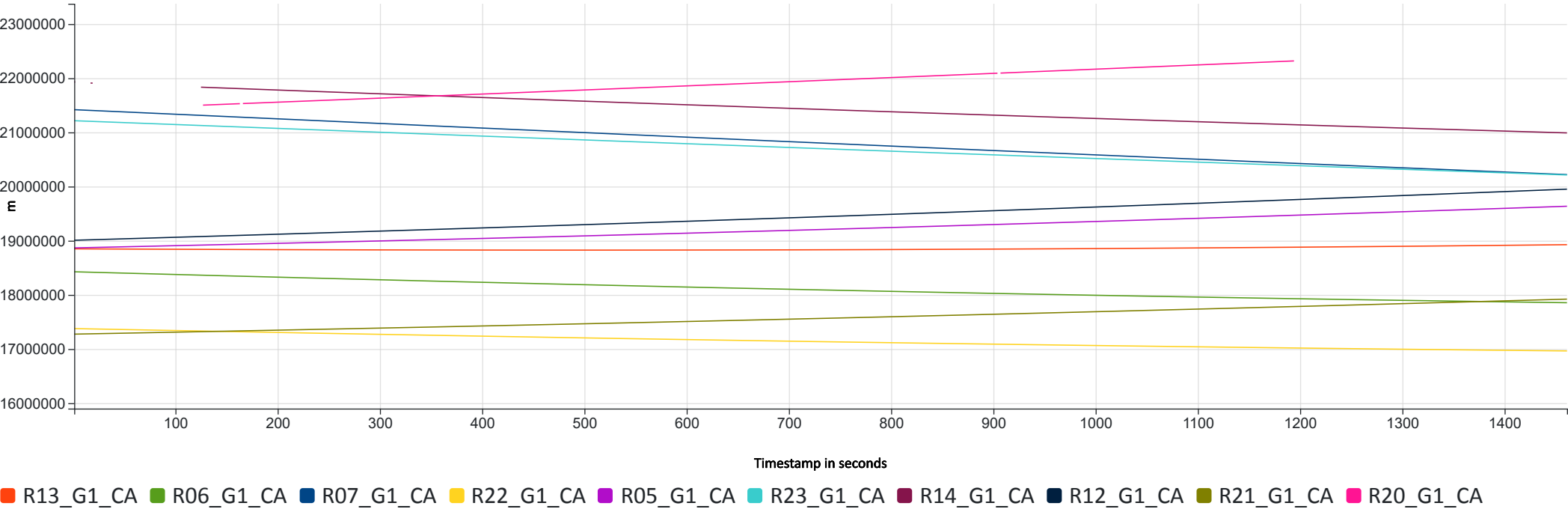
Pseudo range (m) - Galileo E5b



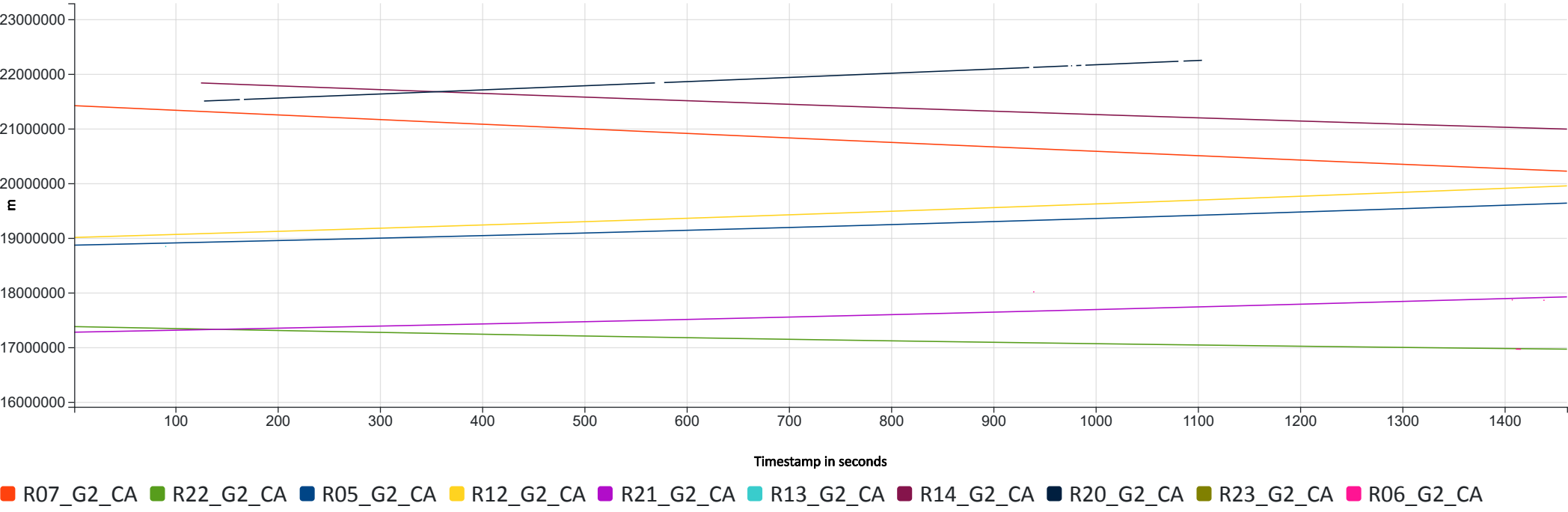
Pseudo range (m) - Galileo E1



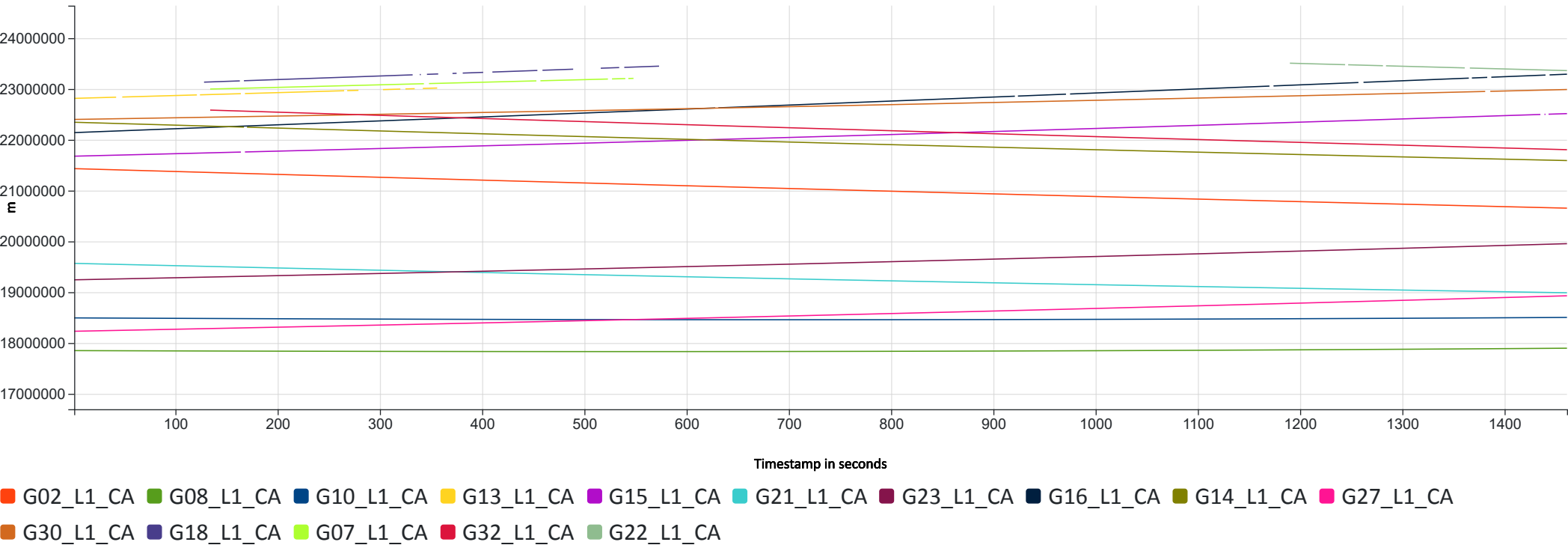
Pseudo range (m) - Glonass G1



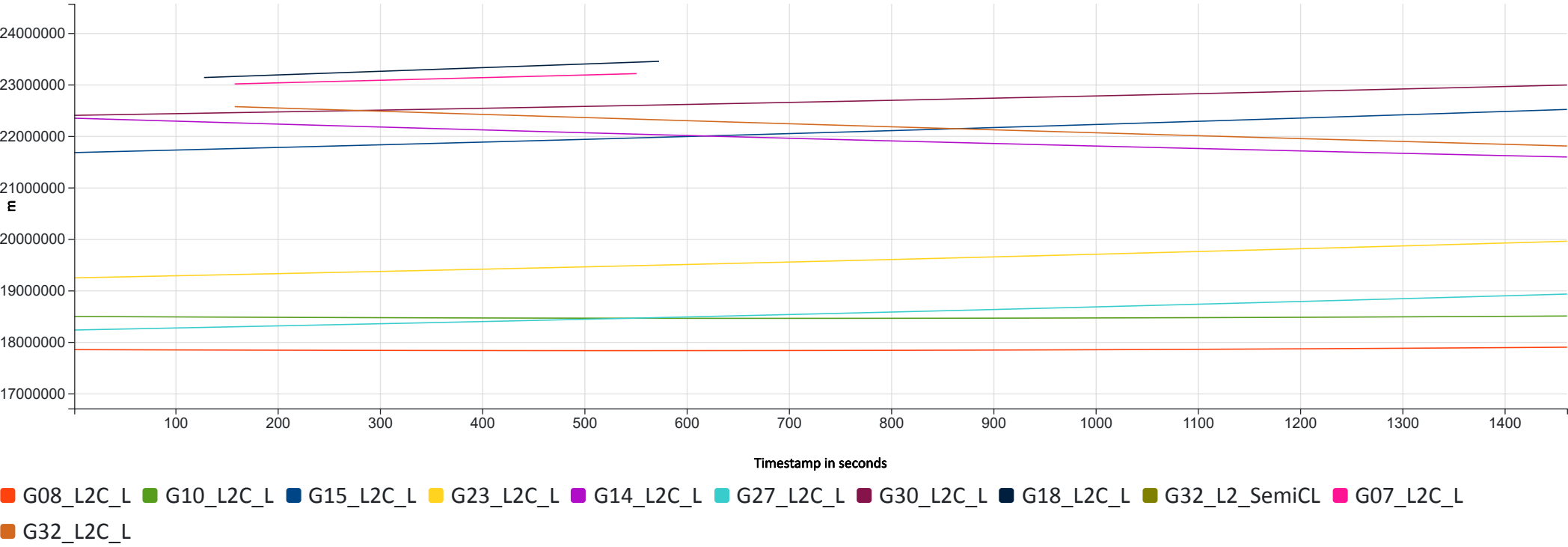
Pseudo range (m) - Glonass G2



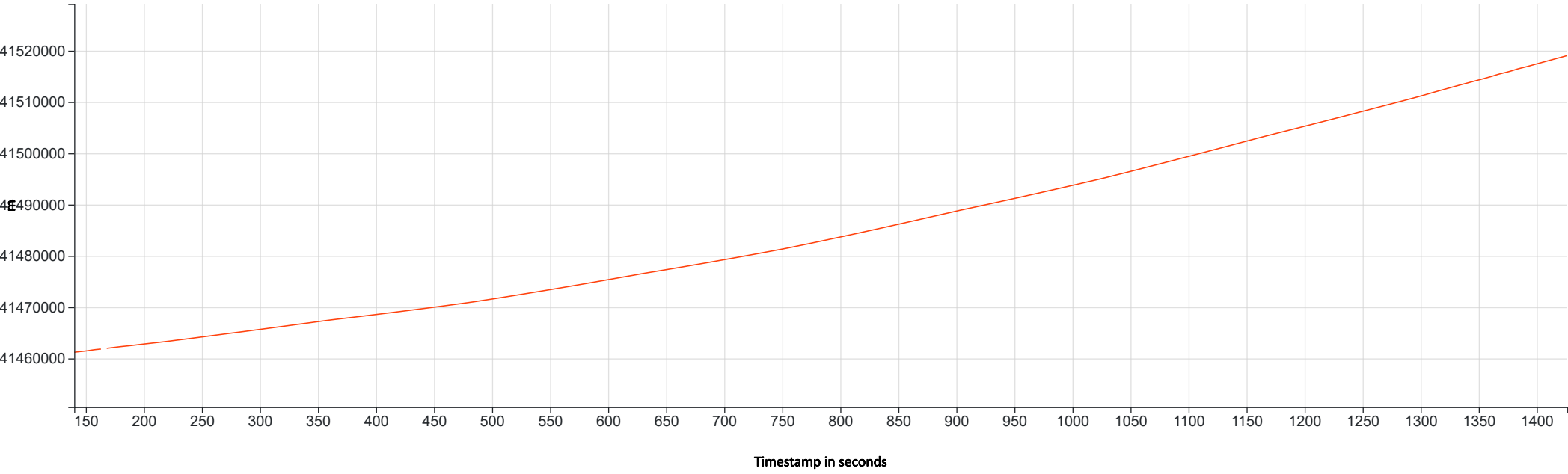
Pseudo range (m) - GPS L1



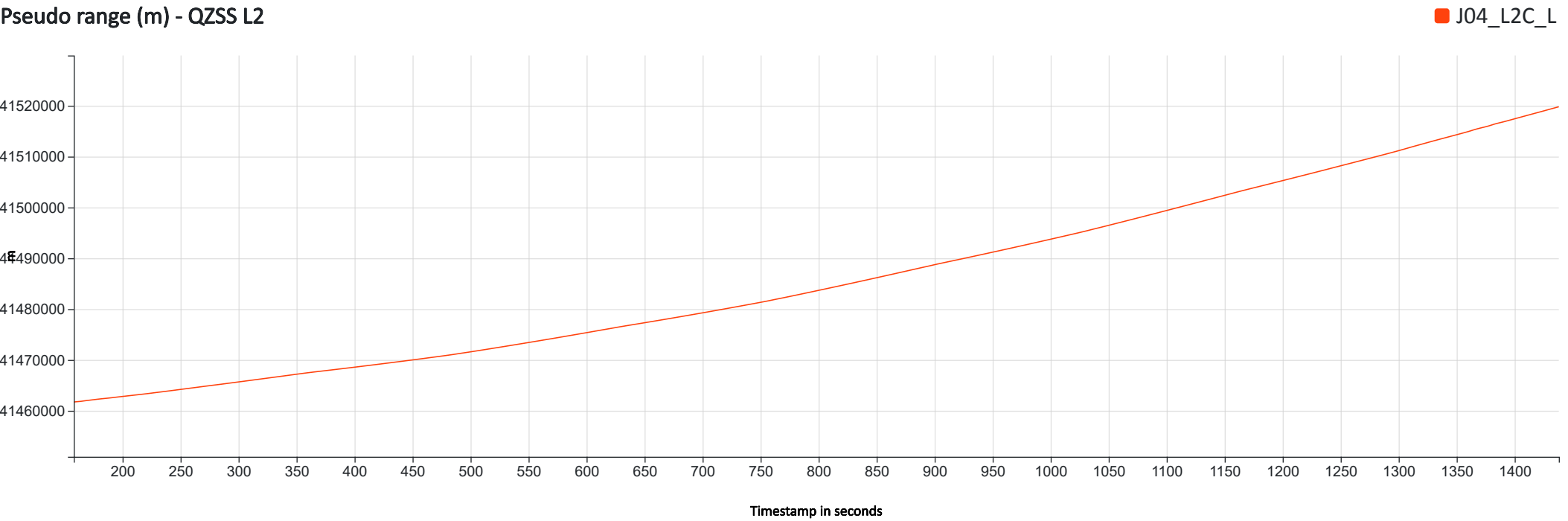
Pseudo range (m) - GPS L2



Pseudo range (m) - QZSS L1

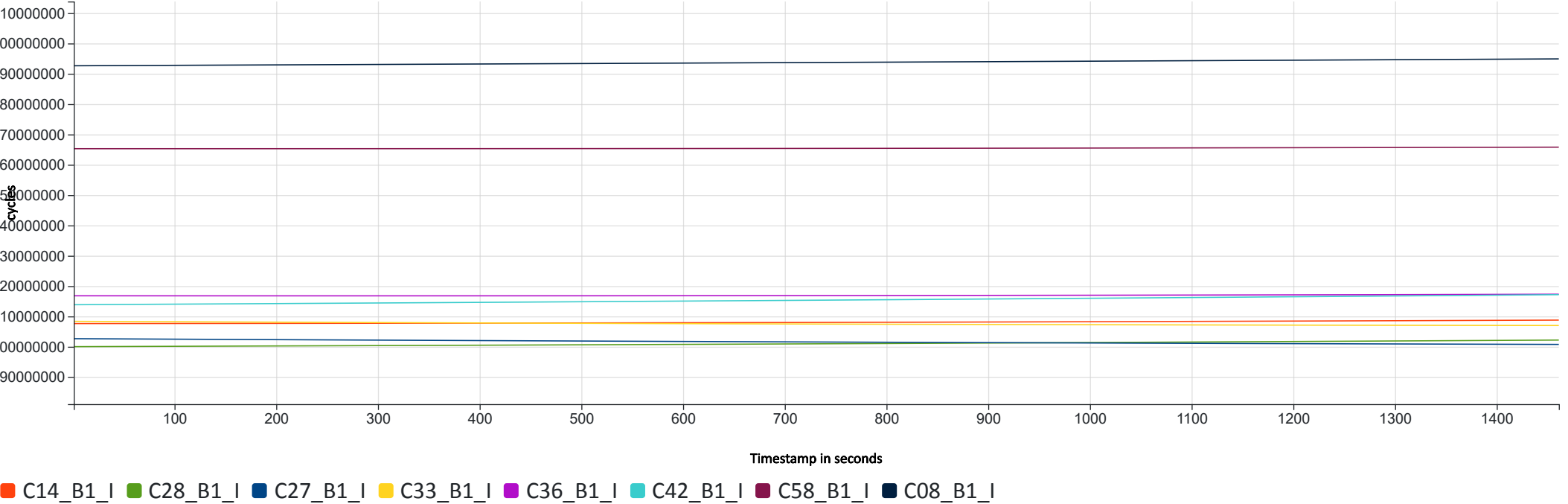


Pseudo range (m) - QZSS L2

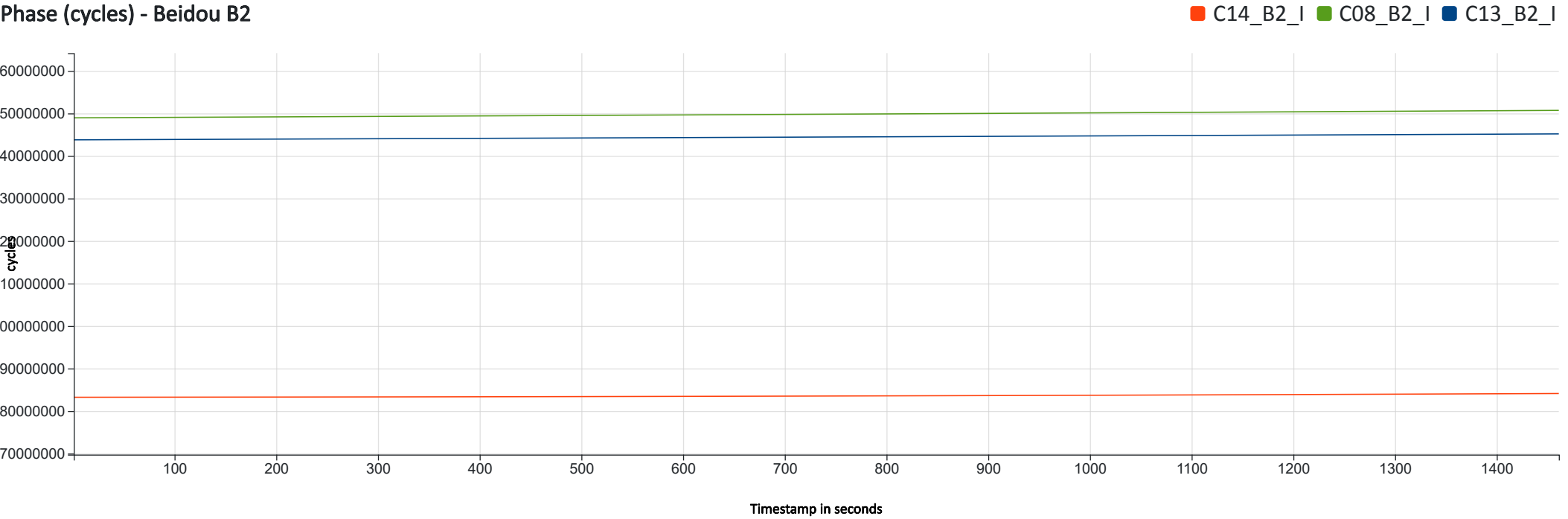


Phase

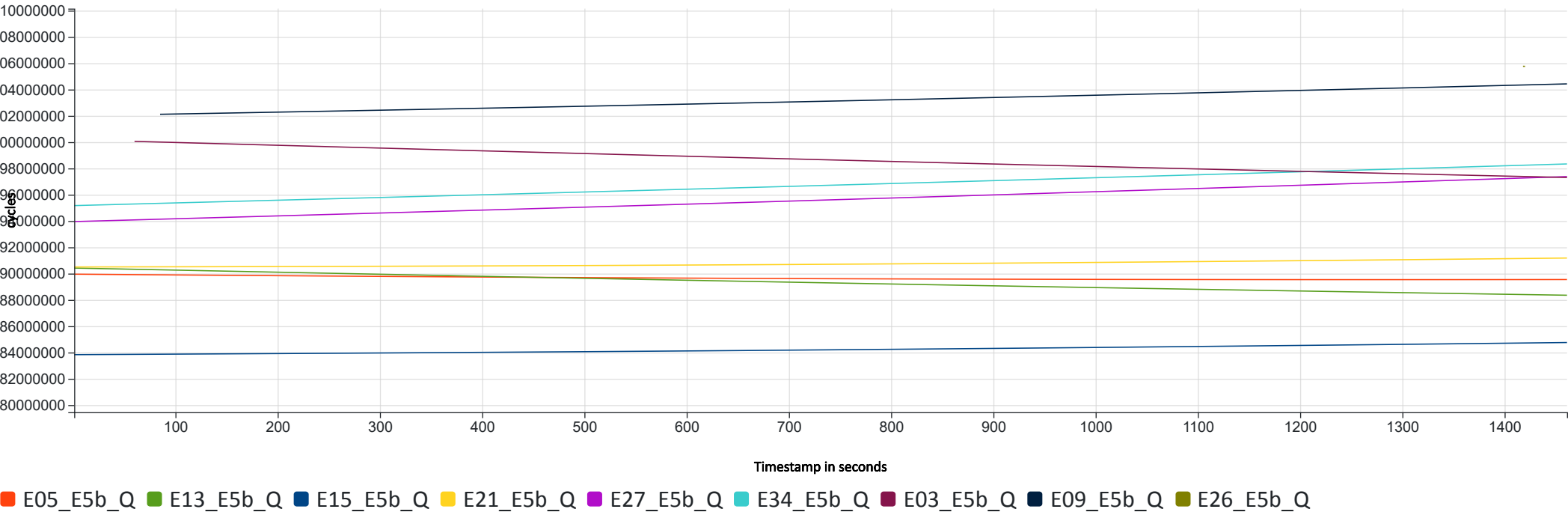
Phase (cycles) - Beidou B1



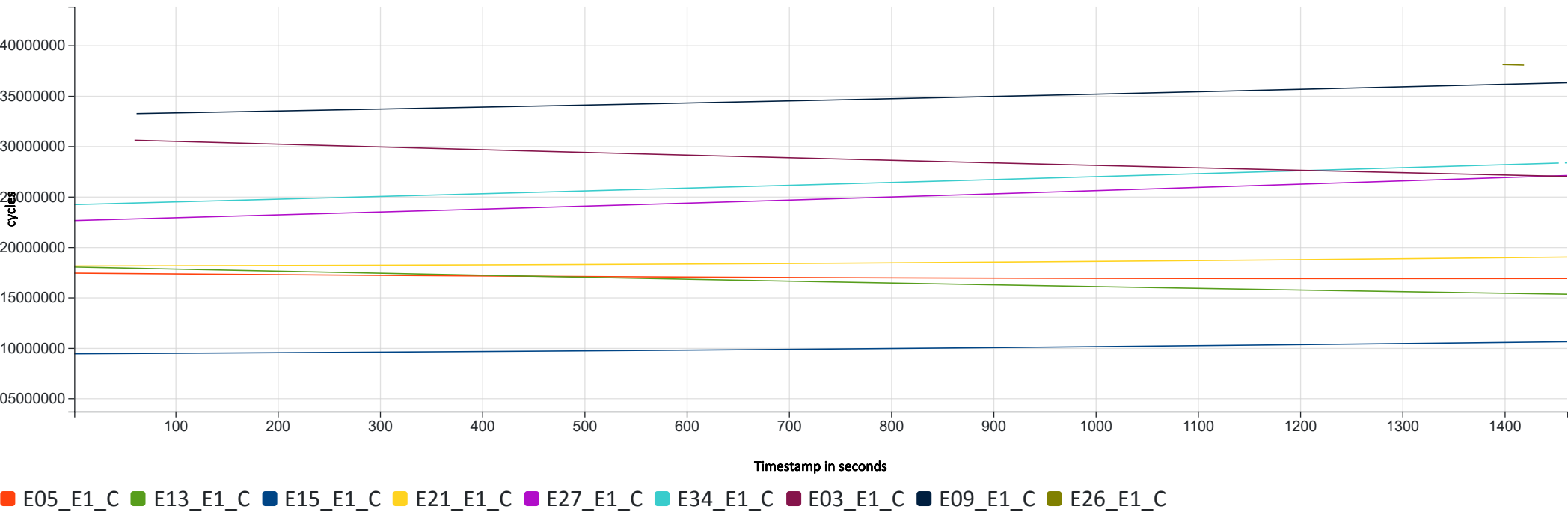
Phase (cycles) - Beidou B2



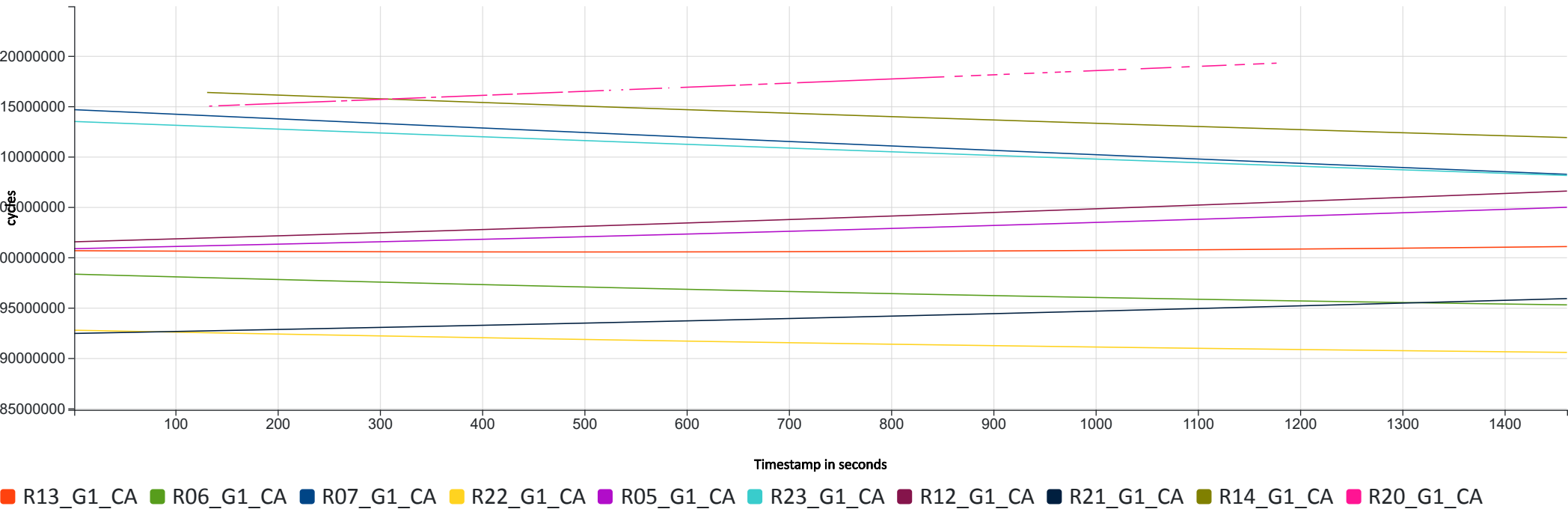
Phase (cycles) - Galileo E5b



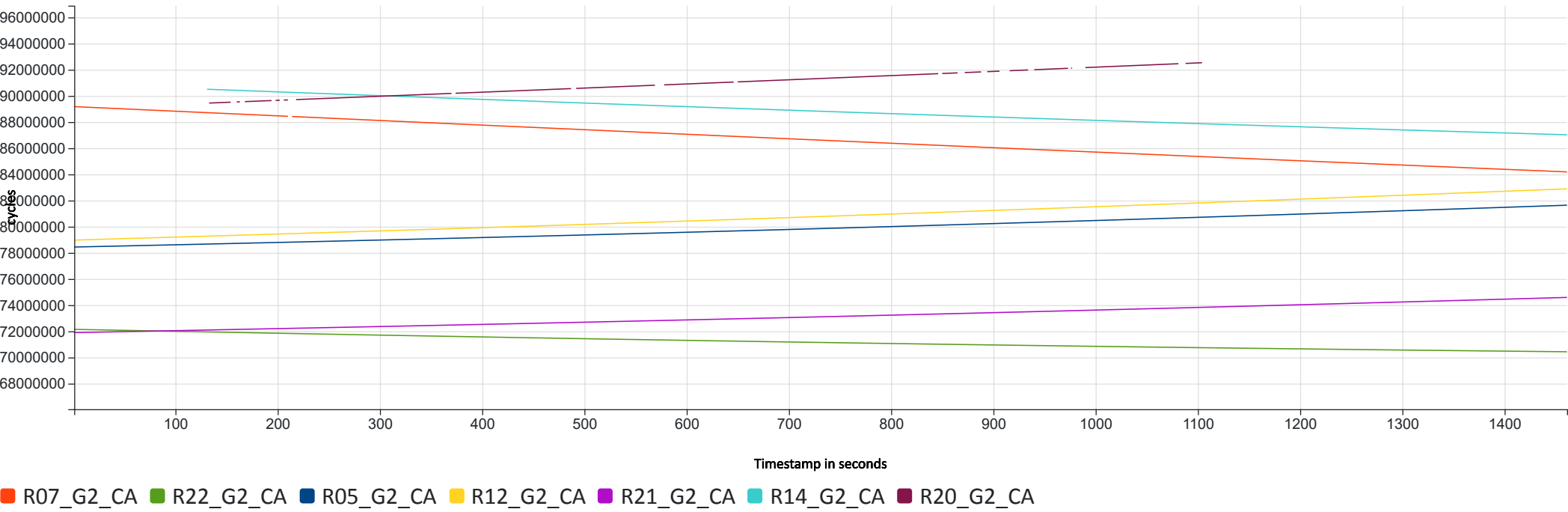
Phase (cycles) - Galileo E1



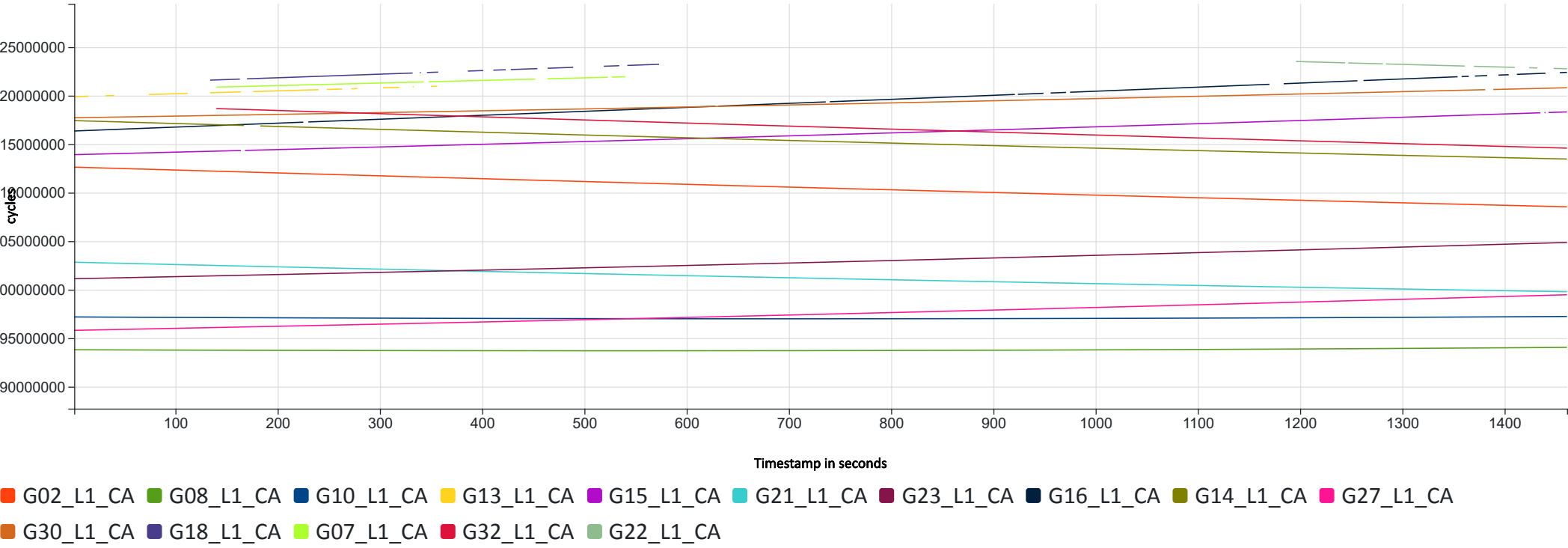
Phase (cycles) - Glonass G1



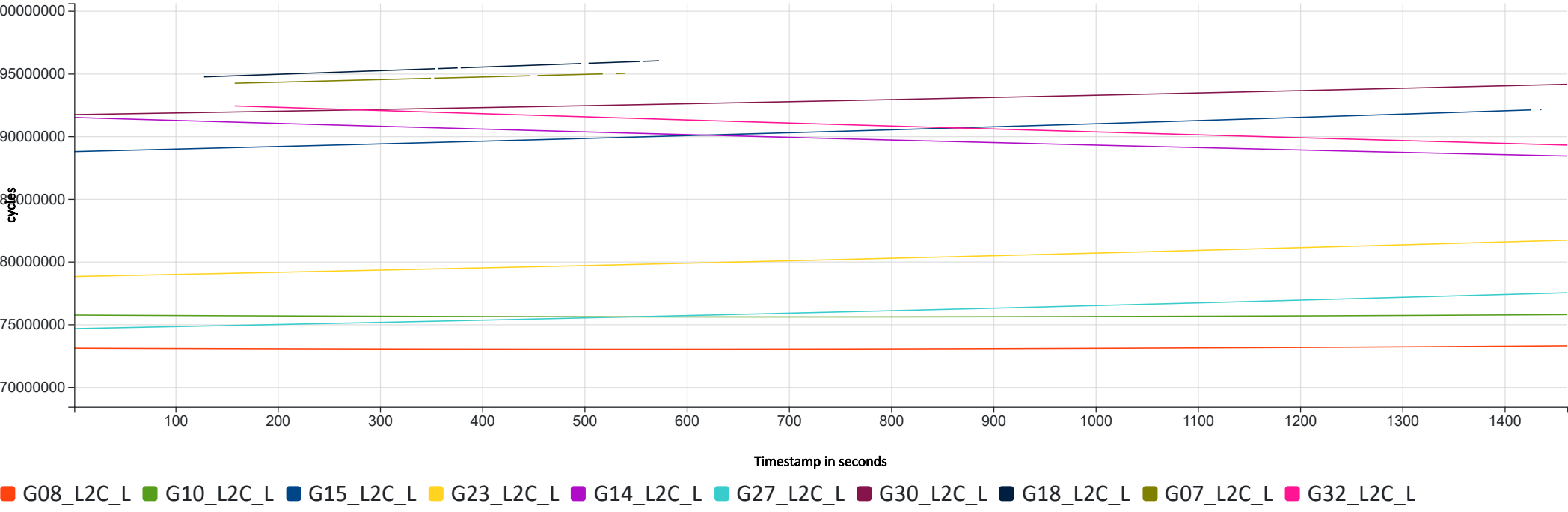
Phase (cycles) - Glonass G2



Phase (cycles) - GPS L1

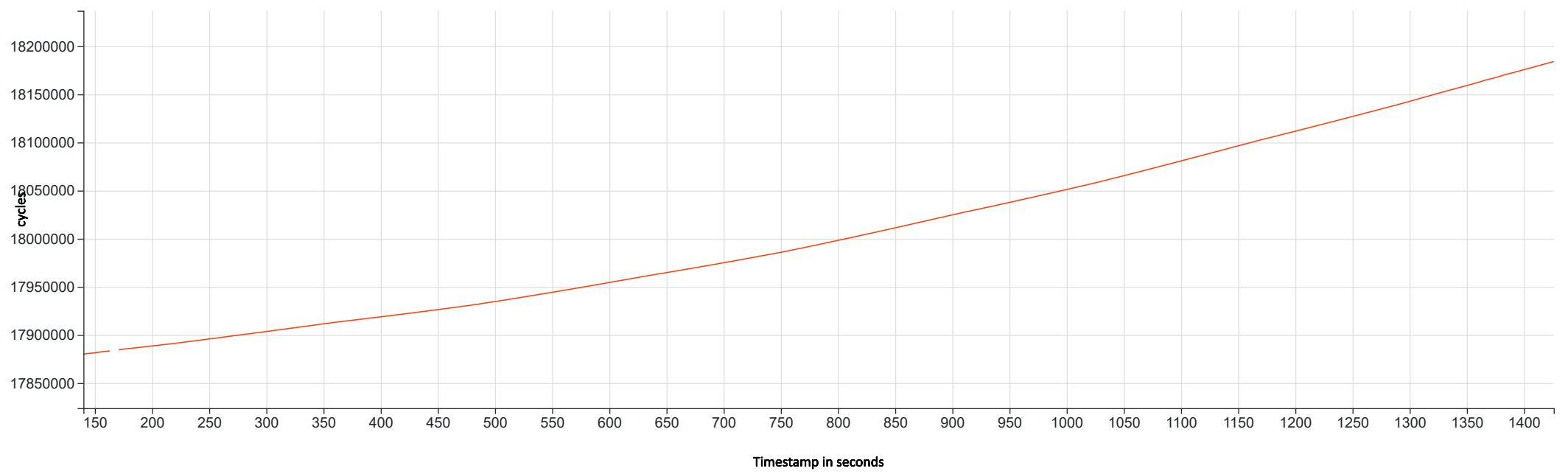


Phase (cycles) - GPS L2



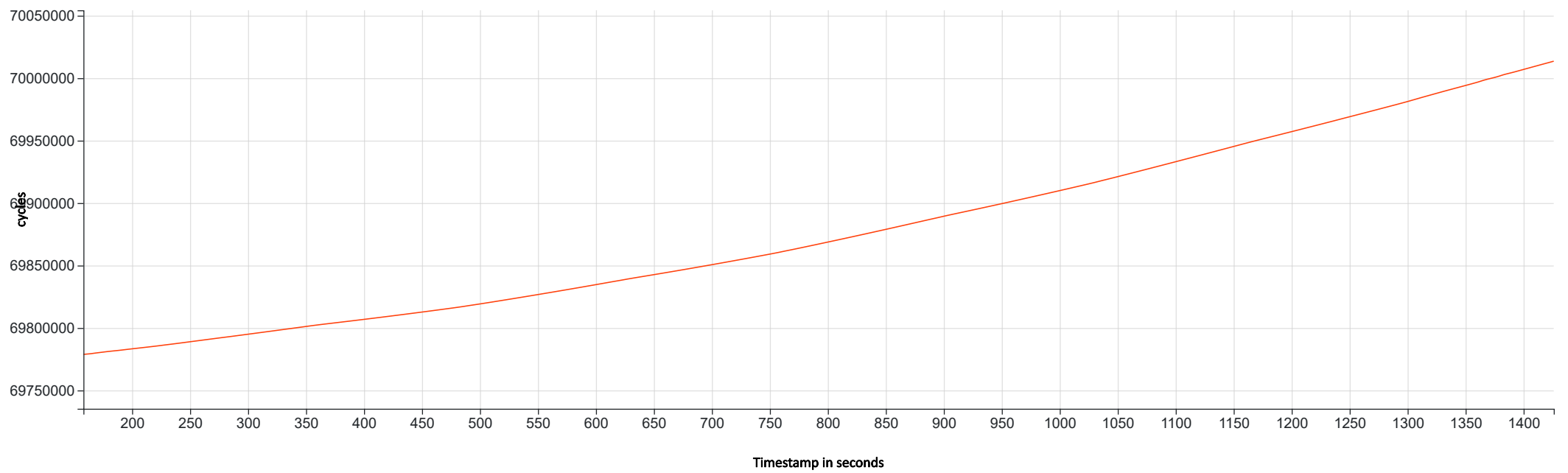
Phase (cycles) - QZSS L1

■ J04_L1_CA



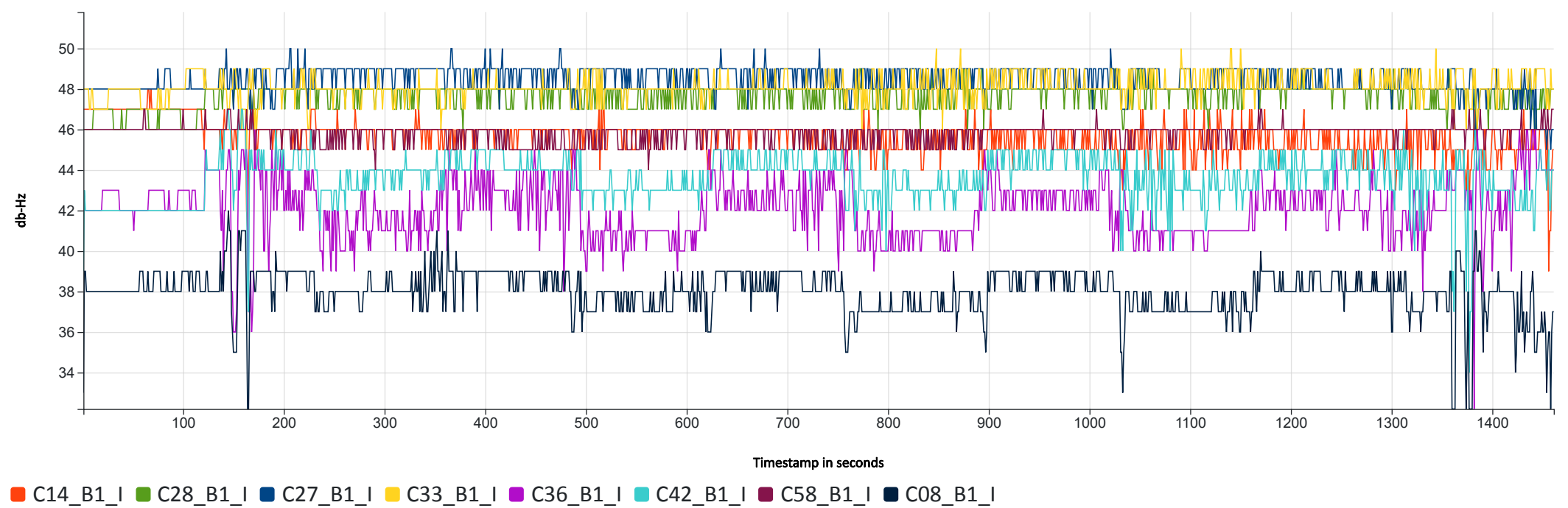
Phase (cycles) - QZSS L2

■ J04_L2C_L

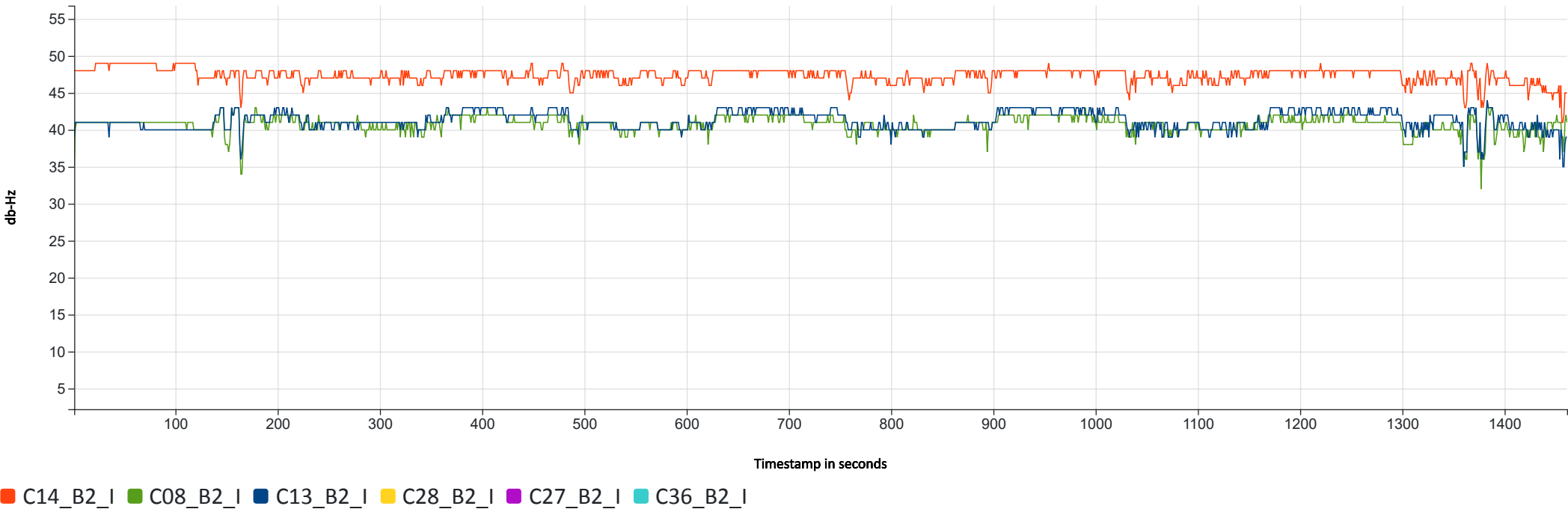


Carrier to noise

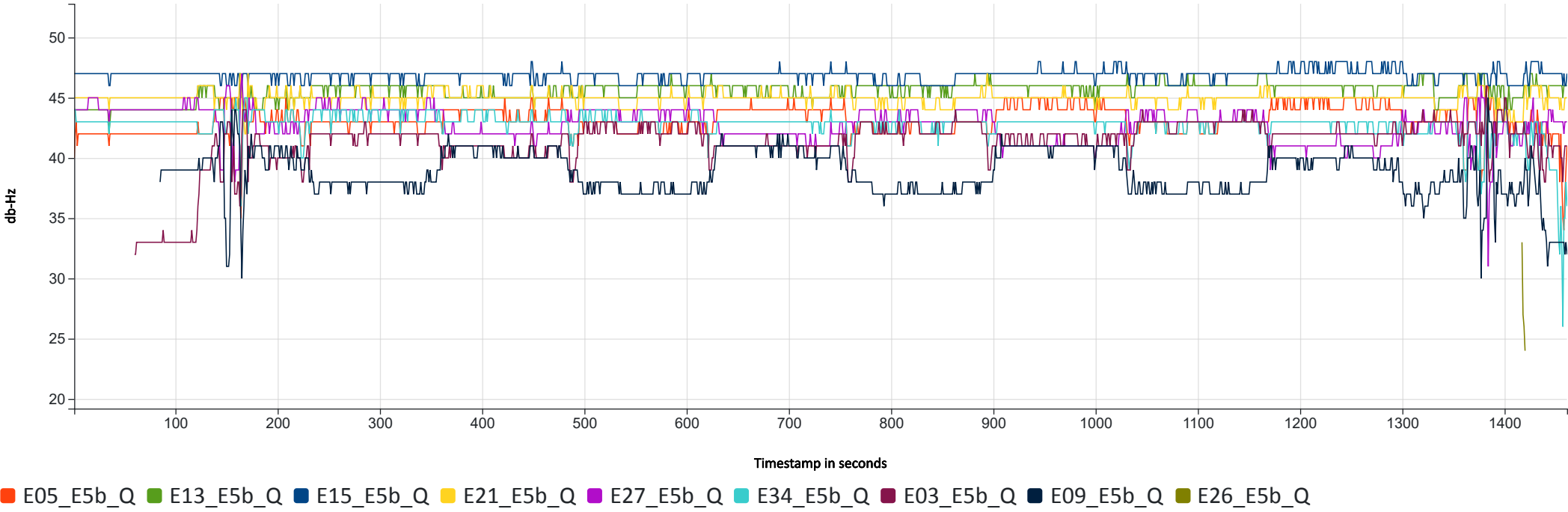
Carrier to noise (dB-hz) - Beidou B1



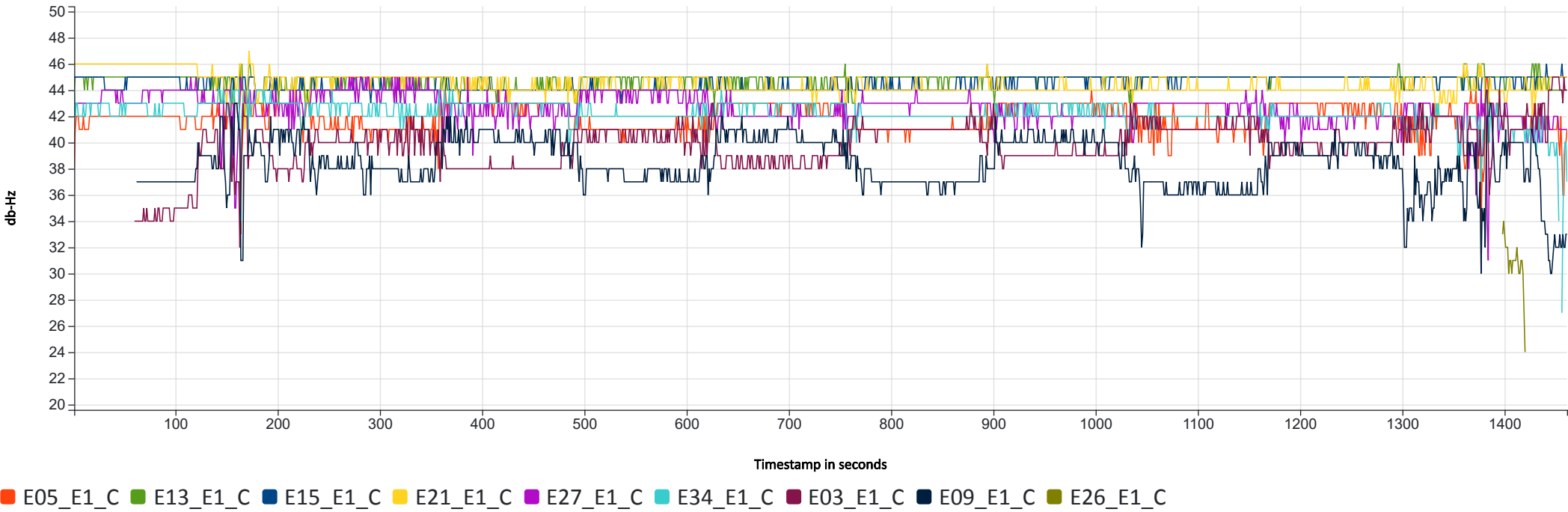
Carrier to noise (dB-hz) - Beidou B2



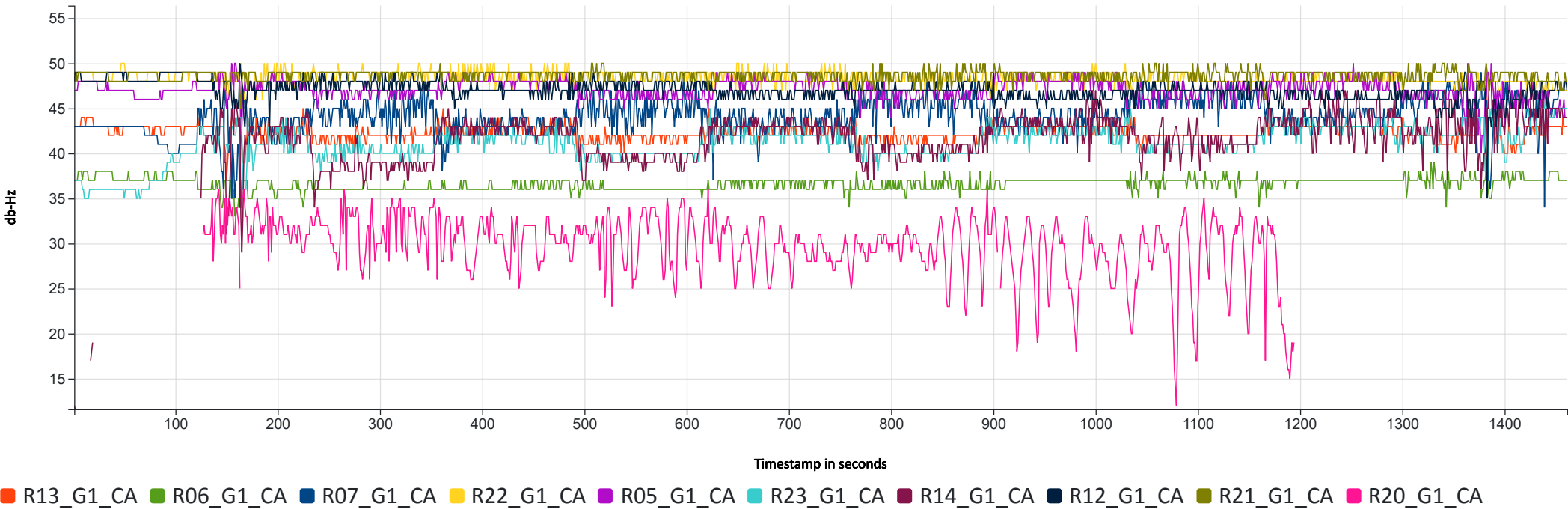
Carrier to noise (dB-hz) - Galileo E5b



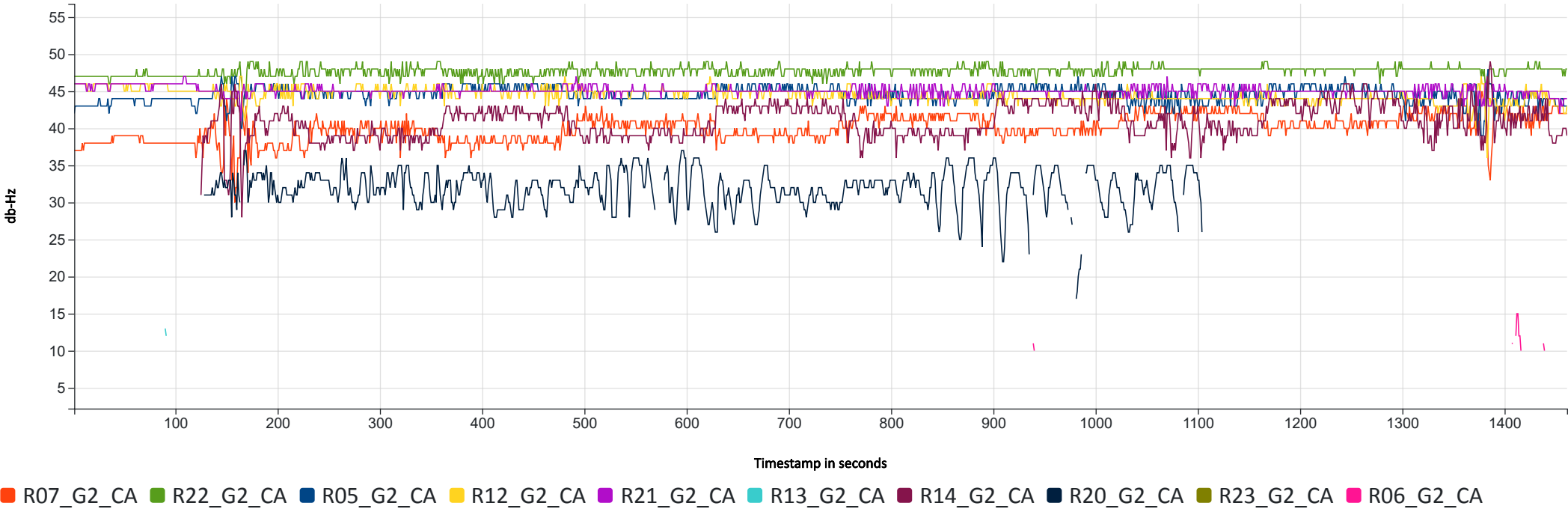
Carrier to noise (dB-hz) - Galileo E1



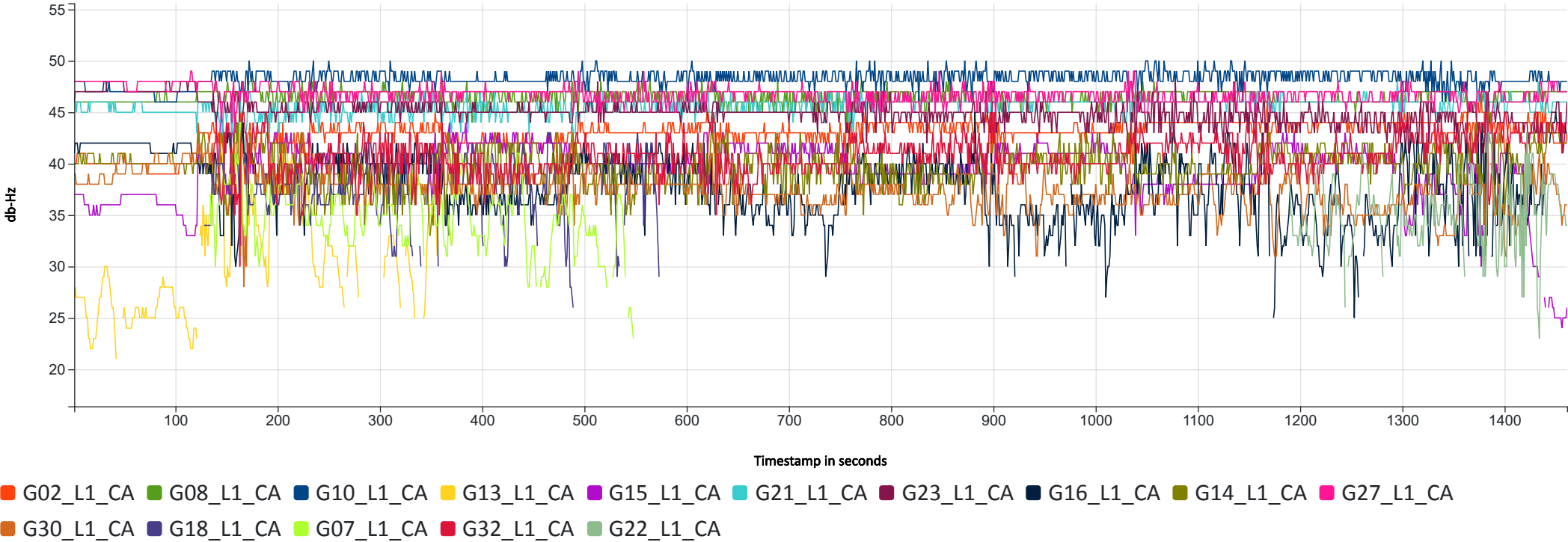
Carrier to noise (dB-hz) - Glonass G1



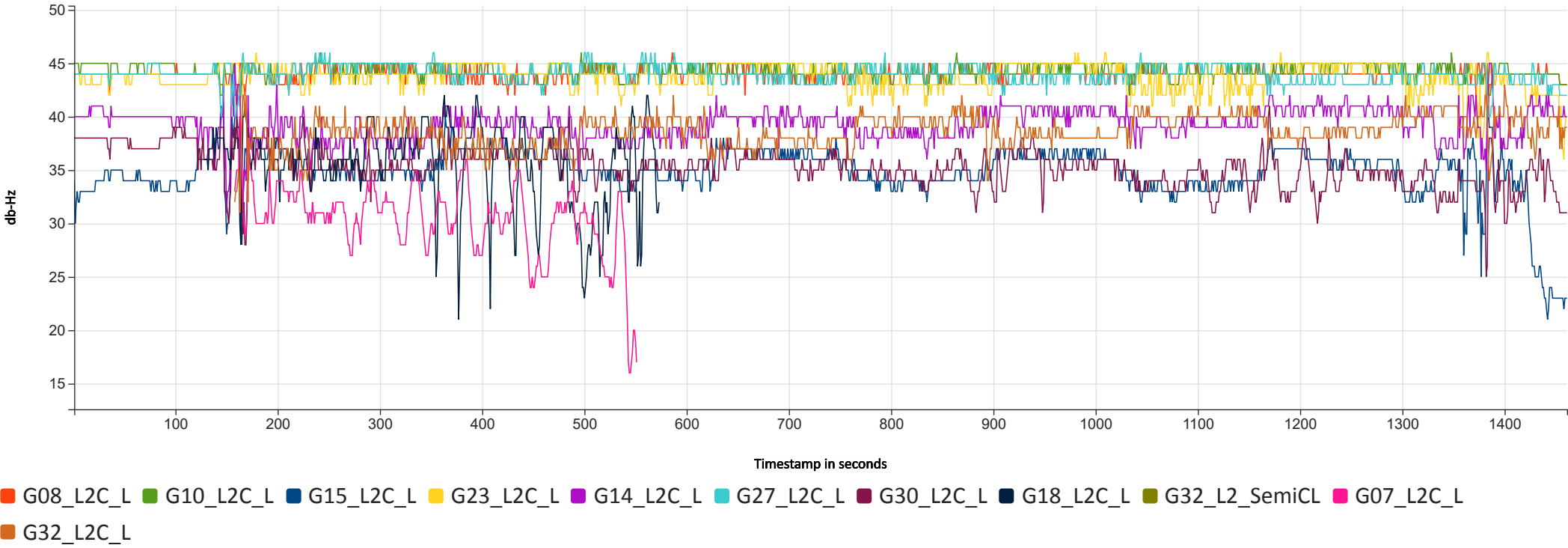
Carrier to noise (dB-hz) - Glonass G2



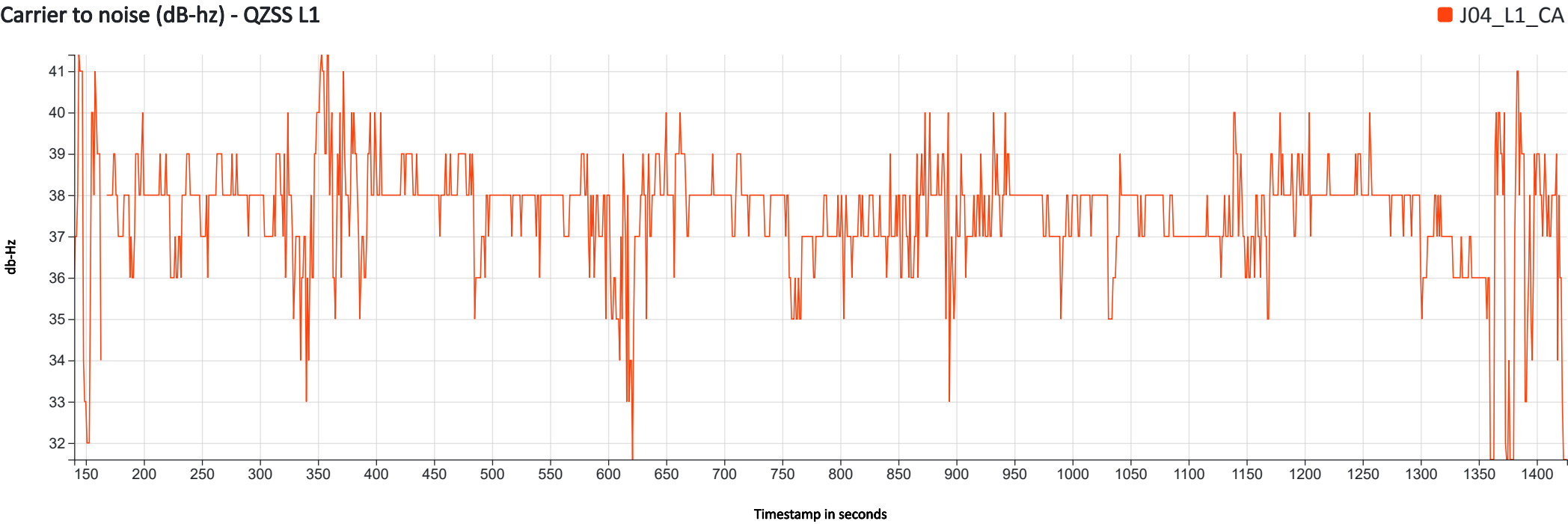
Carrier to noise (dB-hz) - GPS L1



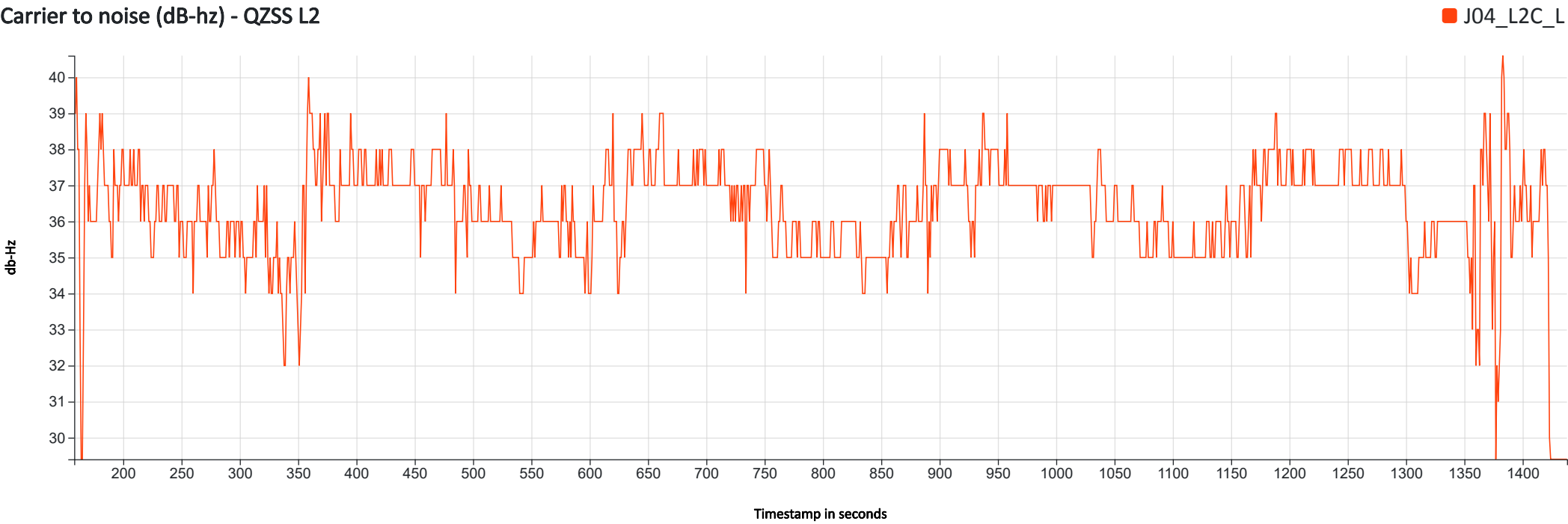
Carrier to noise (dB-hz) - GPS L2



Carrier to noise (dB-hz) - QZSS L1

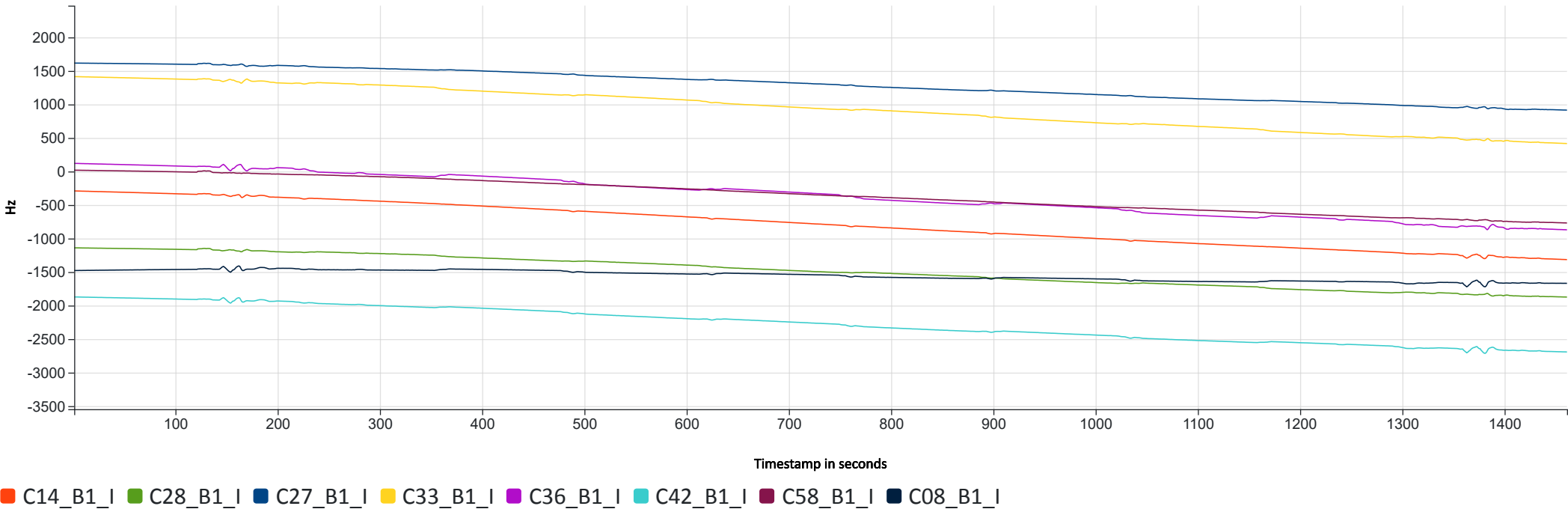


Carrier to noise (dB-hz) - QZSS L2

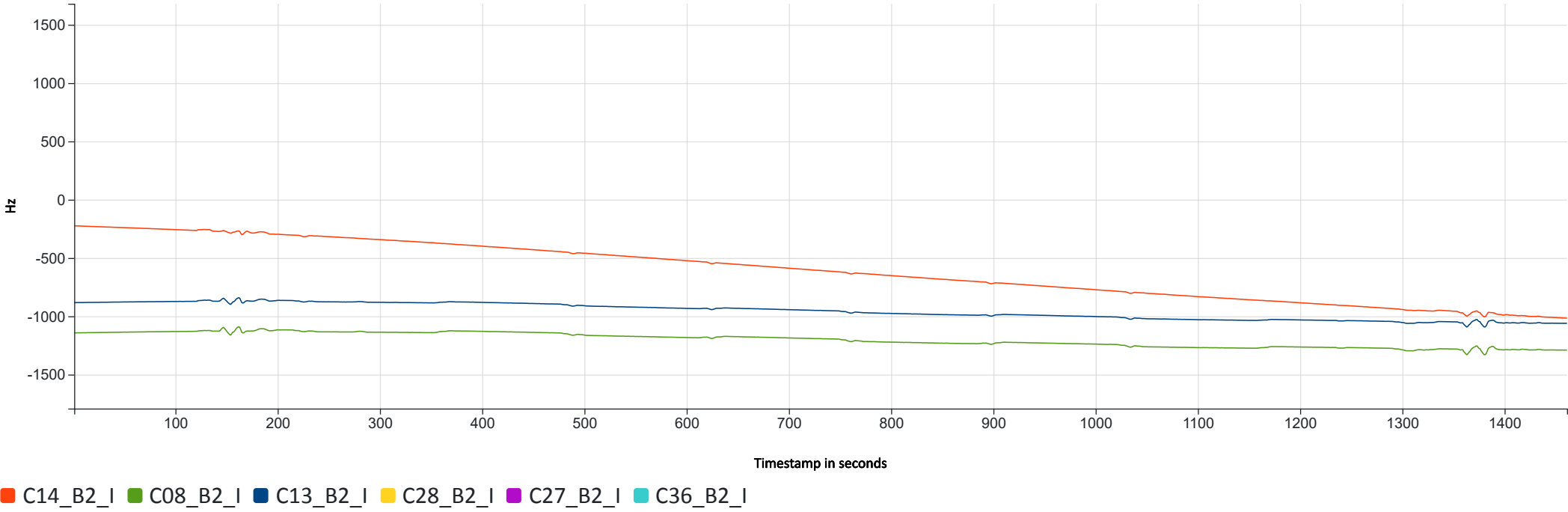


Doppler

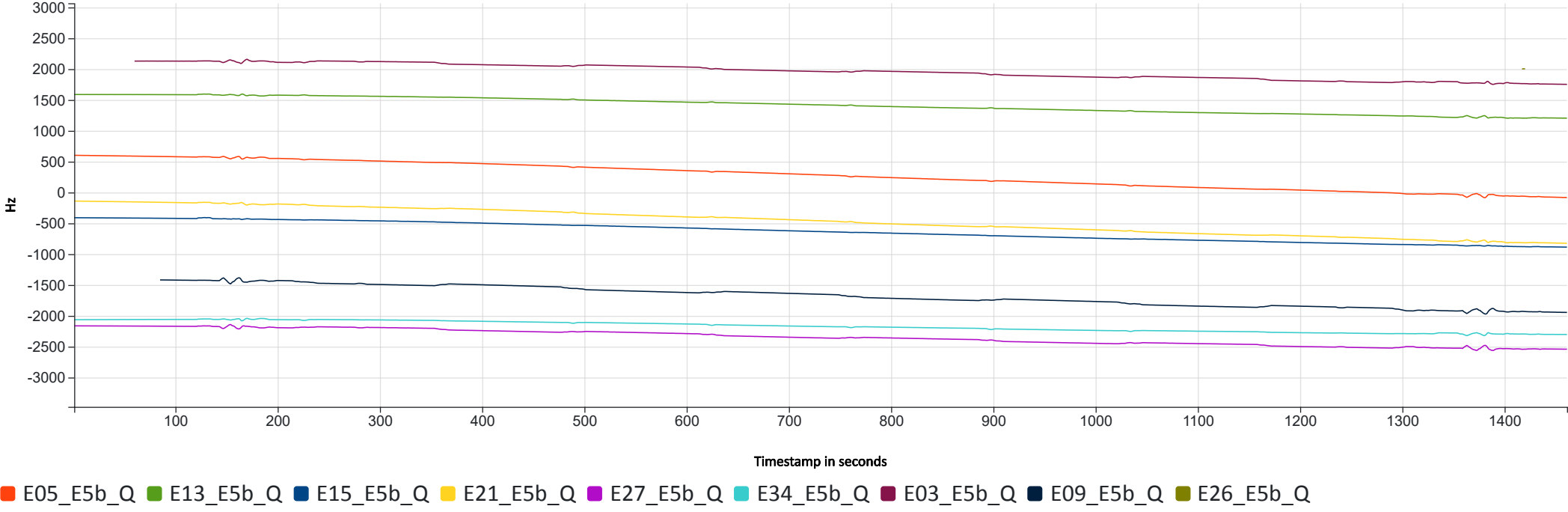
Doppler (hz) - Beidou B1



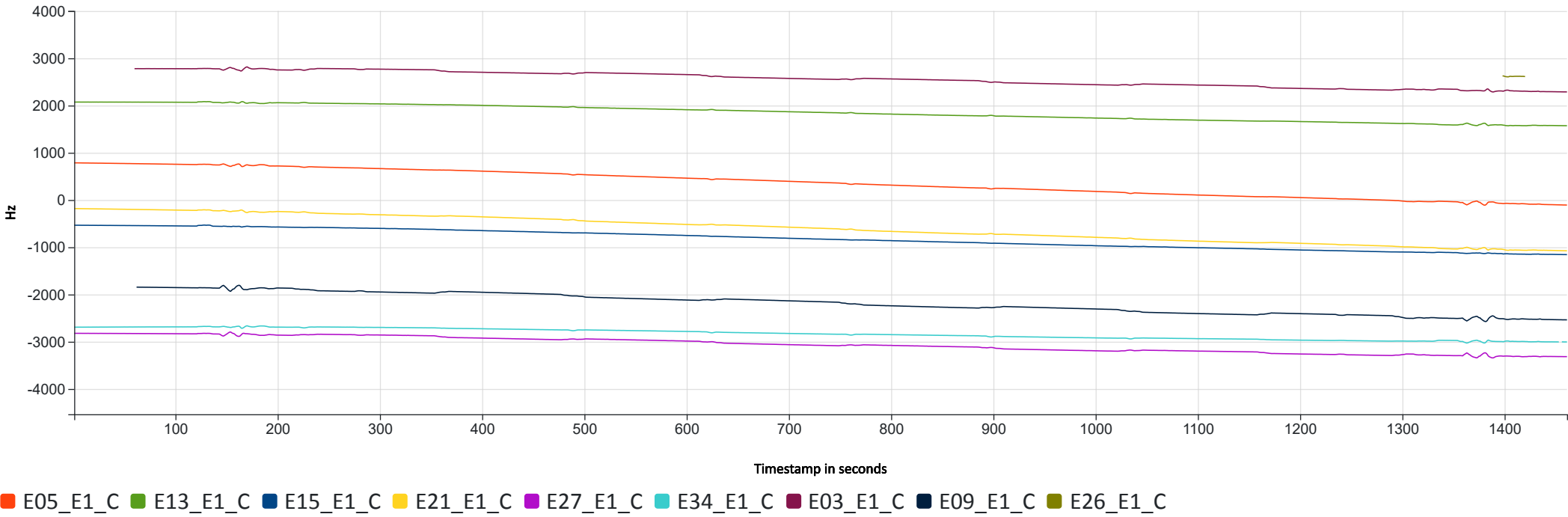
Doppler (hz) - Beidou B2



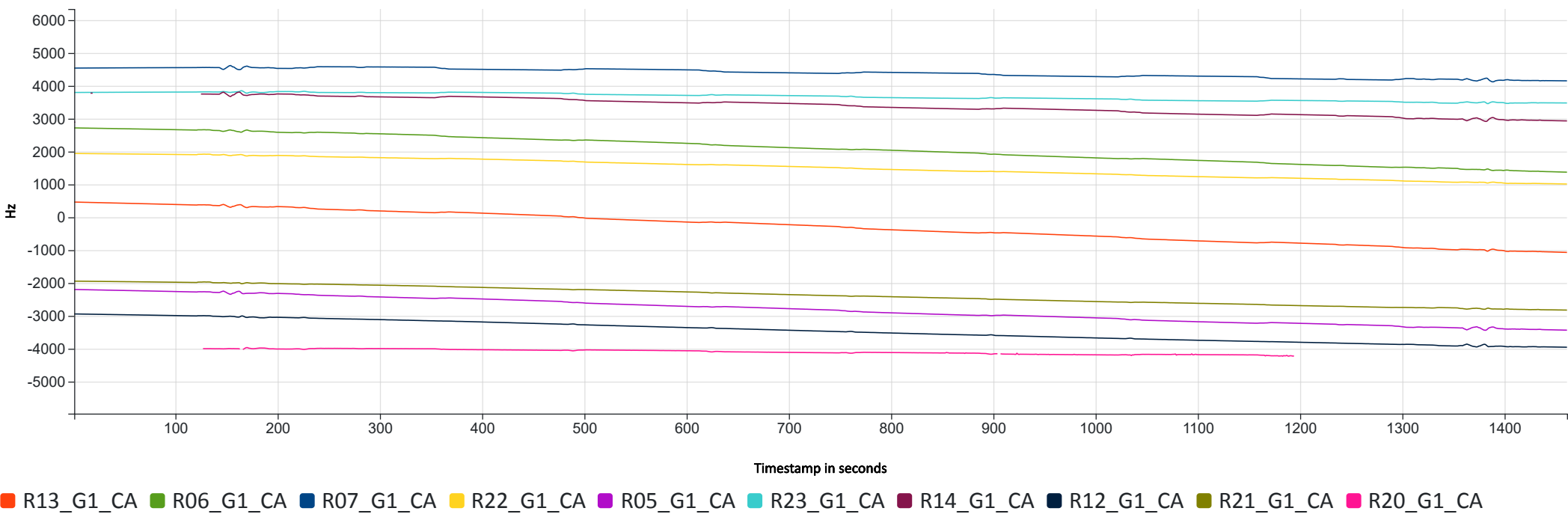
Doppler (hz) - Galileo E5b



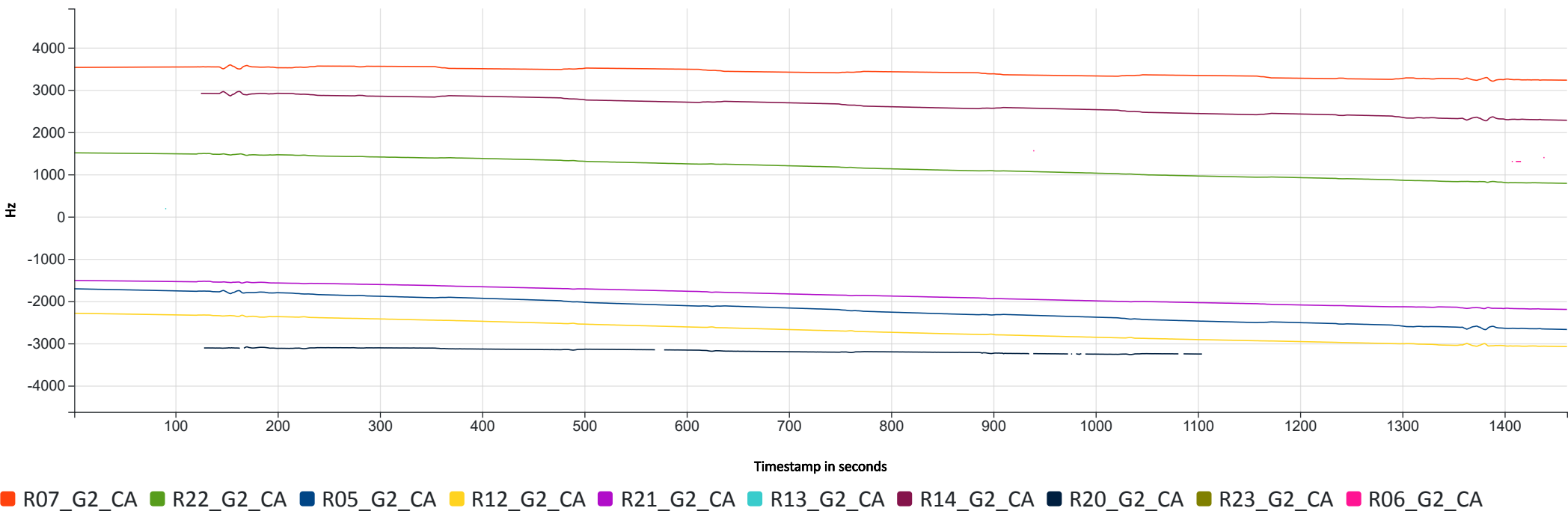
Doppler (hz) - Galileo E1



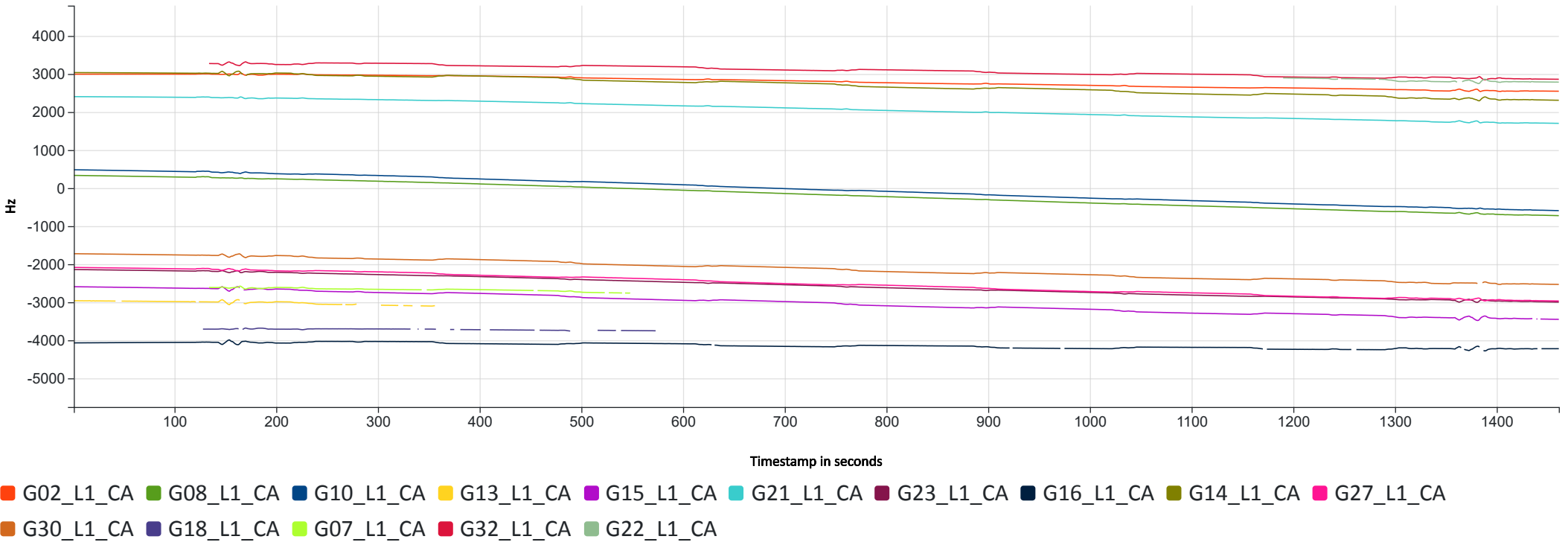
Doppler (hz) - Glonass G1



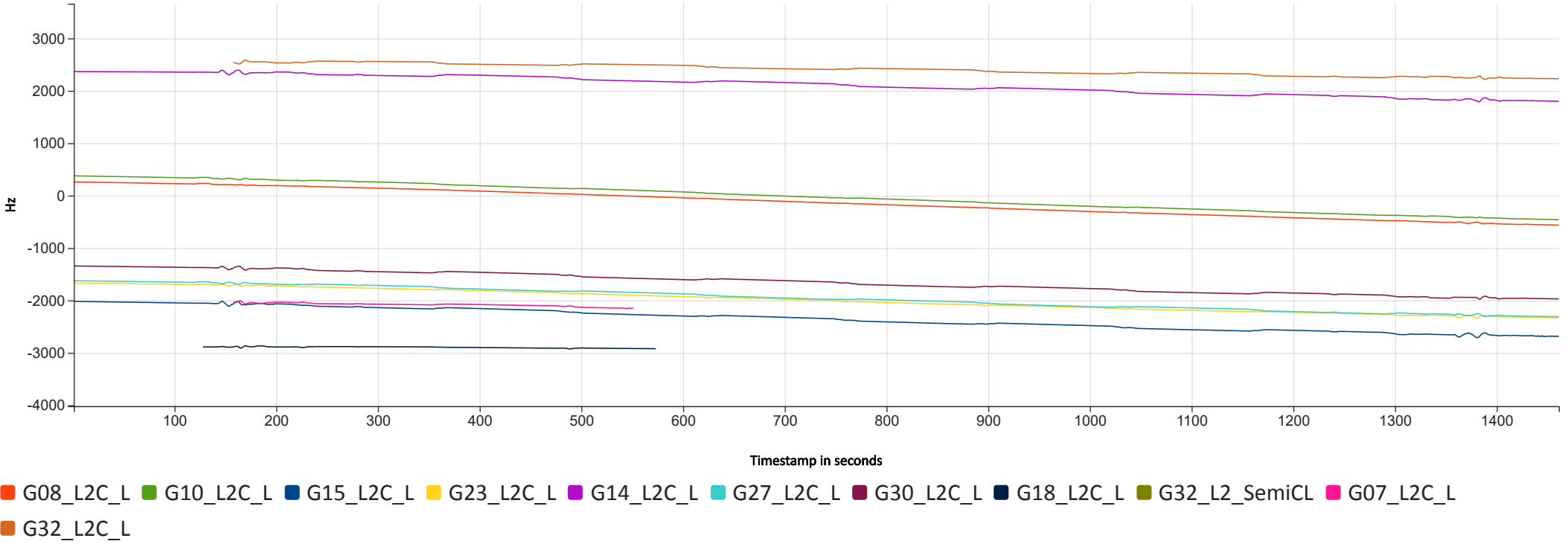
Doppler (hz) - Glonass G2



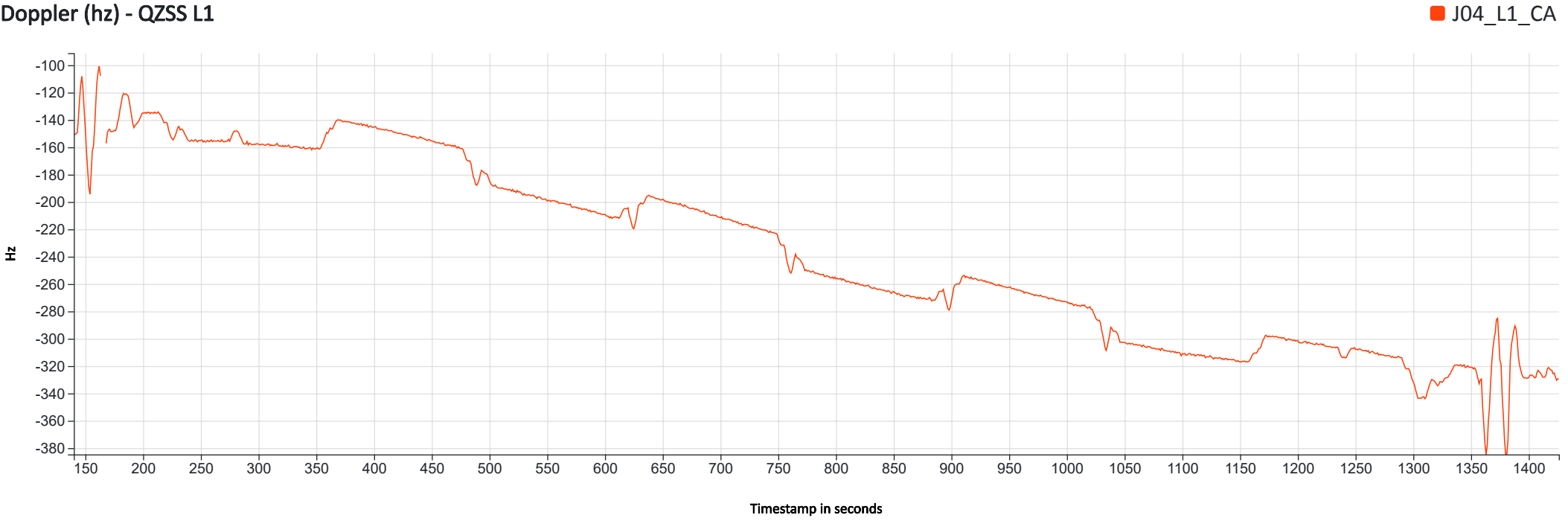
Doppler (hz) - GPS L1



Doppler (hz) - GPS L2

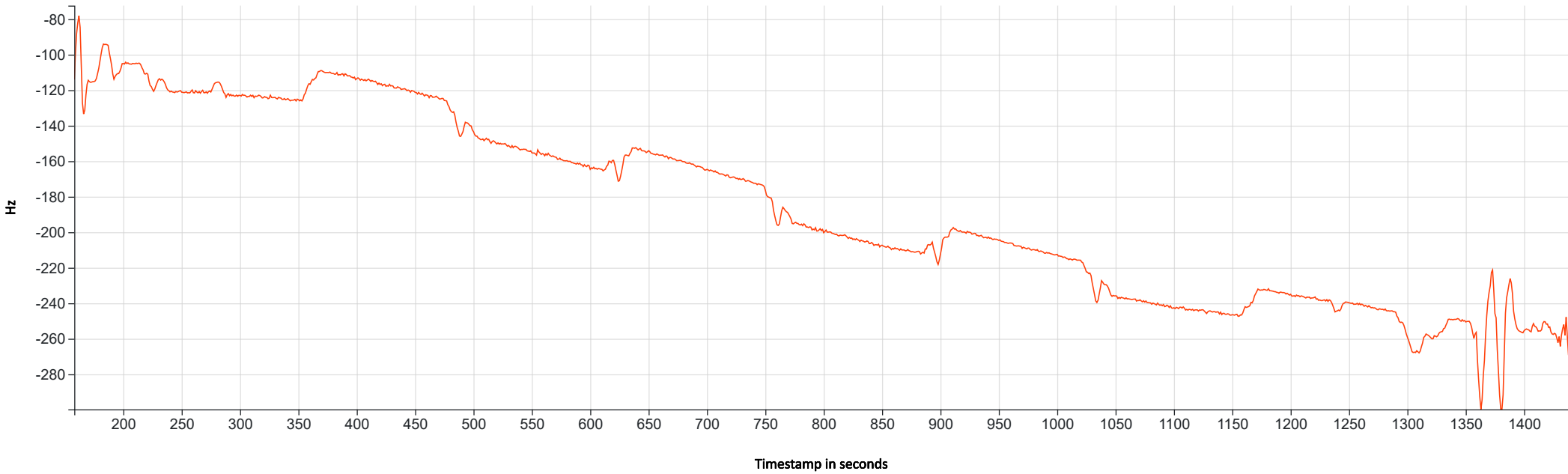


Doppler (hz) - QZSS L1



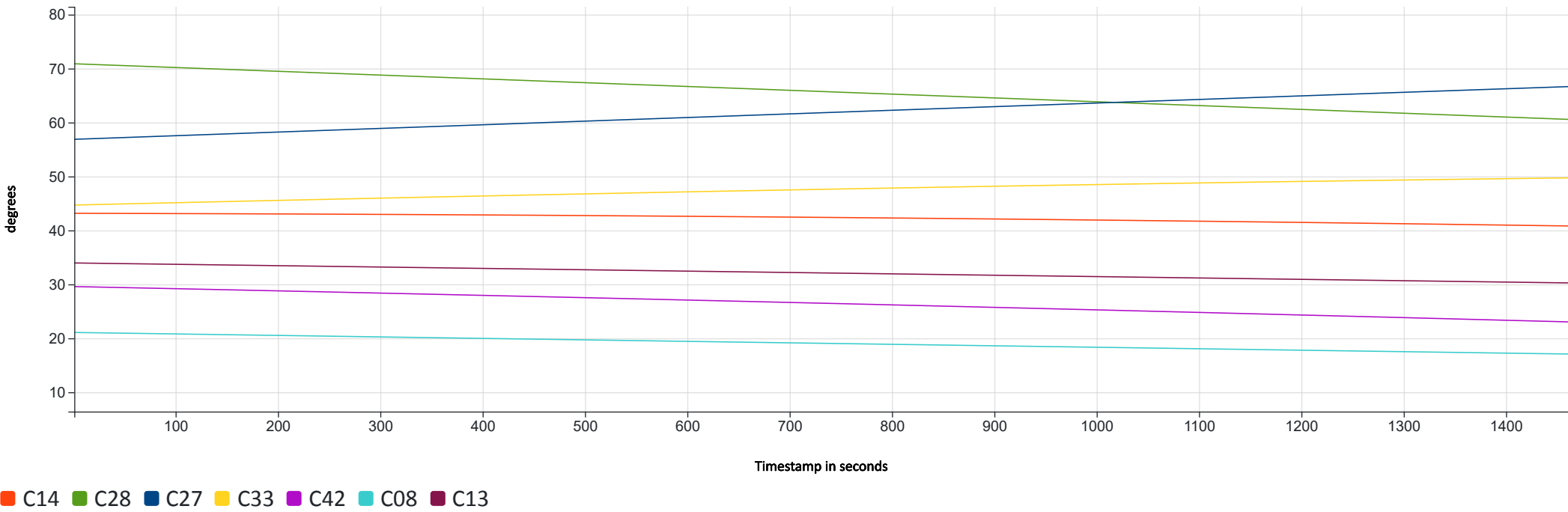
Doppler (hz) - QZSS L2

J04_L2C_L

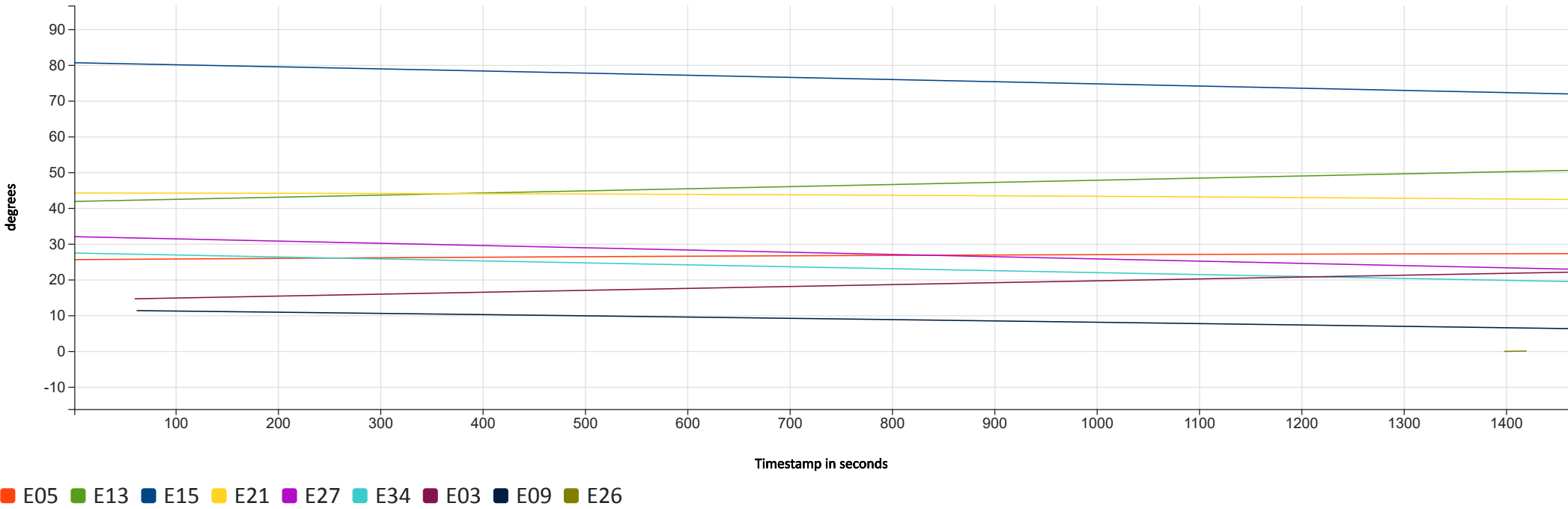


Elevation

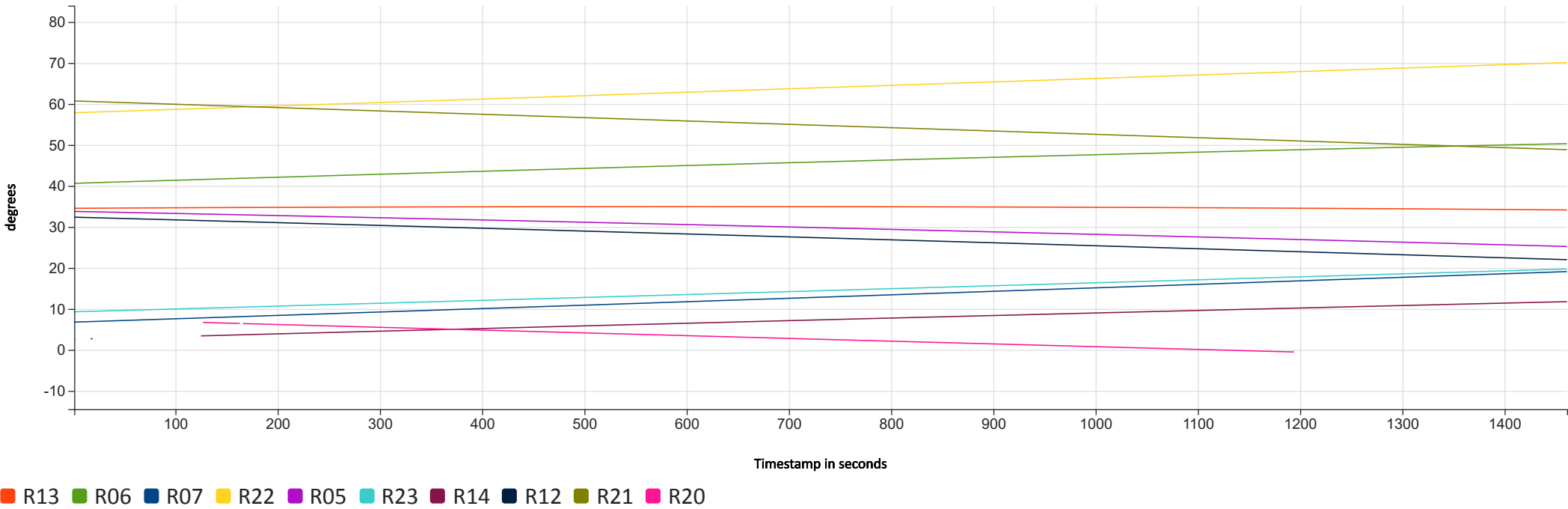
Elevation (degrees) - BEIDOU



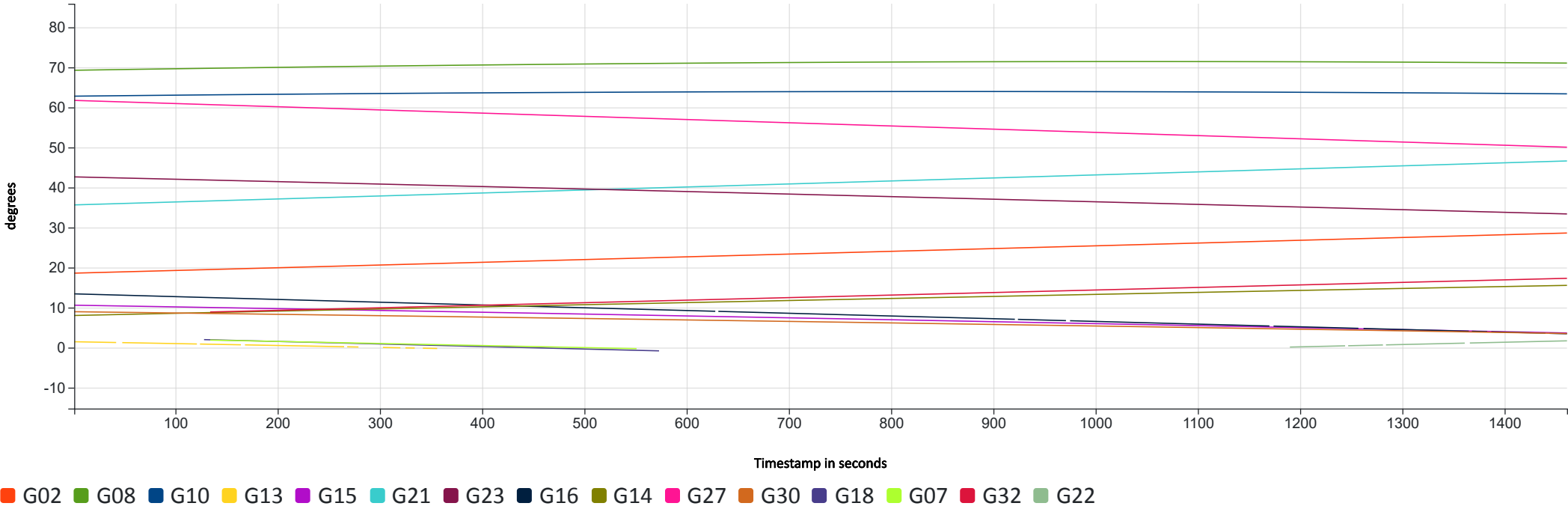
Elevation (degrees) - GALILEO



Elevation (degrees) - GLONASS

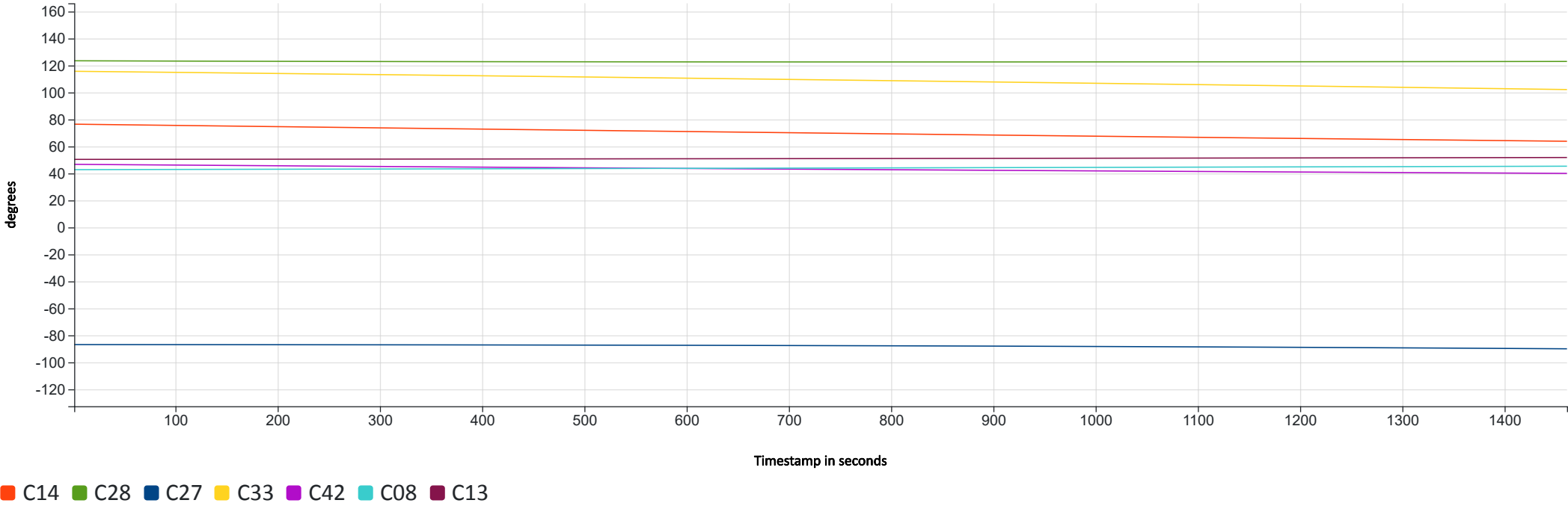


Elevation (degrees) - GPS

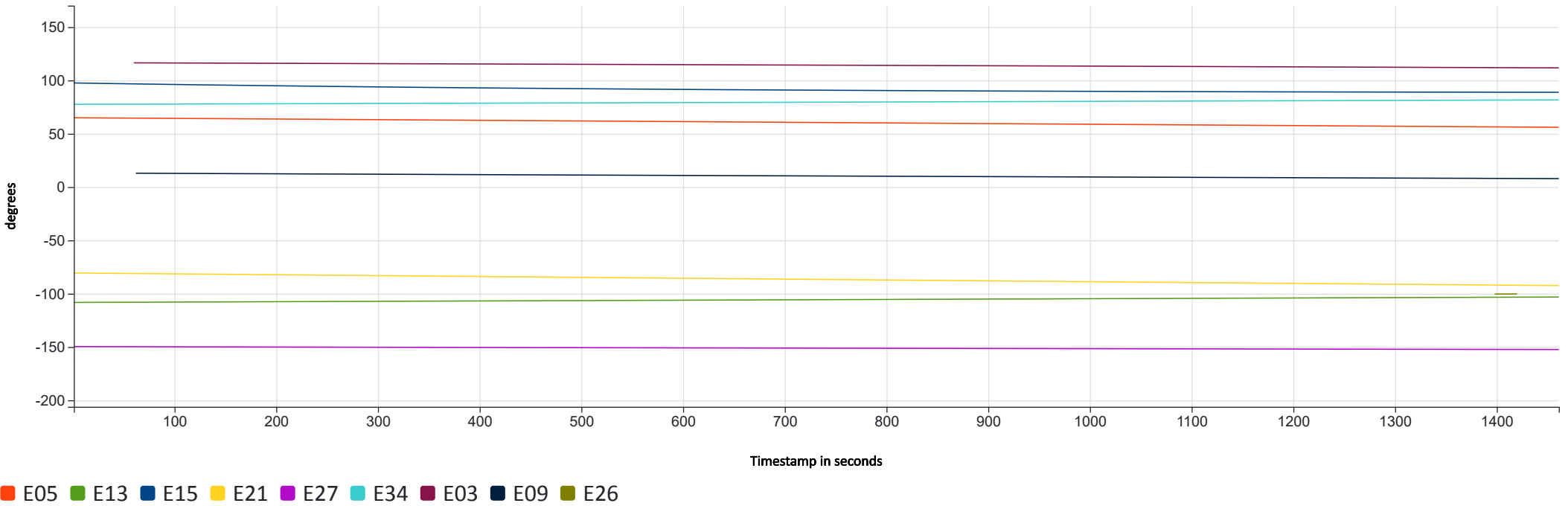


Azimuth

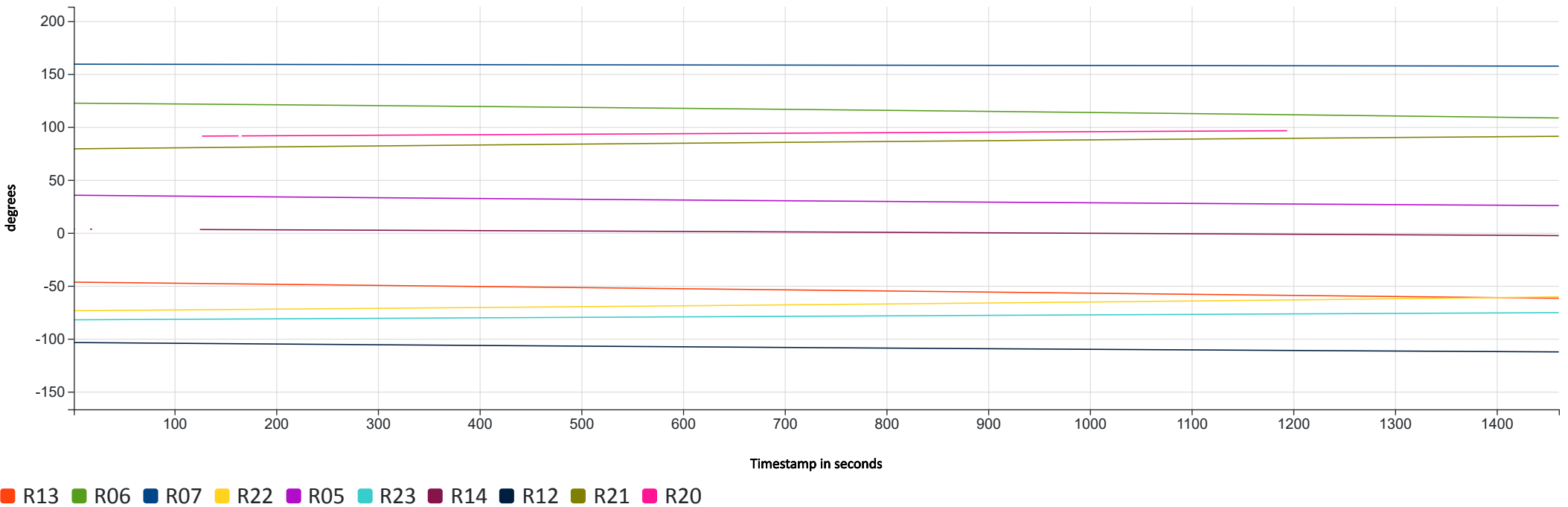
Azimuth (degrees) - BEIDOU



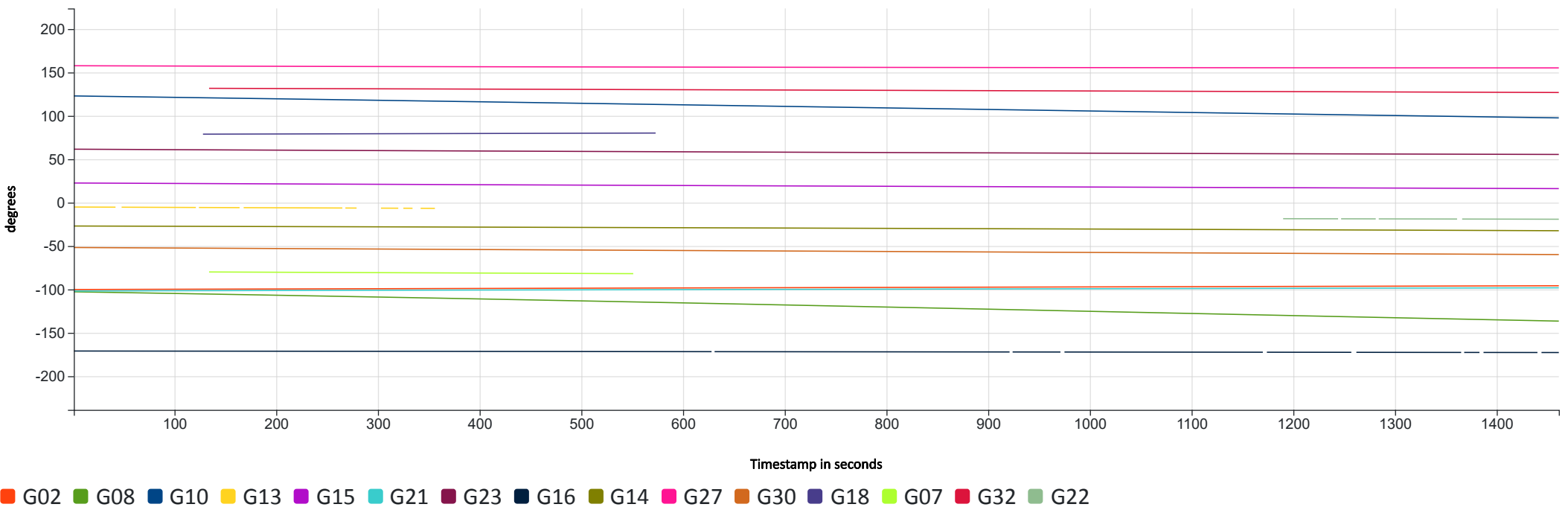
Azimuth (degrees) - GALILEO



Azimuth (degrees) - GLONASS



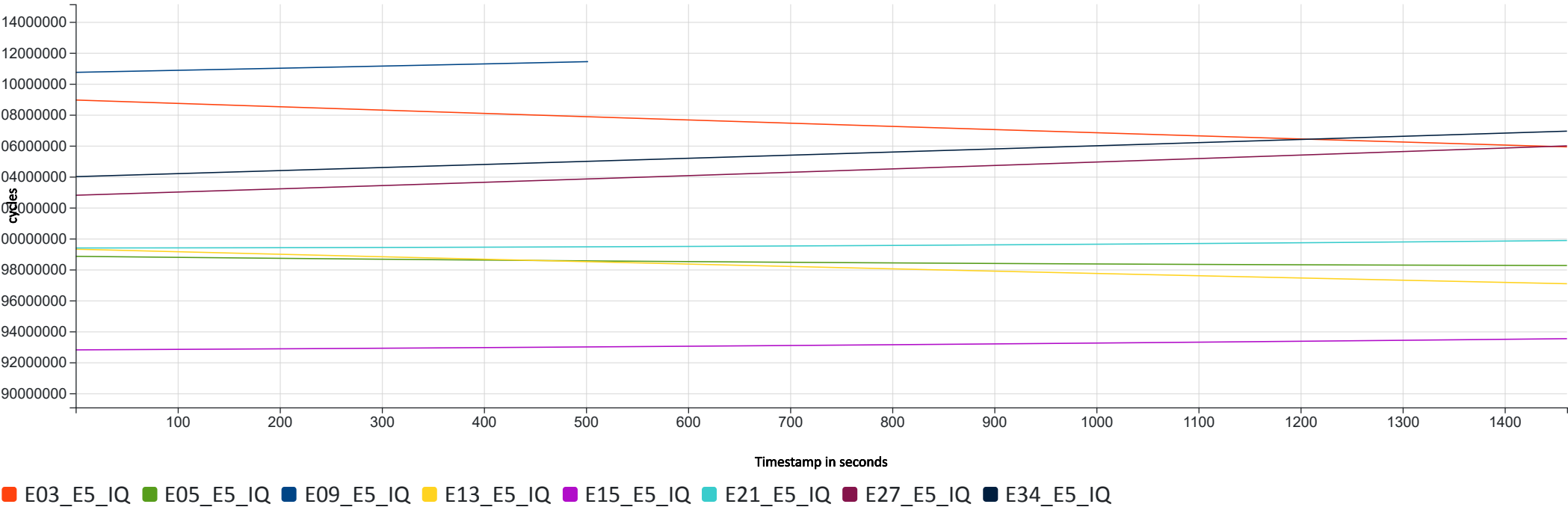
Azimuth (degrees) - GPS



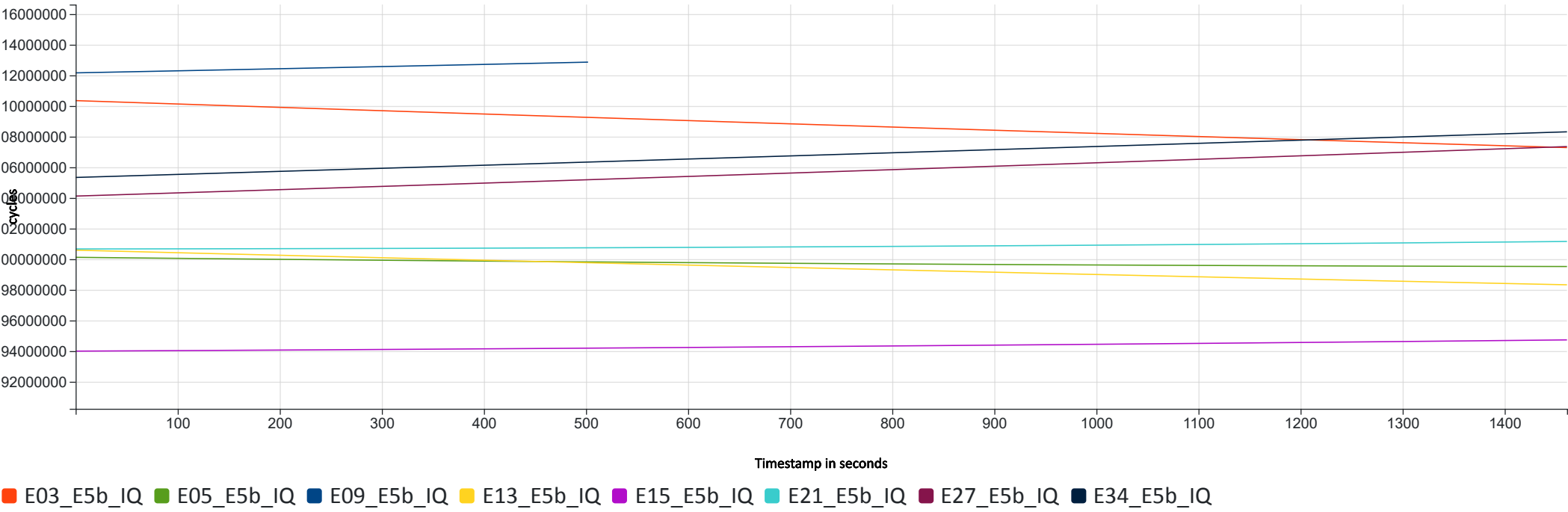
Advanced Base Station GNSS QC

Phase

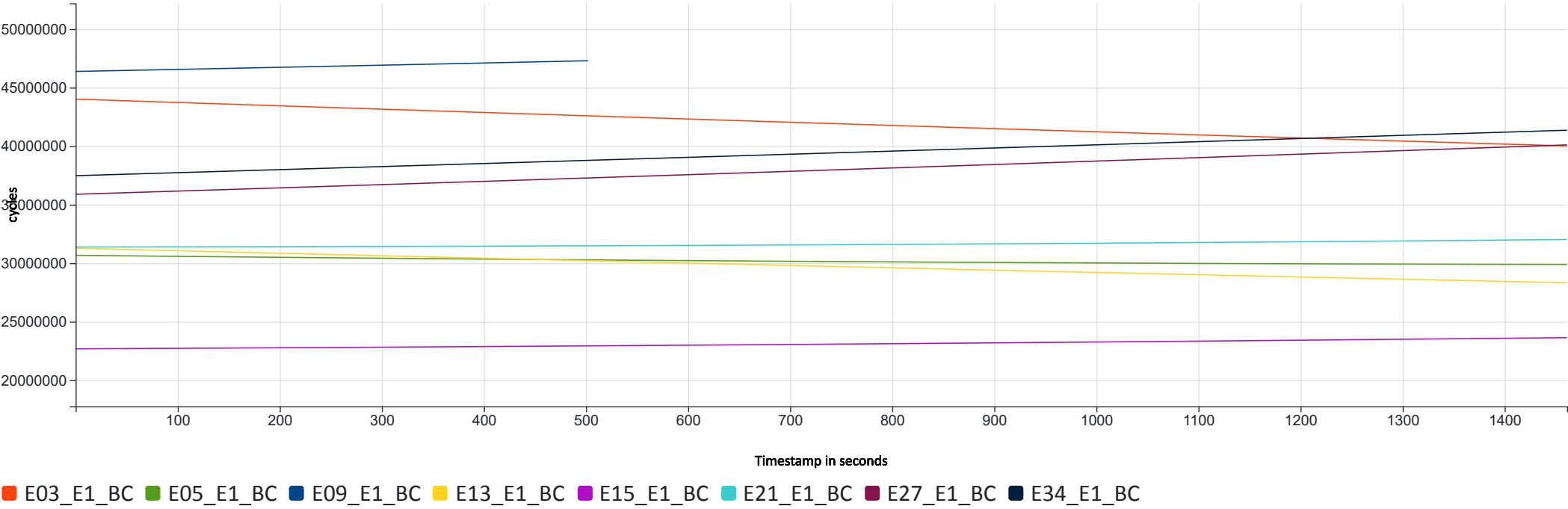
Phase (cycles) - Galileo E5a+b



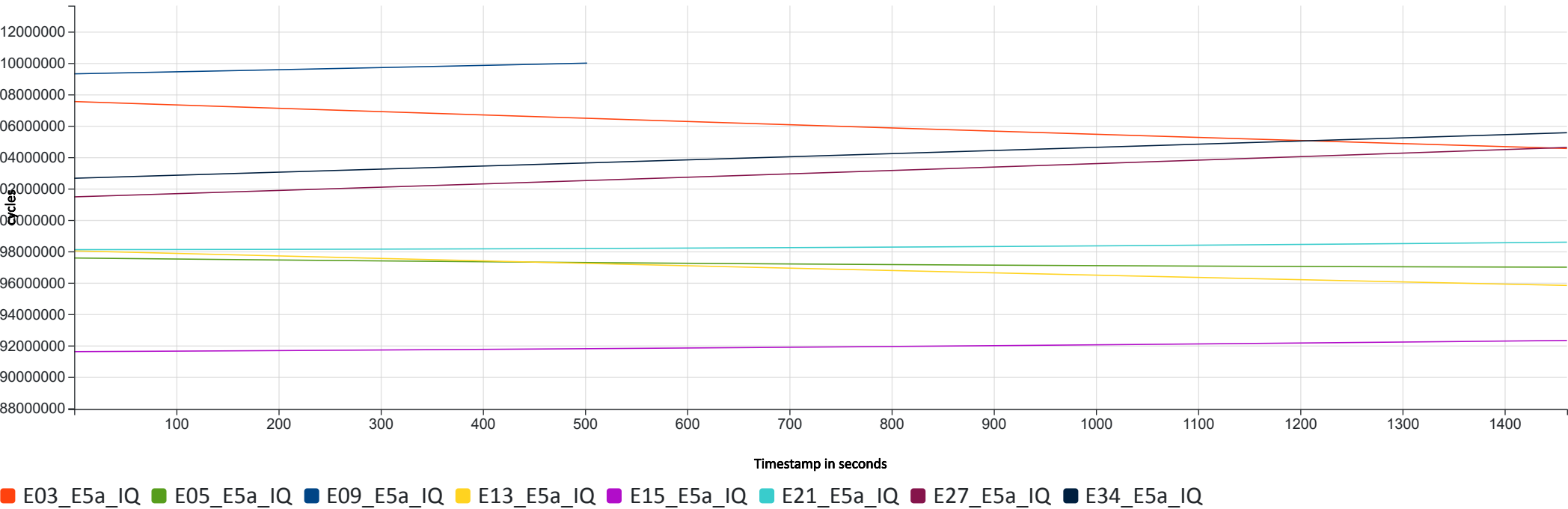
Phase (cycles) - Galileo E5b



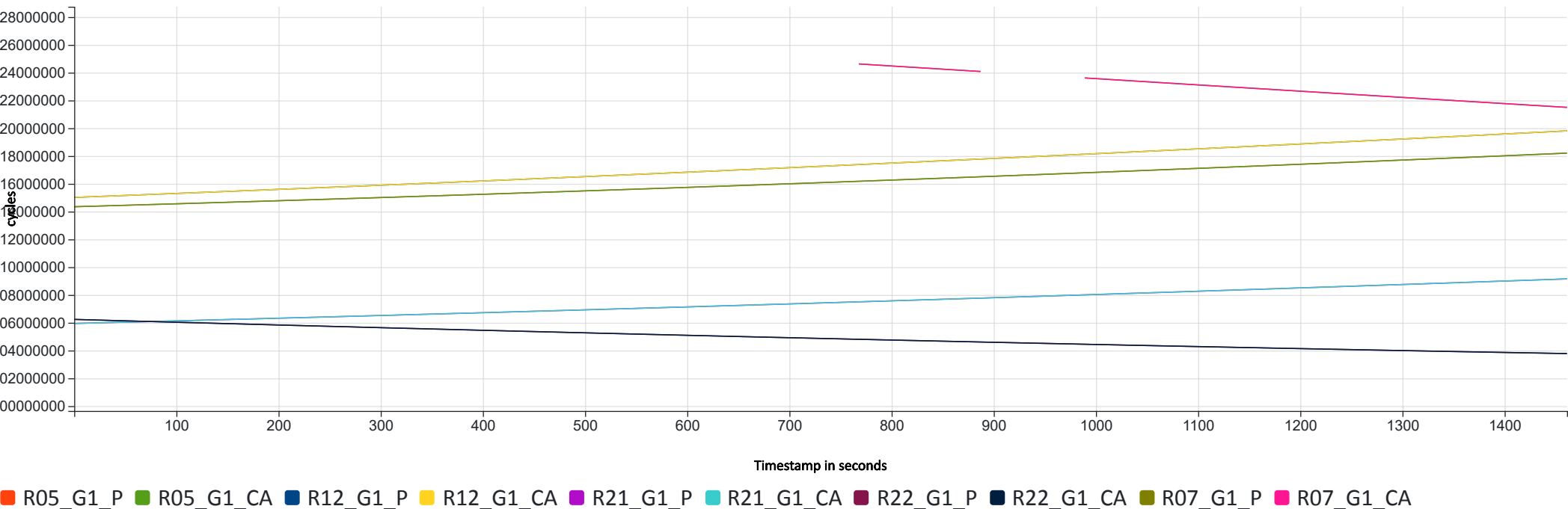
Phase (cycles) - Galileo E1



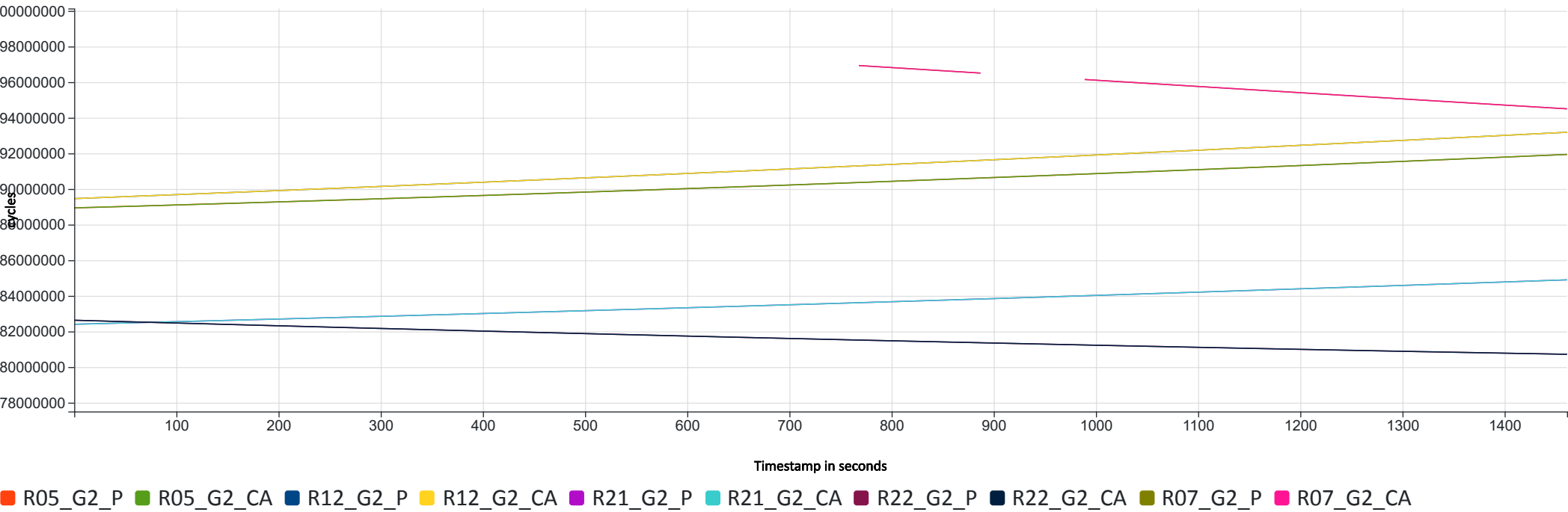
Phase (cycles) - Galileo E5a



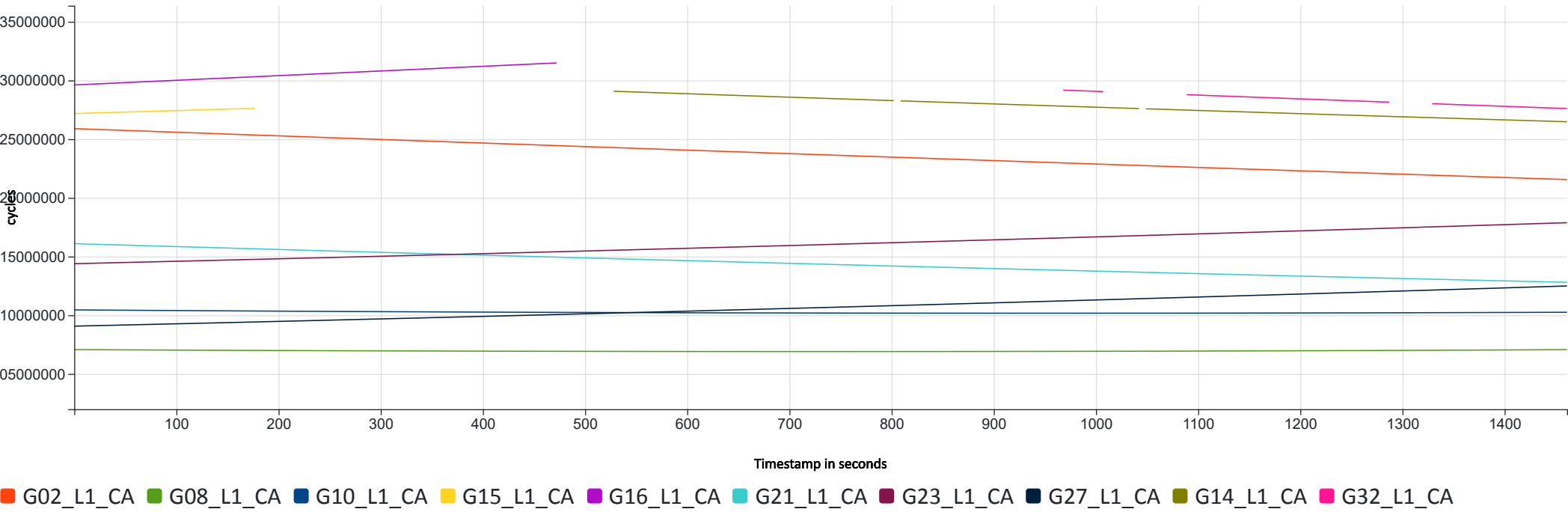
Phase (cycles) - Glonass G1



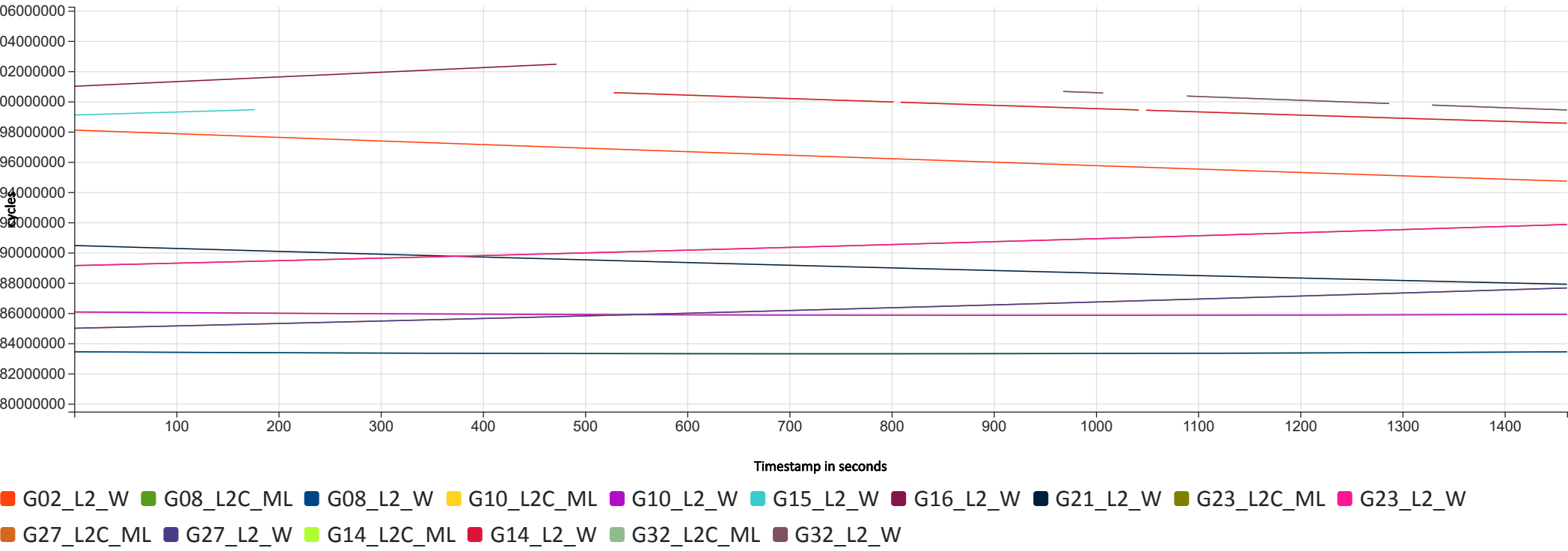
Phase (cycles) - Glonass G2



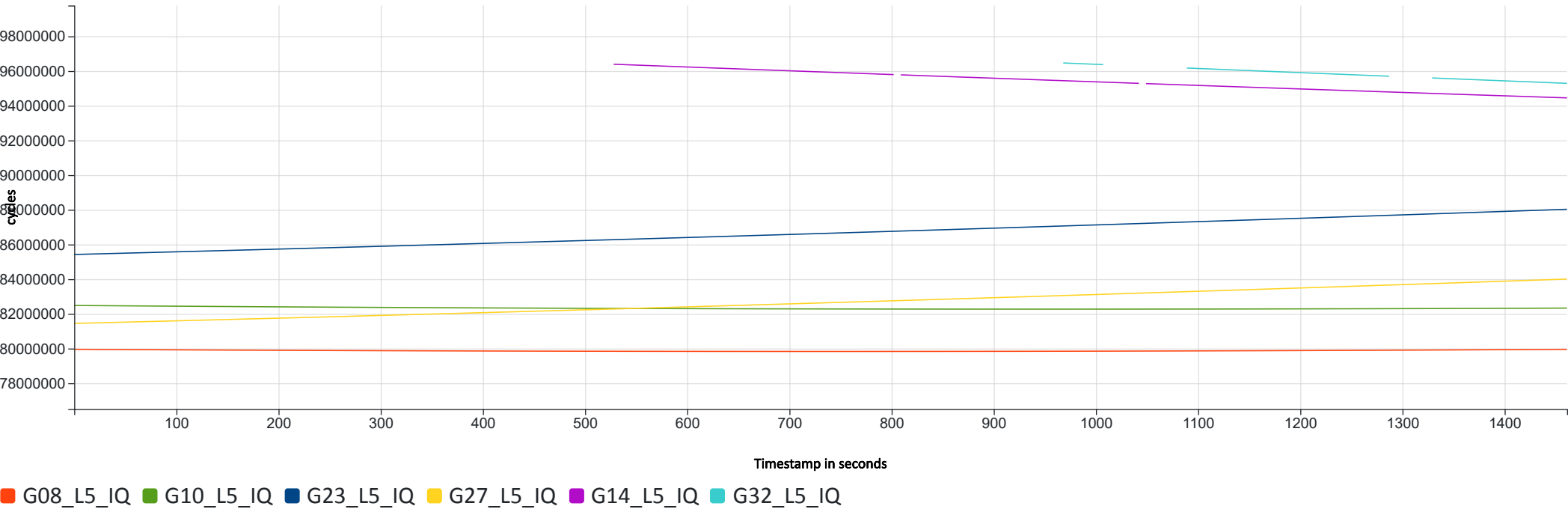
Phase (cycles) - GPS L1



Phase (cycles) - GPS L2

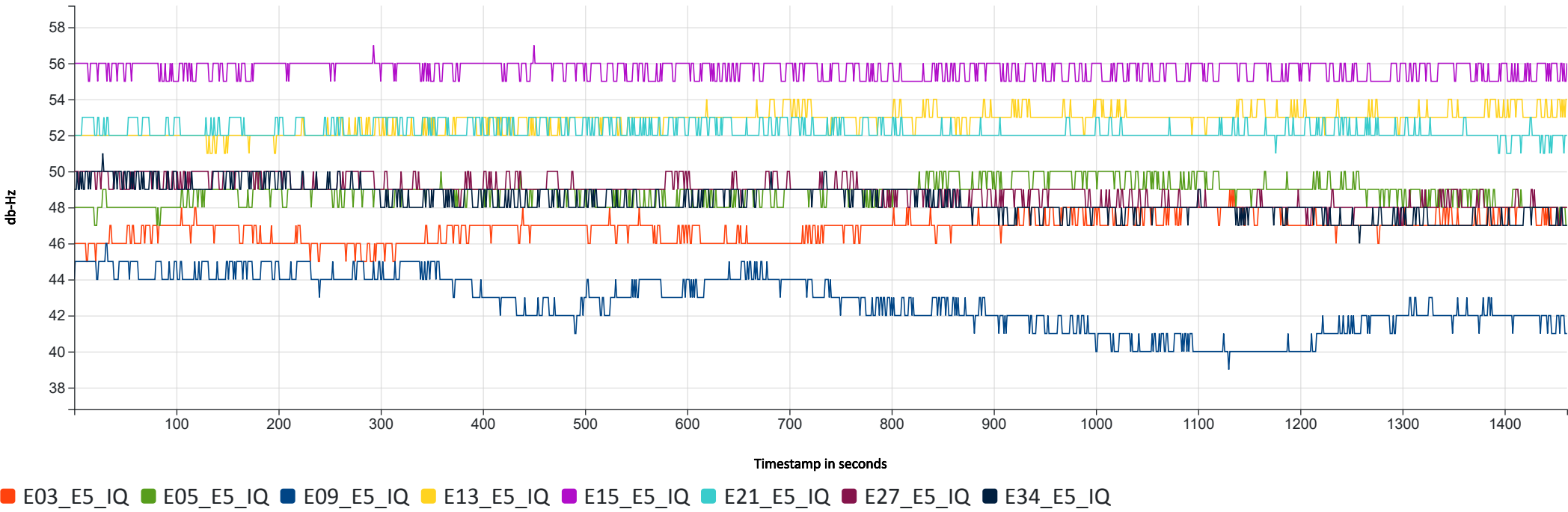


Phase (cycles) - GPS L5

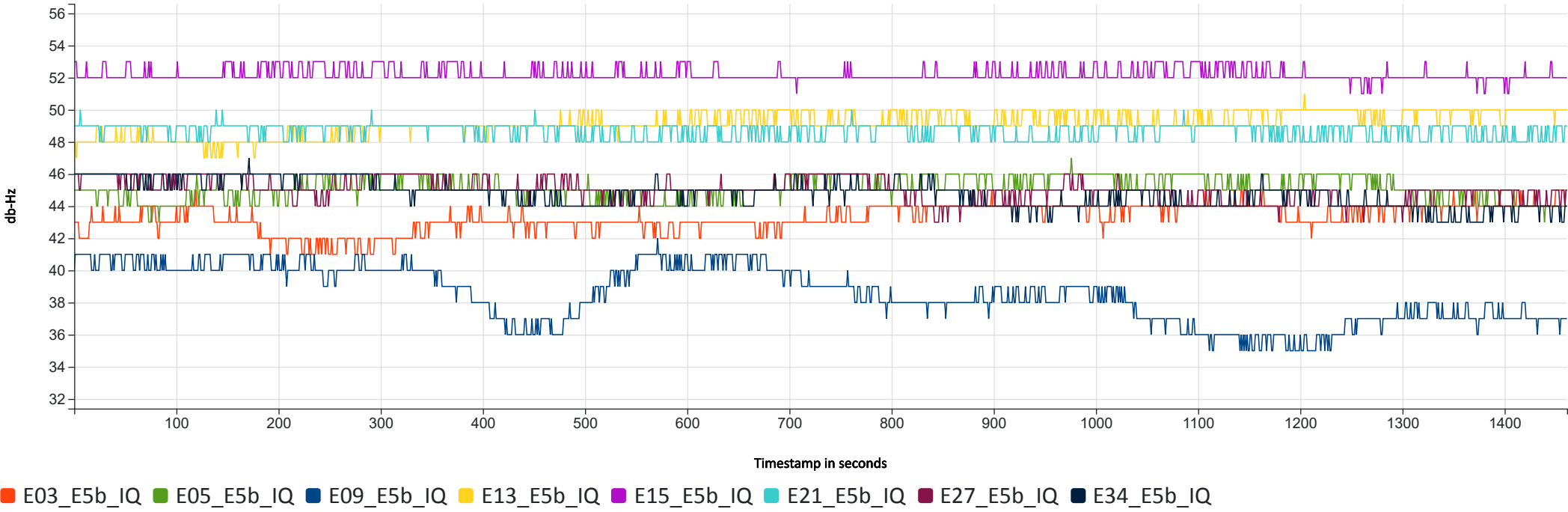


Carrier to noise

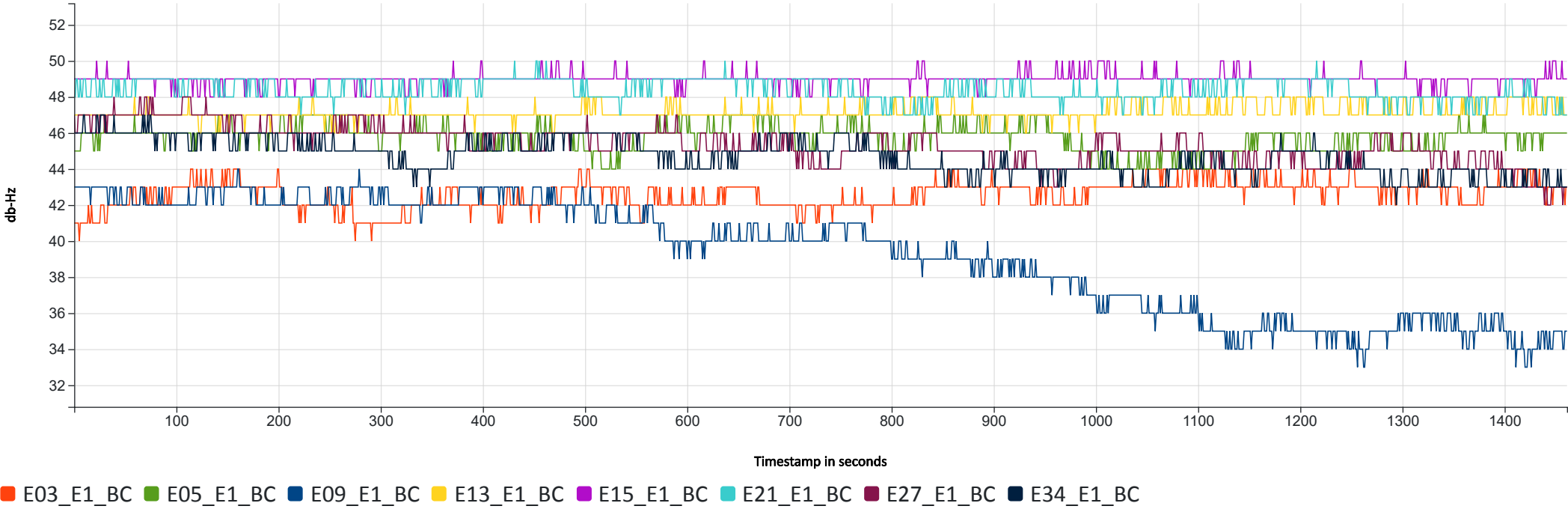
Carrier to noise (dB-hz) - Galileo E5a+b



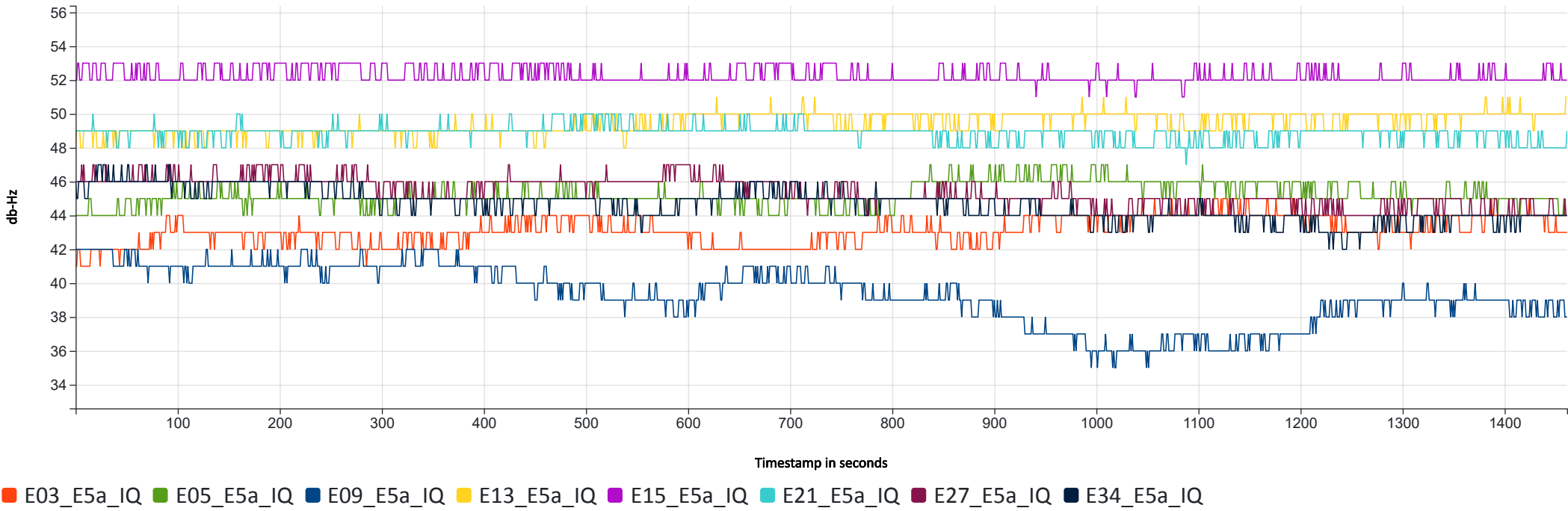
Carrier to noise (dB-hz) - Galileo E5b



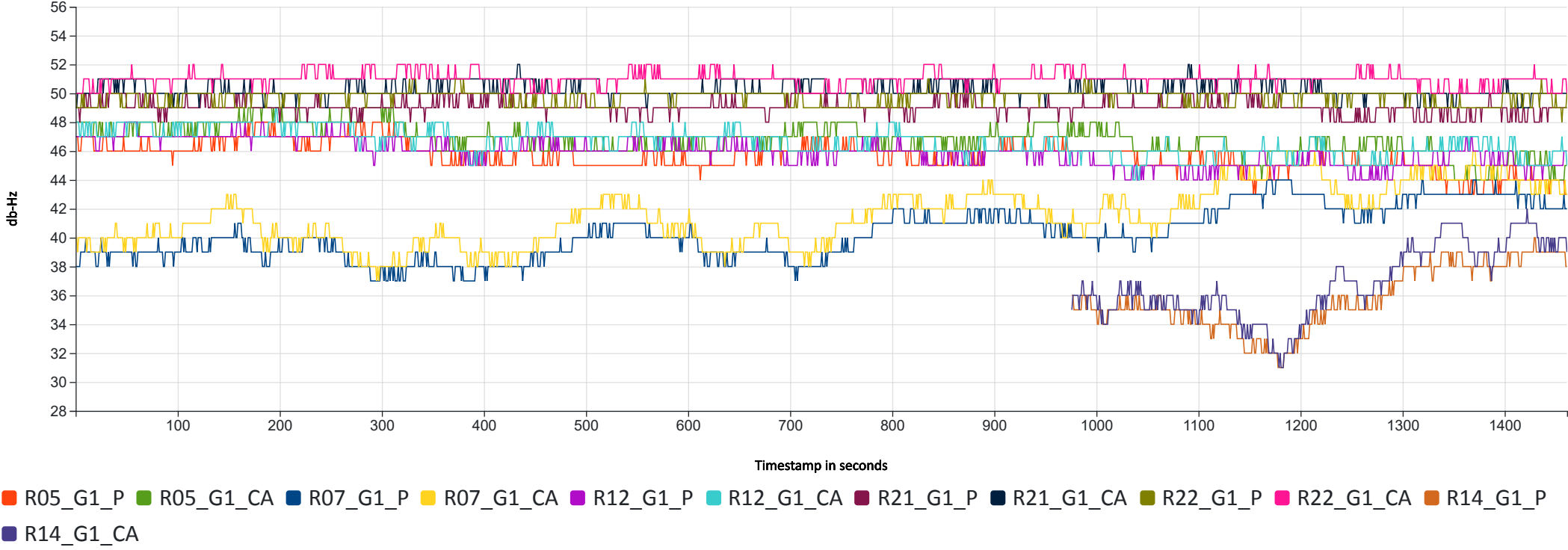
Carrier to noise (dB-hz) - Galileo E1



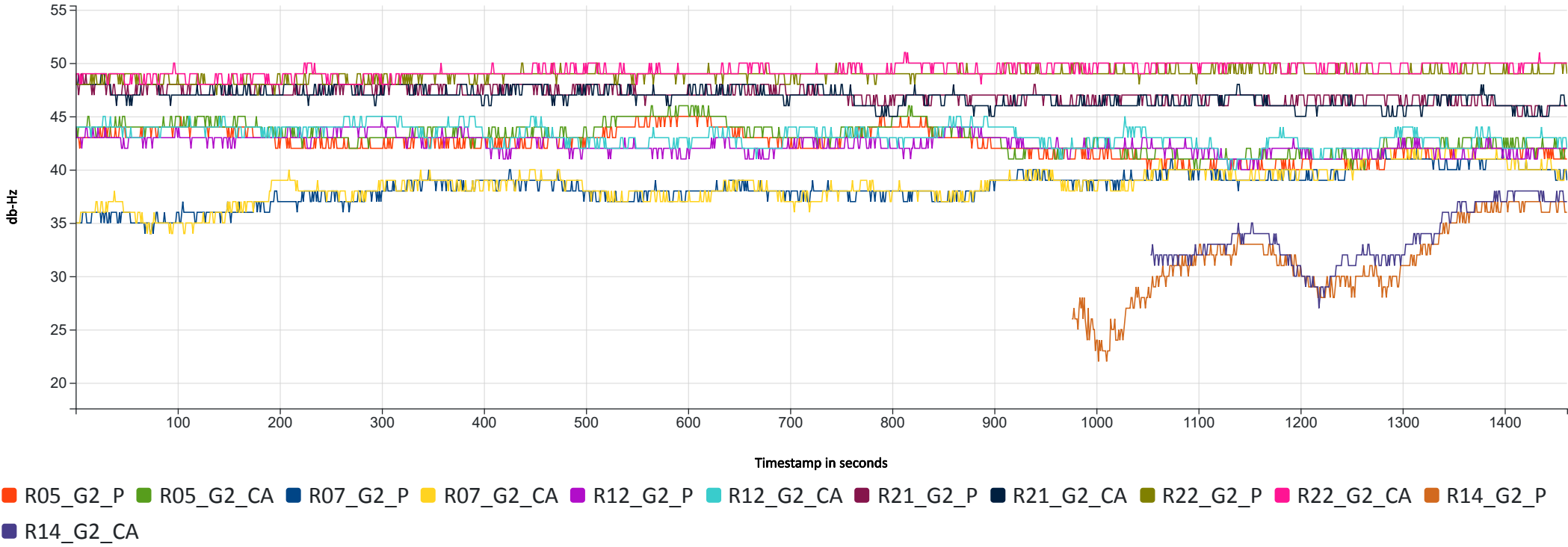
Carrier to noise (dB-hz) - Galileo E5a



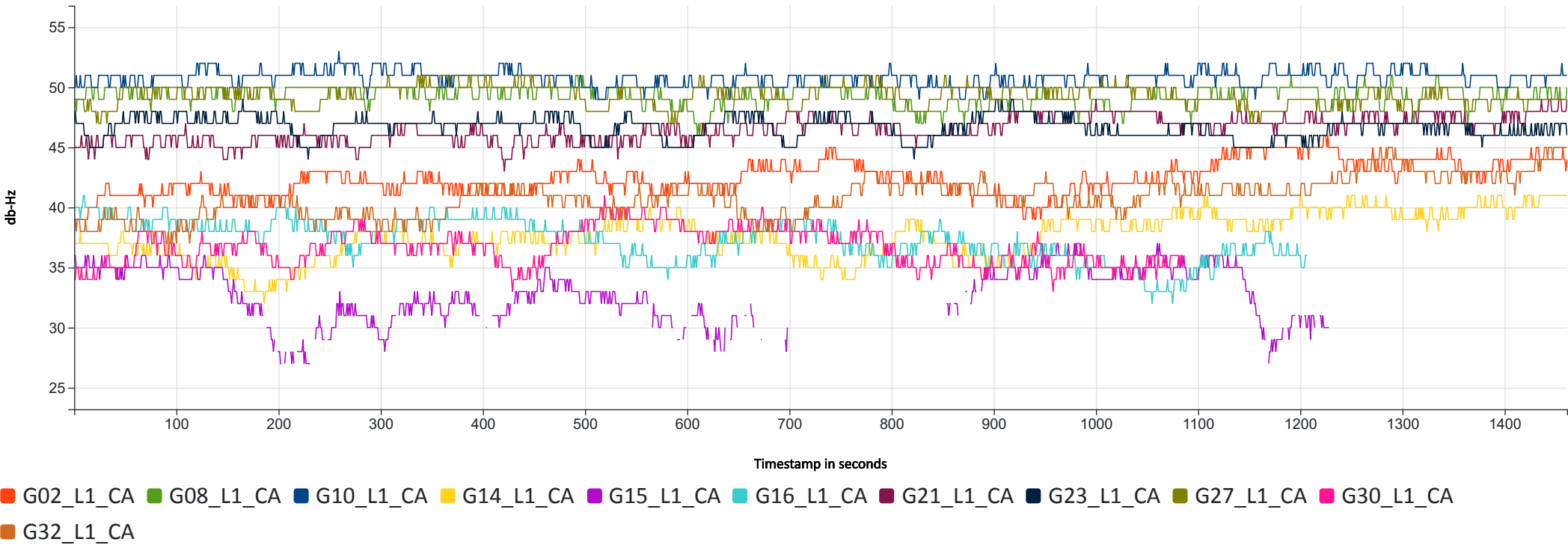
Carrier to noise (dB-hz) - Glonass G1



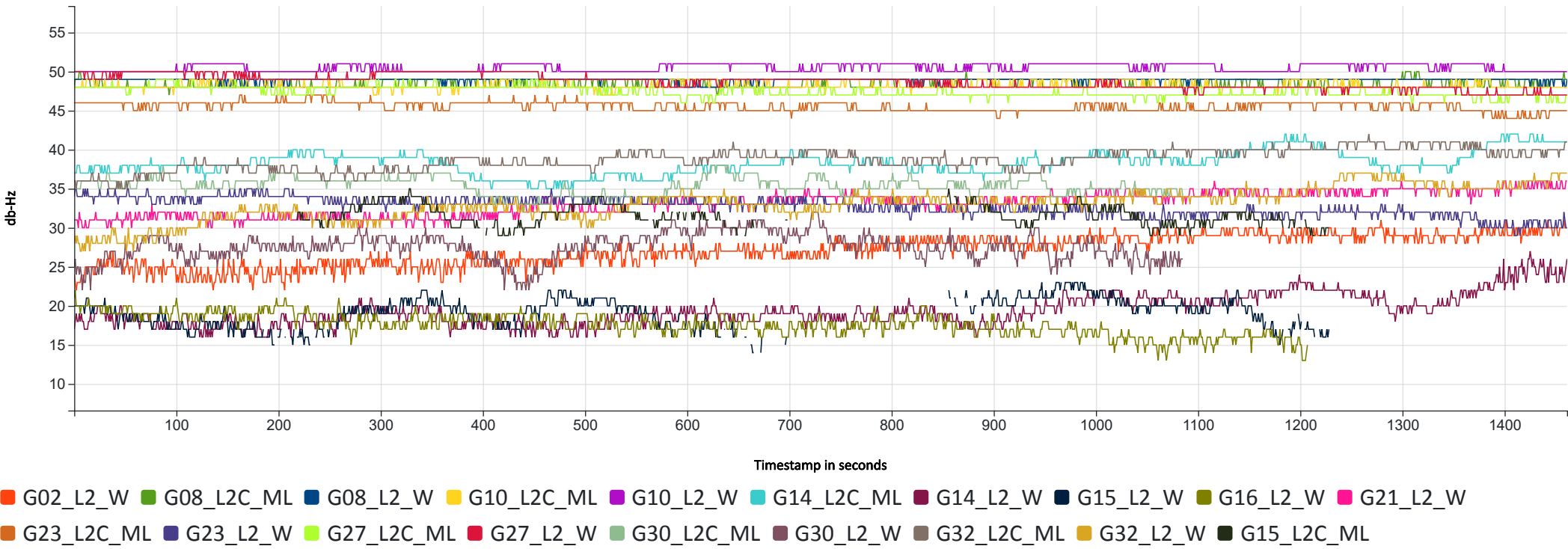
Carrier to noise (dB-hz) - Glonass G2



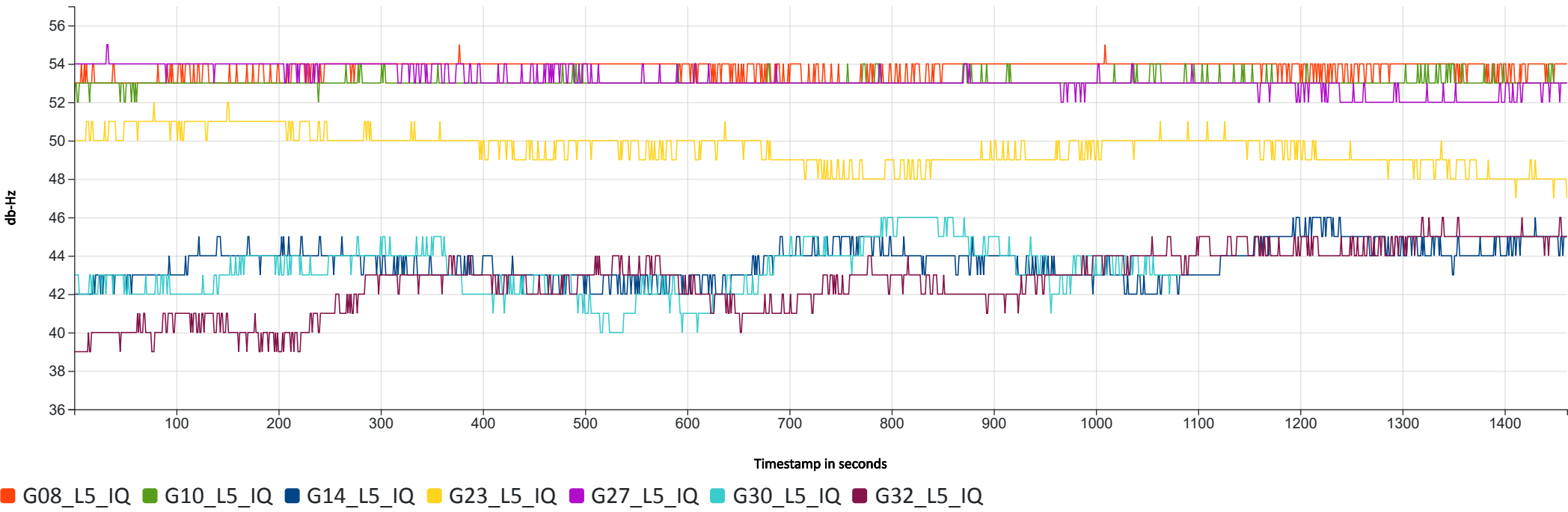
Carrier to noise (dB-hz) - GPS L1



Carrier to noise (dB-hz) - GPS L2

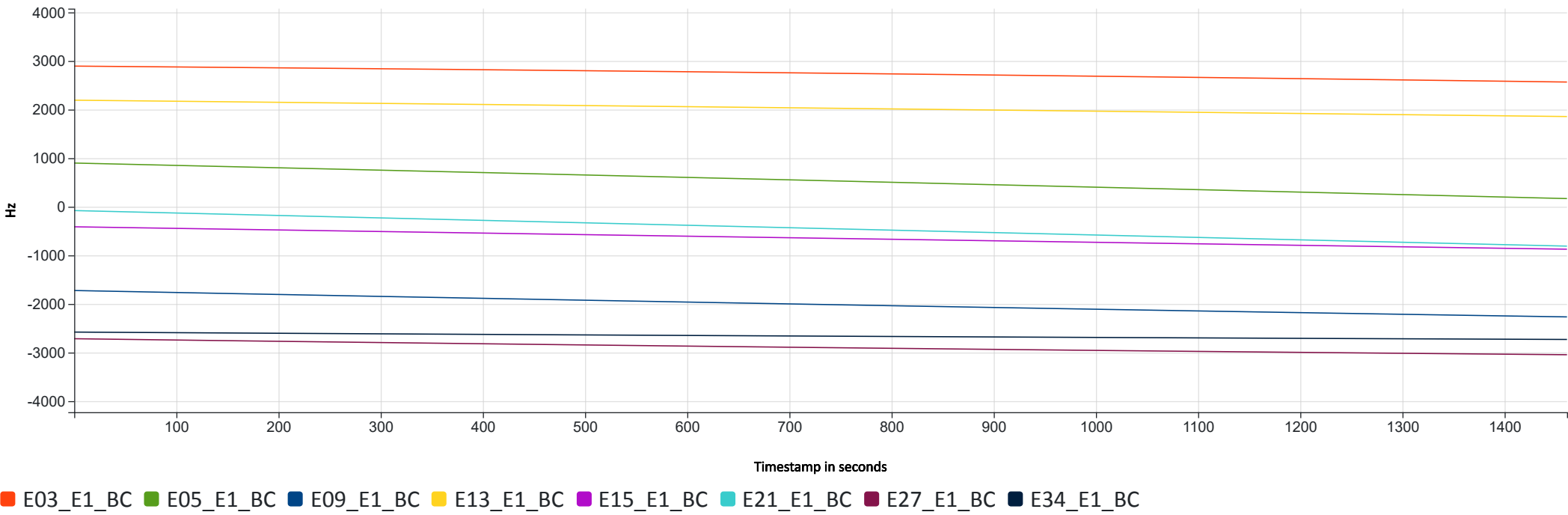


Carrier to noise (dB-hz) - GPS L5

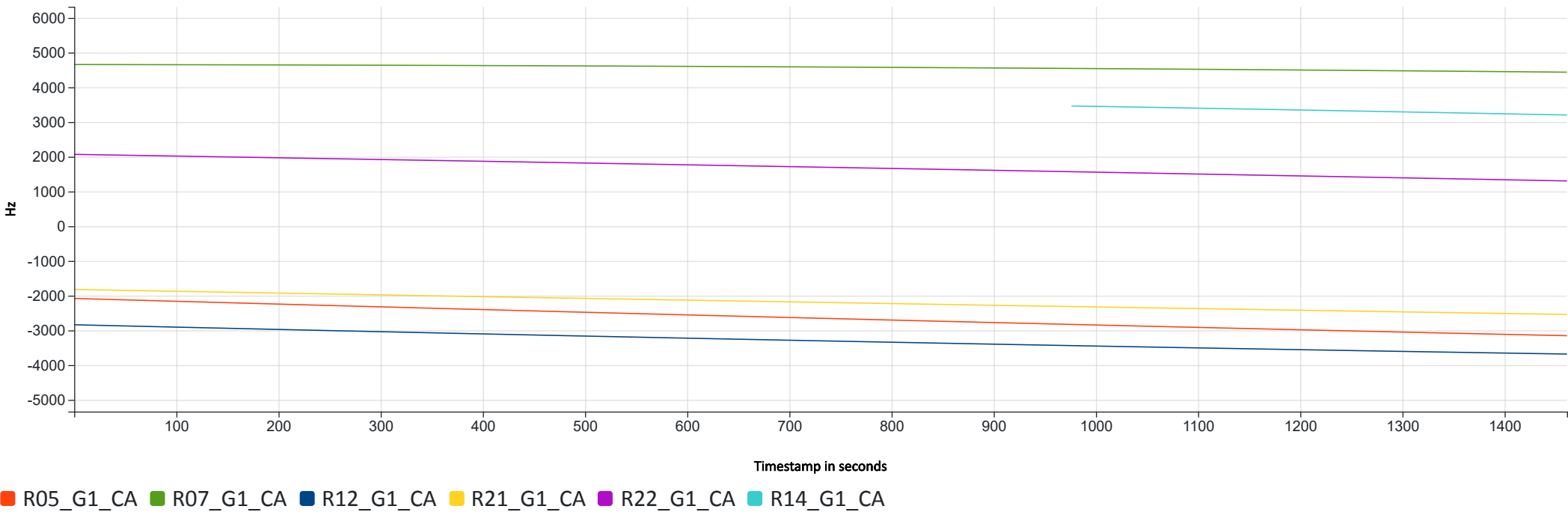


Doppler

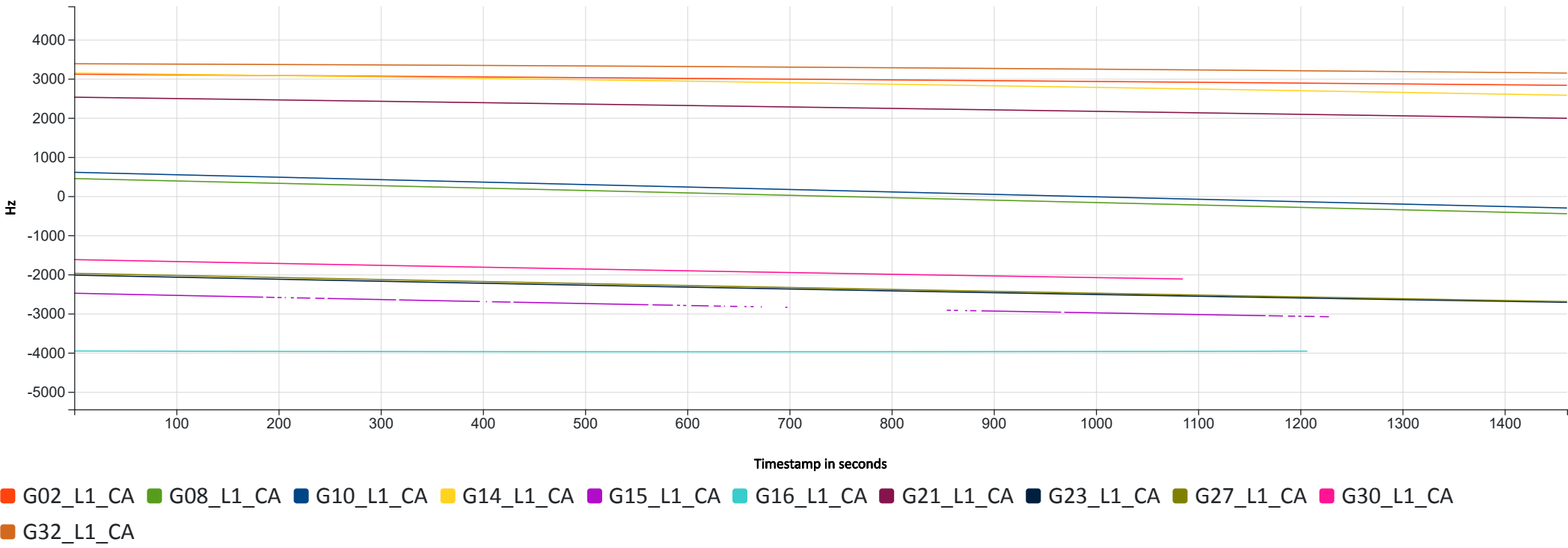
Doppler (hz) - Galileo E1



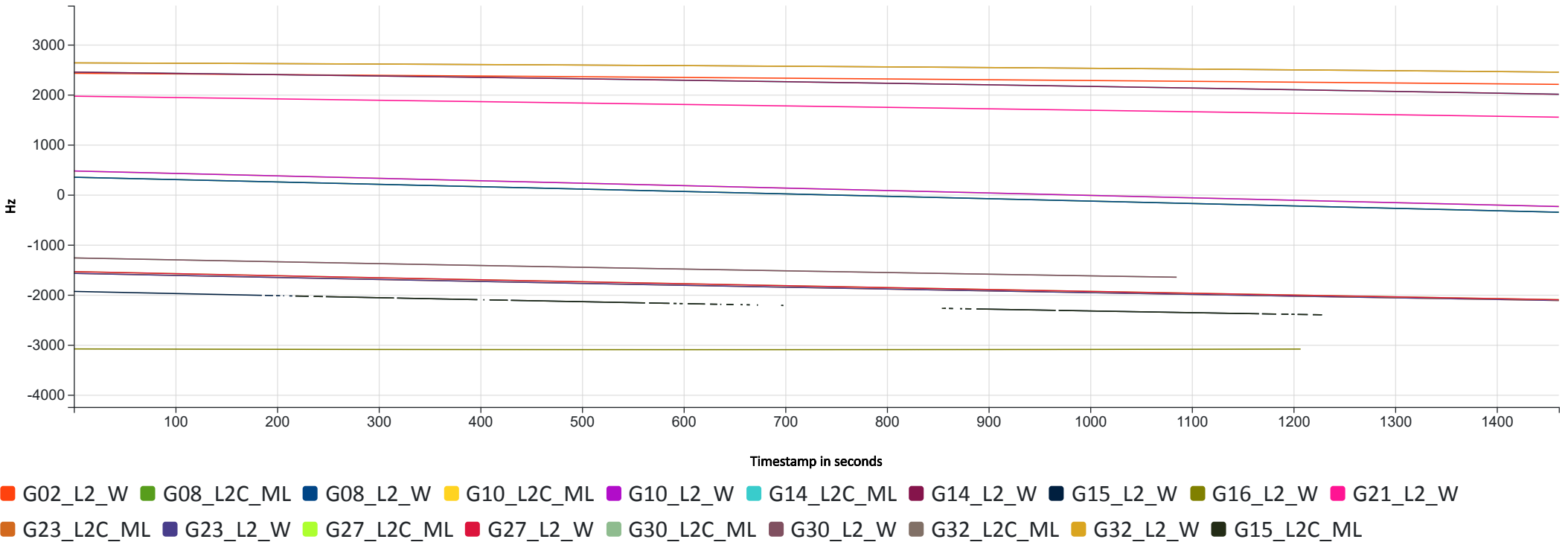
Doppler (hz) - Glonass G1



Doppler (hz) - GPS L1

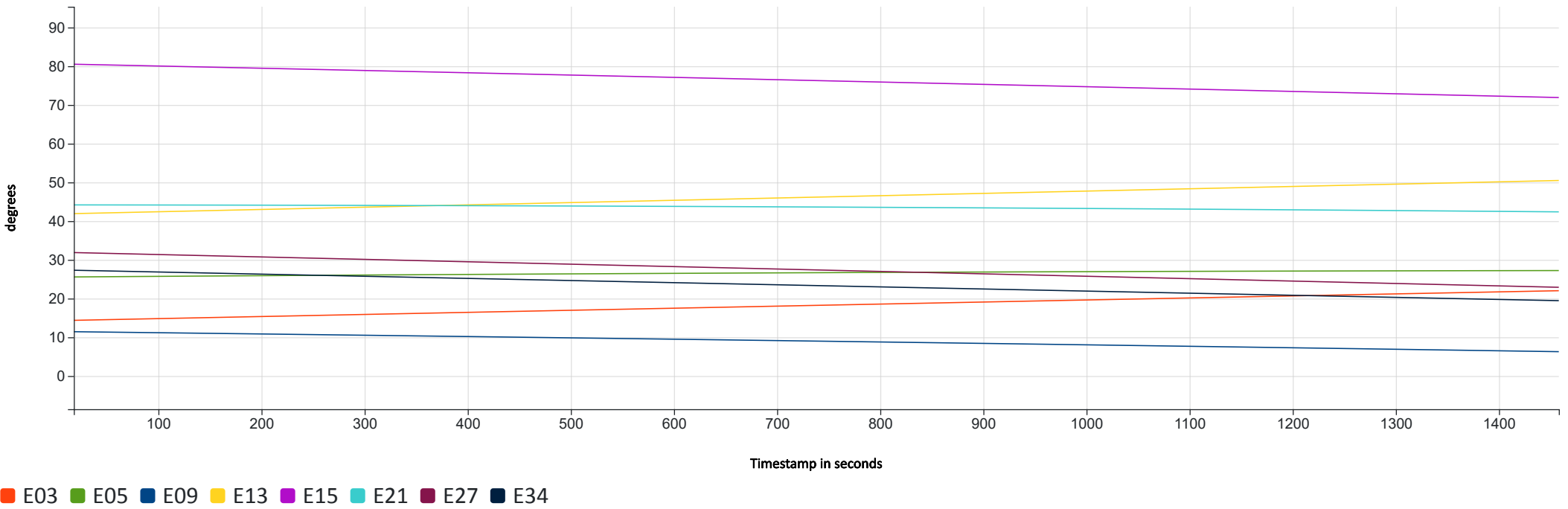


Doppler (hz) - GPS L2

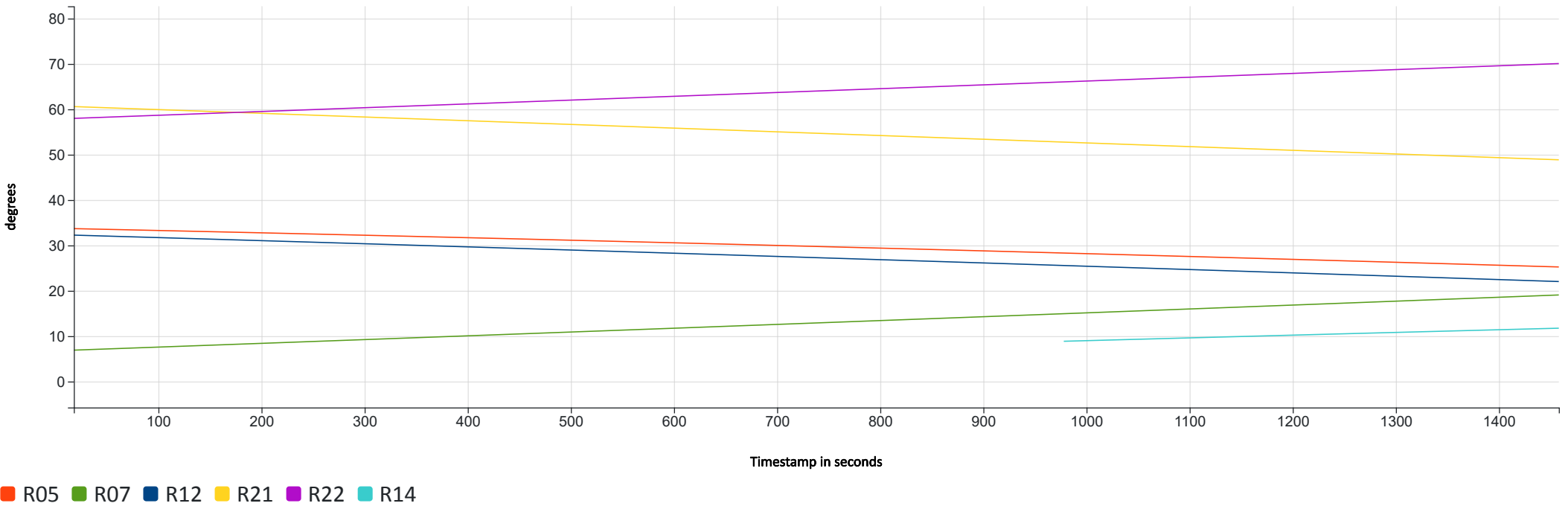


Elevation

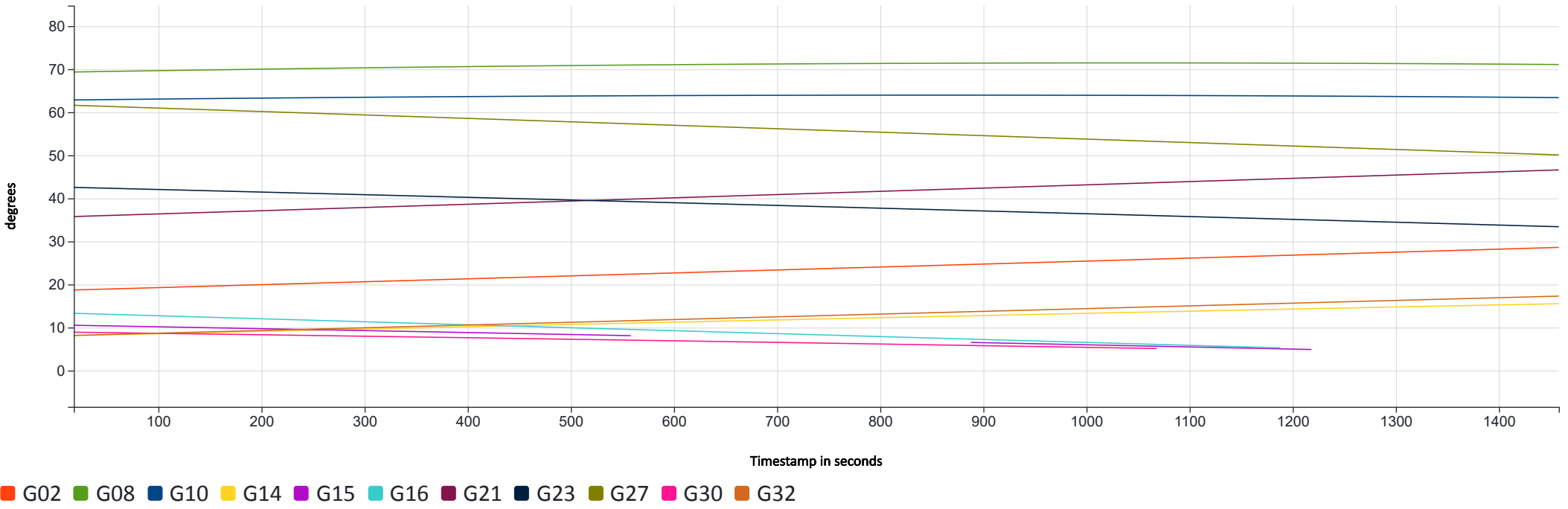
Elevation (degrees) - GALILEO



Elevation (degrees) - GLONASS

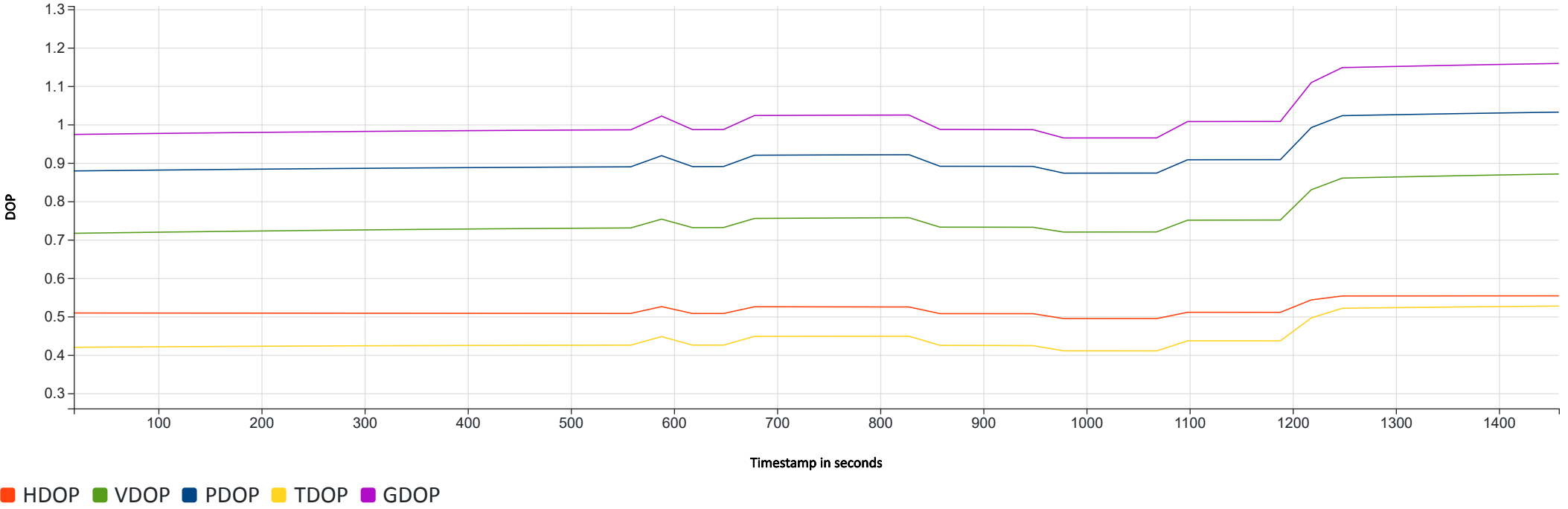


Elevation (degrees) - GPS



Dilution of precision

Dilution of precision (DOP)



Export Jobs

This section lists all export tasks that have been executed to review generated output files and general export status.

JOB #1 - SBET TXT 200Hz - MSL

INFO

Profile Name	SBET TXT 200Hz - MSL
Plugin	Text Exporter
Dataset	merged
Number of Exported Files	1
Export Status	Success

EXPORTED FILES

File path
E:/20240829_Olberg/LIDAR/YS-20240829-053522/GNSS PostProcess/postprocess_20240830-033503/YS2024-08-2905_35_42sbet.txt